

深度学习在网络安全应用中的应用：综合调查

Vinayakumar Ravi, Mamoun Alazab, Soman KP, Sriram Srinivasan, Sitalakshmi Venkatraman, Quoc-Viet Pham, 和 Simran K

摘要—深度学习（DL），一种新型机器学习（ML），由于其成功应用于许多经典人工智能（AI）任务，与经典机器学习算法（CMLAs）相比，正获得越来越多的研究兴趣。最近，深度学习架构正被创新性地建模用于网络安全领域的多样化应用。文献中现在正增长着深度学习架构及其变体，用于探索不同创新深度学习模型和原型，这些模型可以定制以适应特定的网络安全应用。然而，文献中缺乏一份综合调查报告此类研究。许多基于调查的研究专注于特定的深度学习架构和某些类型的恶意攻击，在有限的过去网络安全问题场景中，缺乏未来展望。本文旨在提供对过去、现在和未来深度学习架构的综合调查，包括与智能自动化、物联网（IoT）、大数据（BD）、区块链、云和边缘技术相关的下一代网络安全场景。

本文通过比较和分析各种近期研究论文的贡献与挑战，以教程风格全面综述了用于网络安全多样化应用的最先进的深度学习架构。首先，本综述的独特之处在于报告了深度学习架构在广泛网络犯罪检测方法中的应用，例如入侵检测、恶意软件和僵尸网络检测、垃圾邮件和网络钓鱼检测、网络流量分析、二进制分析、内部威胁检测、CAPTCHA分析和隐写术。其次，本综述涵盖了网络安全应用领域中的关键深度学习架构，如密码学、云安全、生物识别安全、物联网和边缘计算。第三，讨论了基于深度学习的研究需求，用于下一代网络物理系统（CPS）中的网络安全应用，这些系统利用行为数据分析、自然语言处理（NLP）、信号与图像处理以及区块链技术，为未来智慧城市和工业4.0提供支持。最后，对开放挑战和提出的深度学习架构的批判性讨论为未来研究方向做出了贡献。

索引词—网络安全，机器学习，神经网络，深度学习，通信网络，云和

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边缘计算。

WITH 互联网已成为每个人的基本资源，由于网络物理系统（CPS）在云计算、移动计算、边缘计算和物联网（IoT）等技术的推动下快速发展，我们正进入工业4.0时代。然而，随着系统变得越来越异构、复杂和互联，CPS中的网络安全问题也日益重要，因为其固有的安全风险和漏洞。2018 [1]中漏洞总数增加了13%。预计在野外出现的零日漏洞将从2015年的每周一个增加到 2021 [2]的每天一个。虽然全球网络安全就业市场正在增长以应对这种情况，但熟练的网络安全研究人员和从业者存在危机，未来可能存在高达25%的培训专业人员短缺。 [3]需要一份作为网络安全专业人员的教程的综述。重要的是要识别文献中的差距，以帮助解决未来ICT系统中的网络安全重要问题。

网络安全这一术语已经演变为一系列概念和程序，旨在保护信息与通信技术（ICT）系统和网络，以在网络空间中保持信息的机密性、完整性和可用性。网络犯罪涉及在CPS中进行的犯罪活动，导致计算机硬件、网络 and 软件遭受恶意攻击。更重要的是，与未经授权的访问、盗窃、披露以及有意或无意的损害相关的数据完整性风险日益重要。尽管网络安全领域的不法分子和对手数量随时间增加，但总体威胁类别并未改变。安全研究的主要目标是攻击者实现其目标设置障碍，因此，了解各种类型的攻击至关重要。为了应对这些威胁，人们正在采用各种网络安全方法，如入侵检测（ID）、社交网络分析、恶意软件分析、高级持续性威胁、Web应用安全和应用密码学。然而，随着CPS向工业4.0的巨大发展，目前缺乏一个能够主动应对系统和物理过程变化的适应性网络安全框架。

当前网络安全工具利用来自网络传感器、日志和端点代理的大量数据，这些数据可以使用数据挖掘（DM）技术高效处理

Deep Learning for Cyber Security Applications: A Comprehensive Survey

Vinayakumar Ravi, Mamoun Alazab, Soman KP, Sriram Srinivasan, Sitalakshmi Venkatraman, Quoc-Viet Pham, and Simran K

Abstract—Deep Learning (DL), a novel form of machine learning (ML) is gaining much research interest due to its successful application in many classical artificial intelligence (AI) tasks as compared to classical ML algorithms (CMLAs). Recently, DL architectures are being innovatively modelled for diverse applications in the area of cyber security. The literature is now growing with DL architectures and their variations for exploring different innovative DL models and prototypes that can be tailored to suit specific cyber security applications. However, there is a gap in literature for a comprehensive survey reporting on such research studies. Many of the survey-based research have a focus on specific DL architectures and certain types of malicious attacks within a limited cyber security problem scenario of the past and lack futuristic review. This paper aims at providing a well-rounded and thorough survey of the past, present, and future DL architectures including next-generation cyber security scenarios related to intelligent automation, Internet of Things (IoT), Big Data (BD), Blockchain, cloud and edge technologies.

This paper presents a tutorial-style comprehensive review of the state-of-the-art DL architectures for diverse applications in cyber security by comparing and analysing the contributions and challenges from various recent research papers. Firstly, the uniqueness of the survey is in reporting the use of DL architectures for an extensive set of cybercrime detection approaches such as intrusion detection, malware and botnet detection, spam and phishing detection, network traffic analysis, binary analysis, insider threat detection, CAPTCHA analysis, and steganography. Secondly, the survey covers key DL architectures in cyber security application domains such as cryptography, cloud security, biometric security, IoT and edge computing. Thirdly, the need for DL based research is discussed for the next generation cyber security applications in cyber physical systems (CPS) that leverage on BD analytics, natural language processing (NLP), signal and image processing and blockchain technology for smart cities and Industry 4.0 of the future. Finally, a critical discussion on open challenges and new proposed DL architecture contributes towards future research directions.

Index Terms—Cyber Security, Machine Learning, Neural Networks, Deep Learning, Communication Networks, Cloud and

This work was supported by the Department of Corporate and Information Services, Northern Territory Government of Australia, the Paramount Computer Systems, and Lakhshya Cyber Security Labs. Mamoun Alazab was with Charles Darwin University, Australia e-mail: (mamoun.alazab@cdu.edu.au). Vinayakumar R was with Division of Biomedical Informatics, Cincinnati Children’s Hospital Medical Center, Cincinnati, OH, United States e-mail: (Vinayakumar.Ravi@cchmc.org, vinayakumarr77@gmail.com) Vinayakumar R, Sriram S, Soman KP and Simran K was with Center for Computational Engineering and Networking(CEN), Amrita School of Engineering, Coimbatore, Amrita Vishwa Vidyapeetham, India e-mail: (r_vinayakumar@cb.amrita.edu, sri27395ram@gmail.com). Quoc-Viet Pham is with Research Institute of Computer, Information and Communication, Pusan National University, 2 Busandaehakro 63beon-gil, Geumjeong-gu, Busan 46241, South Korea e-mail: (vietpq@pusan.ac.kr). Sitalakshmi Venkatraman is with Department of IT, Melbourne Polytechnic, Prahran Campus, VIC 3181, Australia e-mail: (sitavenkat@melbournepolytechnic.edu.au)

Edge Computing.

I. INTRODUCTION

WITH Internet becoming an essential resource for everyone, we are entering Industry 4.0 due to a rapid advancement in cyber physical systems (CPS) geared by technologies such as cloud computing, mobile computing, edge computing, and Internet of Things (IoT). However, the topic of cyber security in CPS is also growing in importance due to the inherent security risks and vulnerabilities as systems become increasingly heterogeneous, complex, and interconnected. The overall number of vulnerabilities has increased by 13% in 2018 [1]. It is predicted that zero-day exploits seen in the wild will grow from one per week in 2015 to one per day by 2021 [2]. While the cyber security job market is growing globally to counter this situation, there is a crisis of skilled cyber security researchers and practitioners with a potential shortage of trained professionals up to 25% [3]. There is a need for a survey that serves as a tutorial for cyber security professionals. It is important to identify the gaps in literature to aid addressing the important problem in cyber security for future ICT systems.

The term cyber security has evolved as a set of concepts and procedures to protect ICT systems and networks with the objective of preserving the confidentiality, integrity, and availability of information in the Cyberspace. Cybercrime deals with criminal activities carried out in the CPS resulting in computer hardware, networks, and software being maliciously attacked. More importantly, risks related to data integrity from unauthorized access, theft, disclosure, and intentional or accidental harm are of growing importance. Even though there is an increase in miscreants and adversaries in the field of cyber security over time, there has been no change in general threat categories. The main objective of security research is to prevent the attackers from achieving their goals and therefore, it is extremely important to have a good knowledge of various types of attacks. To counter such threats, various cyber security approaches such as intrusion detection (ID), social network analysis, malware analysis, advanced persistent threats, web application security, and applied cryptography are being adopted. However, with the huge emergence of CPS towards Industry 4.0, there is lack of an adaptive cyber security framework that can react and respond to changes in systems and physical processes proactively.

Current cyber security tools make use of huge amounts of data from network sensors, logs, and endpoint agents that can be efficiently processed using data mining (DM) techniques

以提供关于恶意活动的及时信息。DM技术提取隐藏特征以区分正常和恶意活动，并成功应用于市场中的传统网络安全解决方案，例如网络级和主机级防火墙、杀毒软件、入侵检测系统（IDSs）和入侵防御系统（IPSs）。然而，这些解决方案仅能检测已知的恶意活动，无法检测与工业4.0和大数据（BD）相关的全新恶意活动。由于BD数据具有量大、格式多样、类型复杂等特点，这些网络安全解决方案因其领域专家的局限性而面临诸多挑战。BD分析能够收集、存储、处理和可视化海量数据。因此，将BD分析应用于网络安全变得至关重要，并形成了最近的研究方向。

随着CPS中新技术的发展，终端用户系统生成数据表现出未知模式，这些模式尚未被建模用于分类为恶意软件或良性软件。此外，资源受限的物联网设备由于其固有的处理限制，存在多个漏洞和安全漏洞。因此，网络安全专家和研究人员正在探索使用机器学习（ML）和深度学习（DL）的认知技术，以便将此类基于人工智能（AI）的技术整合到系统中，使其判断接近领域专家 [4]。然而，根据现有文献的讨论，人们认识到存在误导ML/DL部署的潜在风险 [5]。

ML和DL已成为计算机视觉和语音处理领域各种应用的基本工具。与经典机器学习算法（CMLAs）相比，DL架构的性能更好，更重要的是，在计算机视觉和健康相关应用中，其表现优于领域专家 [6]。CMLAs的主要缺点之一是它们依赖于特征工程方法，这些方法通常由领域专家指导。因此，安全研究人员正在探索DL方法来处理动态演变的恶意活动。与CMLAs相比，DL能够从复杂系统（包括自然语言处理（NLP））中捕获重要特征，因此非常适合处理大量数据样本[5]。随着技术的不断发展，网络安全数据每天都在持续增长，基于深度学习的解决方案的性能也随之提高。最近，提出了几种DL架构，在进入工业4.0时代时，对它们相对适用性进行调研以应对日益增长的网络安全问题至关重要。

A. 现有关于机器学习和深度学习的网络安全调查

过去十年中为解决网络安全问题而开发的传统机器学习框架已被广泛调查 [7], [8], [9], [10], [11], [12]。然而，这些不包括深度学习方法。关于深度学习框架的调查仅限于网络安全中一个非常狭窄的应用领域。大多数研究集中在特定

网络安全技术，例如入侵检测 [13], [14],垃圾邮件检测、异常检测和恶意软件分析 [15]。最近的研究集中于提供与防御CPS [16], 相关工作的综述，并且一些研究讨论了用于保护物联网技术的各种机器学习和深度学习方法 [17]。某些调查包括简短的教程类型，要么专注于深度强化学习（DRL）在网络安全中的应用，要么在检测各种攻击（包括恶意软件、垃圾邮件、内部威胁、网络入侵、虚假数据注入和僵尸网络使用的恶意域名）中包含通用深度学习框架 [18][19], [20]。然而，以前的调查只关注某些网络安全场景，从而限制了范围。特别是，它们没有讨论使用深度学习来保护下一代通信网络（如自动驾驶汽车网络、云/边缘计算和区块链）的情况，这些是向5G网络和工业4.0发展的下一代技术。尽管关于将机器学习和深度学习应用于网络安全的调查很多，据我们所知，还没有在如此广泛和全面的范围内进行过关于深度学习所进行的各种研究的详细和全面的调查，以包括下一代计算的复杂性。本文旨在填补文献中的这一空白。

B. 研究贡献

考虑到现有文献中关于下一代计算所用深度学习架构的调查的局限性，我们这项工作的目标是为深度学习在网络安全中的应用提供广泛而全面的调查。我们这项调查的主要贡献可以总结如下：

1) 本文回顾了部署于各种网络安全应用中的深度学习架构，并对其演进过程进行了概述。此外，本调查总结了、比较了和对比了各种深度学习架构，为读者提供了对过去、现在和未来网络安全中深度学习应用的详细理解。

2) 本调查根据多种属性（如架构类型及其应用、研究年份、文本表示、数据集类型以及与CMLA的性能比较）对已报告将深度学习应用于网络安全的各种研究进行了分类。

3) 提供了关于离线与实时部署涉及的网络安全应用的各种问题及主要挑战的概述。此外，解释了网络安全领域共享任务的重要性。4) 研究了网络安全领域中先进的强化学习（RL）与对抗性机器学习应用。5) 讨论了深度学习在生物识别、自然语言处理、信号与图像处理中对于网络安全的重要性。6) 涵盖了基于深度学习架构的网络安全在智慧城市、普适计算、生物识别、物联网、雾计算与云计算以及自动驾驶车辆领域的作用。

to provide timely information about malicious activities. DM techniques extract the hidden features to differentiate between normal and malicious activities and are successfully adopted resulting in legacy cyber security solutions in the market such as network-level and host-level firewalls, antivirus software, Intrusion Detection Systems (IDSs), and Intrusion Protection Systems (IPSs). However, such solutions are effective in detecting only the known malicious activities and fail to detect new types of malicious activities associated with Industry 4.0 and big data (BD). Due to the BD characteristics of voluminous data in different formats, types and modes, many challenging issues exist in such cyber security solutions due to their limitations of domain experts. BD analytics has the capability to collect, store, process, and visualize very large volume of data. Therefore, applying BD analytics to cyber security becomes critical and forms a new research direction recently.

With the evolution of new technologies in CPS, data generated by the end-user systems exhibit unknown patterns that are yet to be modelled for classification as malware or benign. Further, the resource constrained IoT devices suffer from several vulnerabilities and security breaches due to their inherent processing limitations. Hence, cyber security specialists and researchers are exploring cognitive technologies using machine learning (ML) and deep learning (DL) for cyber security so that such artificial intelligence (AI) based techniques are incorporated into systems to make judgements as close to domain experts [4]. However, it is understood that there is a potential to mislead ML/DL deployment as discussed in existing literature [5].

ML and DL have become an essential tool for various applications in the field of computer vision and speech processing. The DL architectures have obtained better performance as compared to classical ML algorithms (CMLAs) and more importantly have outperformed domain experts in several computer vision and health related applications [6]. One of the major disadvantages of CMLAs is their reliance on the feature engineering methods that are usually dictated by domain experts. Hence, security researchers are exploring DL methods to handle the dynamically evolving malicious activities. DL scales well for very large amount of data samples compared to CMLAs since it can capture the important features from complex systems including natural language processing (NLP) [5]. As the data of cyber security continuously grows day by day with the evolution of technology, the performance of DL based solutions also improves. Recently, several DL architectures are proposed, and it is important to conduct a survey of their relative suitability to address the growing cyber security issues as we enter Industry 4.0.

A. Existing Surveys on ML and DL on Cyber Security

Classical ML frameworks that were developed in the past decade for solving cyber security problems have been surveyed extensively [7], [8], [9], [10], [11], [12]. However, these do not include DL methodologies. Surveys on DL frameworks have been confined to a very narrow set of applications in cyber security. Majority of research studies focus on a specific

cyber security technique, such as intrusion detection [13], [14], spam detection, anomaly detection, and malware analysis [15]. Recent studies have focused on providing a summary review of the work related to defending CPS [16], and some have discussed various ML and DL methodologies for securing IoT technologies [17]. Certain surveys including short tutorial types have either a specific focus on the application of deep reinforcement learning (DRL) to cyber security or generic DL frameworks in the detection of various attacks included malware, spam, insider threats, network intrusions, false data injection, and malicious domain names used by botnets [18] [19], [20]. However, previous surveys have focused on only certain cyber security scenarios forming limitation in the scope. In particular, they do not discuss the use of DL for securing next-generation communication networks such as autonomous vehicle networks, cloud/edge computing, and blockchain, which are next-generation technologies proceeding towards 5G network and Industry 4.0. Though many surveys on the application of ML and DL to cyber security exist, to the best of our knowledge, a detailed and comprehensive survey on various research undertaken in DL has not yet been performed in such breadth and scope to include the complexity of next generation computing. This paper aims to fill this gap in literature.

B. Research Contribution

Considering the limitations of the state-of-the-art surveys in literature pertaining to DL architectures for next generation computing, our target in this work is to provide an extensive and comprehensive survey on DL for cyber security. The major contributions of our survey can be summarized as follows:

- 1) This paper reviews the DL architectures deployed for various cyber security applications and provides a walk-through of their evolution. Additionally, this survey summarizes, compares and contrasts the various DL architectures providing a detailed understanding of the past, present, and future DL applications in cyber security.
- 2) This survey presents the classification of a variety of studies that have reported on the application of DL to cyber security based on several attributes such as type of architecture and its application, year of study, text representation, type of dataset and performance comparison with CMLA.
- 3) An overview of various issues and major challenges of cyber-security applications involved in off-line and real-time deployment is provided. Further, the importance of shared tasks in the field of cyber security is explained.
- 4) The state-of-the-art reinforcement learning (RL) and adversarial ML applications in cyber security are examined.
- 5) The importance of DL in BD, NLP, signal, and image processing for cyber security is discussed.
- 6) The role of DL architecture-based cyber security in the field of smart cities, pervasive computing, biometric, IoT, fog and cloud computing, and autonomous vehicle is covered.



图1. 本论文组织结构的示意图。

- 7) 探讨了无监督学习相对于半监督学习与监督学习在网络安全中的重要性。此外，总结了可解释人工智能、迁移学习、可视化及混合框架在网络安全中的重要性。
- 8) 审查了用于各种网络安全研究的许多公开数据集，并提供了未来研究方向的建议。

C. 论文组织

本综述文章的结构安排如图1所示。第二节介绍了各种CMLA和深度学习架构的基础知识，并进一步讨论了网络安全中存在的主要问题。第三节探讨了深度学习架构在大数据、信号与图像处理以及适用于网络安全的自然语言处理技术中的重要性。第四节介绍了网络安全中的对抗性深度学习，随后第五节对强化学习在网络安全中的应用进行了全面综述。第六节描述了适用于各种网络安全技术的最新深度学习架构，包括入侵检测、网络安全

利用领域生成算法（DGA）、统一资源分配器（URL）、电子邮件和安全日志数据分析、网络流量分析、Windows/Android恶意软件分析、侧信道攻击检测、内部威胁检测、功能识别、隐写分析和隐写术，以及社交媒体数据用于网络安全。在第七节，我们概述了深度学习在下一代通信网络、智慧城市、区块链、密码学、云计算、边缘计算、自动驾驶车辆网络、普适计算和生物识别安全等各项技术中的应用。第八节讨论了深度学习应用于网络安全的相关杂项问题，包括迁移学习（TL）在网络安全应用中的重要性、网络安全中的无监督学习、离线和实时部署、可解释人工智能在网络安全中的作用，以及深度学习的因果理论用于网络安全。深度学习在网络安全中应用的详细统计数据在第九节报告。最后，在第十节，我们提出了一种由最佳理论和深度学习模型子模块组成的混合网络安全框架，并在第十一节，我们提供结论以及未来研究方向。

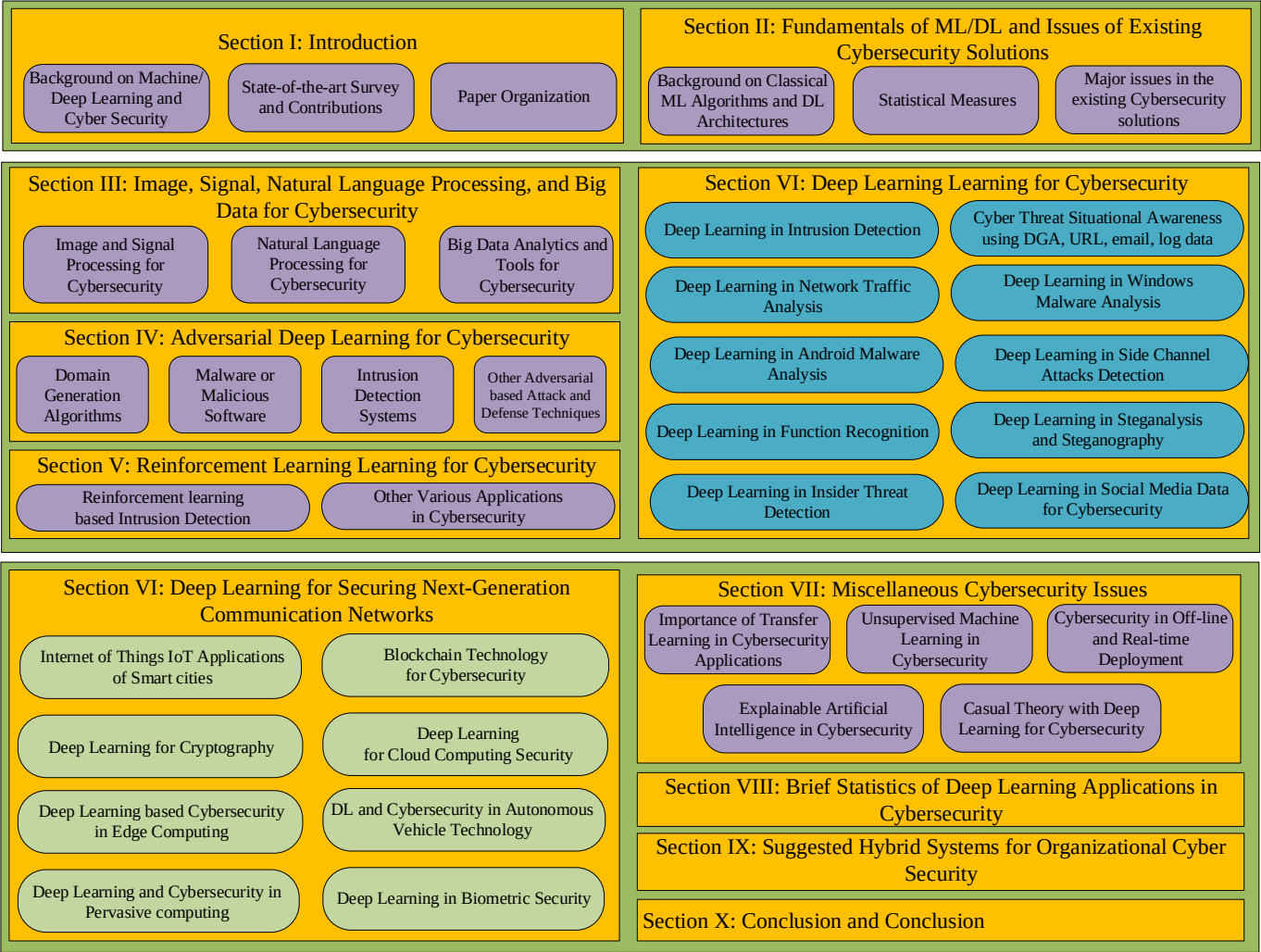


Fig. 1. Diagrammatic view of the organization of this paper.

- 7) Significance of unsupervised learning for cyber security over semi-supervised and supervised learning is explored. Moreover, importance of explainable AI, transfer learning, visualization and hybrid framework in cyber security is summarized.
- 8) Many publicly available datasets used for various cyber security studies are reviewed and suggestions for future research directions are provided.

C. Paper Organization

The structure of this survey article is organized as shown in Fig. 1. Section II presents the basics of various CMLAs and DL architectures and further discusses the major issues that exist in cyber security. Section III explores the importance of DL architectures in big data, signal and image processing, and natural language processing techniques suitable for cyber security. Section IV presents adversarial DL in cyber security, followed which Section V provides a comprehensive review on the applications of RL in cyber security. Section VI describes the state-of-the-art DL architectures for various cyber security techniques, including intrusion detection, cyber

threat situational awareness using domain generation algorithm (DGA), uniform resource allocator (URL), email and security log data analysis, network traffic analysis, Windows/Android malware analysis, side channel attacks detection, insider threat detection, function recognition, steganalysis and steganography, and social media data for cyber security. In Section VII, we provide an overview of the application of DL for various technologies in next-generation communication networks, smart city, blockchain, cryptography, cloud computing, edge computing, autonomous vehicle networks, pervasive computing, and biometric security. Section VIII discusses miscellaneous issues related to the application of DL to cyber security, including the importance of transfer learning (TL) in cyber security applications, unsupervised learning in cyber security, off-line and real-time deployment, role of explainable AI in cyber security, and causal theory with DL for cyber security. The detailed statistics of DL applications in cyber security is reported in Section IX. Finally, in Section X, we propose a hybrid cyber security framework of submodules of best theories and DL models and in Section XI, we provide conclusions along with future research directions.

II. 深度学习基础及现有网络安全解决方案问题

A. 经典机器学习算法和深度学习基础

1955年，约翰·麦卡锡创造了“人工智能（AI）”这一术语，并将AI定义为“制造智能机器的科学和工程”。机器学习（ML）是AI的一个子集，它也在同一十年中引入，但在20世纪90年代随着计算技术的演进和数字数据的指数级增长而变得更加流行。在机器学习中，数学和统计概念构成了核心基础，催生了一系列复杂的算法，主要用于执行模式发现、数据相关性以及异常检测。机器学习算法的输出以概率和置信区间表示。由于缺乏专家分析大量数据的能力，机器学习算法被用于成功自动化AI驱动的学习过程。

在过去十年中，机器学习在网络安全 [6]应用方面已逐渐成熟。一般来说，机器学习中的算法可以分为五种不同类型，即监督学习、半监督学习、无监督学习、强化学习和主动学习。监督学习算法是任务驱动的，依赖于将样本文件标记为恶意软件或非恶意软件。监督型机器学习算法需要预处理和特征工程。常用的CMLA包括朴素贝叶斯（NB）、逻辑回归（LR）、决策树（DT）、Ada Boost（AB）、随机森林（RF）和支持向量机（SVM）。无监督学习是一种数据驱动的方法，只需要样本数据，算法基于数据分布进行隐式学习和标记。虽然无监督模型的性能低于监督模型，但在实时网络安全应用中更受欢迎，因为手动标记样本数据是一项繁琐的任务。半监督学习结合了监督学习和无监督学习的优点，以从两种方法中获益。强化学习是一种环境驱动的方法，基于奖励进行工作，并通过试错方法进行改进。目前大多数基于深度学习的实时系统都基于强化学习。这是网络安全领域中用于恶意软件和僵尸网络检测的一种合适方法。主动学习是强化学习的一个子方法，当看到新的数据样本时会联系用户。

CMLA由3个主要步骤组成：1)原始数据收集，2)特征提取，3)分类。特征提取是特征工程中的一个重要步骤，需要了解相关主题。分类器的性能隐含地依赖于特征提取。神经网络（NN）能够自动进行特征提取和分类，无需人工干预。经典NN的性能在一定程度上相当可观。然而，通过使用通常被称为深度学习（DL）的高级NN，可以完全避免特征工程阶段。这使得DL在长期的人工智能应用中，特别是在涉及各个领域的应用中，实现了最佳性能。

深度学习已成为安全研究人员和安全行业从业者的关注焦点。深度学习现已被应用于网络安全中存在的各种问题

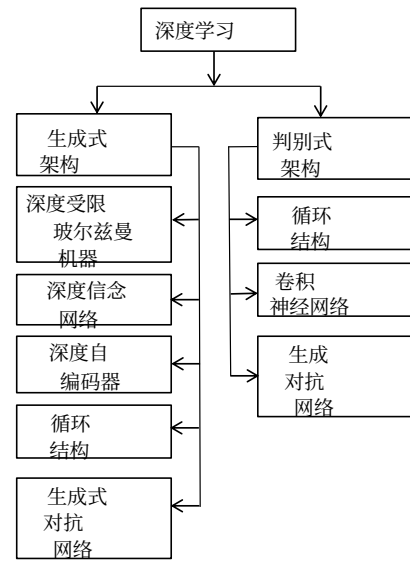


图2。深度学习（DL）架构分类。

，并且在所有用例中表现优于CMLA。深度学习架构可分为生成型和判别型，如图2所示。生成型包括深度玻尔兹曼机（DBM）、深度自编码器（DAE）、深度信念网络（DBN）和循环结构，而判别型包括循环结构和卷积神经网络（CNN）。循环结构和CNN最常使用深度学习架构。DBN和DBM都基于受限玻尔兹曼机（RBM）。生成对抗网络（GAN）属于生成型和判别型深度学习架构。深度学习和人工神经网络（ANN）的结合称为深度神经网络（DNN）。虽然我们在图2中提供了各种深度学习架构分类的概述，但详细的描述和调查报告可在文献 [13], [14], [15]中找到。

为了评估各种可用的深度学习模型的性能，使用了多种统计指标。其中最重要和标准的指标之一是混淆矩阵，它提供了分类结果的详细信息，包括最终区分到的各个类别。分类正确性通过真阳性（TP）和真阴性（TN）来衡量，其中正/负样本被深度学习模型正确分类。此外，假阳性（FP）和假阴性（FN）指标分别表示深度学习模型对正/负样本的不正确预测。利用混淆矩阵，可以估计一些指标，如准确率、精确率、召回率/真阳性率（TPR）/灵敏度、F1-measure/F1分数、假阳性率（FPR）、真阴性率（TNR）和假阴性率（FNR） [19]。准确率、精确率、召回率、F1分数、FPR、TNR和FNR的值范围在0到1之间，值越大表示性能越好。由于这些指标是相关的，提高其中一个指标（如TPR）可能会导致另一个指标（如FPR）的不期望增加。因此，在设计阶段，通常会根据反映TPR对FPR依赖性的分类阈值，评估最优检测准确率，该阈值由受试者工作特征（ROC）曲线表示。

II. FUNDAMENTALS OF DL AND ISSUES OF EXISTING CYBER SECURITY SOLUTIONS

A. Basics of Classical ML Algorithms and Deep Learning

In 1955, John McCarthy coined the term Artificial intelligence (AI) and defined AI as “the science and engineering of making intelligent machines”. ML is a subset of AI which was also introduced during the same decade but became more popular in 1990s with the evolution of computing technologies and the exponential increase in digital data. In ML, the mathematical and statistics concepts form the core foundation in the emergence of a variety of complex algorithms for primarily performing pattern discovery, correlations and anomalies of data. The outputs of ML algorithms are represented in terms of probabilities and confidence intervals. Due to the limitations in having experts to analyse huge amounts of data, ML algorithms are employed to successfully automate the learning process for AI based automation.

Over the past decade, ML has matured in its application to cyber security [6]. In general, algorithms in ML can be grouped into five different types namely, supervised, semi-supervised, unsupervised, reinforcement, and active learning. Supervised learning algorithms are task-driven and rely on labelling of the sample files as malware or not. Supervised ML algorithms require preprocessing and feature engineering. Such commonly used CMLAs are Naive Bayes (NB), Logistic regression (LR), Decision Tree (DT), Ada Boost (AB), Random Forest (RF), and Support vector machine (SVM). Unsupervised learning is a data driven approach that requires only the sample data and the algorithm uses implicit learning and labelling based on the distribution of the data. While the performance of unsupervised models is lower as compared to supervised models, they are preferred in real-time cyber security applications as manual labelling of sample data is a tedious task. Semi-supervised learning combines both the supervised and unsupervised learning to get benefits from both approaches. RL is an environment driven approach which works based on rewards and is improved by a trial and error approach. Most of the DL based real-time systems in current days are based on RL. This is a suitable method for malware and botnet detection in the domain of cyber security. Active learning is a sub method of RL that contacts the user whenever a new data sample is seen.

CMLAs are composed of 3 main steps: 1) raw data collection, 2) feature extraction, and 3) classification. Feature extraction is an important step in feature engineering which requires knowledge about the subject. The performance of the classifier implicitly relies on the feature extraction. Neural network (NN) is capable of automatic feature extraction and classification without human intervention. The performance of the classical NN is considerable to a certain extent. However, feature engineering phase can be completely avoided by using advanced NN typically named as DL. This made the DL to achieve the best performance in long-standing AI applications related to various domains.

DL has become a focal point for both the security researchers and people from security industries. DL is now being employed in various problems existing in cyber security

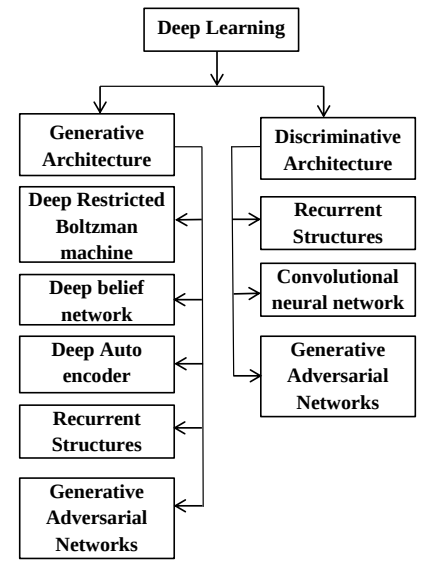


Fig. 2. Classification of DL architecture.

and performs well in all use cases compared to CMLAs. DL architectures can be classified into generative and discriminative, as shown in Fig. 2. Generative category is composed of deep Boltzmann machine (DBM), deep autoencoder (DAE), deep belief network (DBN), and recurrent structures, whereas discriminative category is composed of recurrent structures and convolutional neural network (CNN). Recurrent structures and CNN most commonly use DL architectures. Both DBN and DBM are based on the restricted Boltzmann machines (RBM). The generative adversarial network (GAN) belongs to both the generative and discriminative DL architecture category. The combination of DL and artificial neural networks (ANN) is called deep neural networks (DNN). While we have provided a pictorial overview of the classification of various DL architectures in Fig. 2, a detailed description and survey reports are available in literature [13], [14], [15].

To evaluate the performance of various DL models available, several statistical measures are used. One of the most important and standard metrics adopted is the confusion matrix which provides the details of the classification results, including the individual distinguished classes arrived at. Classification correctness is measured in terms of true positive (TP) and true negative (TN), where the positive/negative samples are correctly classified by the DL model. In addition, false positive (FP) and false negative (FN) measures indicate the incorrect prediction of the positive and negative samples respectively by the DL model. Using the confusion matrix, some metrics can be estimated such as accuracy, precision, recall/true positive rate (TPR)/sensitivity, F1-measure/F1-score, false positive rate (FPR), true negative rate (TNR), and false negative rate (FNR) [19]. The values of Accuracy, Precision, Recall, F1-score, FPR, TNR and FNR range from 0 to 1 with larger values representing better performance. Since these measures are correlated, any desire to increase one measure such as TPR may result in an undesired increase of another measure such as FPR. Therefore, during the design phase, an optimal detection accuracy is usually assessed based on a discrimi-

为了基准测试，估计了ROC曲线下面积（AUC）。AUC值通常介于0.5到1.0之间，AUC值越大表示性能越好。

B. 关键深度学习架构

本节总结了近期文献中报道的、在网络安全领域应用广泛的四种关键深度学习（基于人工神经网络/深度神经网络）架构的显著特征

1) 深度信念网络 (DBN) 和深度玻尔兹曼机 (DBM) : DBN或深度网络基于与经典人工神经网络的生成式工程相关。它包含一个输入层，至少一个隐藏层和一个输出层。需要注意的是，单层DBN与前馈网络（FFN）相同。输入层和隐藏层都应至少有一个神经元，科学上称为处理单元。输出层有一个单元对应每个需要网络分类的输入类别。此外，具有多个隐藏层的网络可能在组装方面提供更多机会。例如，在准备阶段，受限玻尔兹曼机（RBM）等无监督学习组件可以通过将输入向量传递通过至少一个RBM隐藏层来接收最小化元素向量。DBN的训练阶段包括预训练和重建两个步骤。对于没有类标签的训练样本，预训练阶段将输入随机传播到RBM层。每一层RBM学习代表前一层数据的特征，并在顶层存在联想记忆。每个隐藏层单元遵循条件分布生成二进制形式特征向量，这些特征向量反向传播以重建训练样本。此过程对所有训练样本迭代执行。

2) 自编码器 (AE): 通过线性或非线性操作学习输入数据别名表示的一组神经网络架构称为自编码器 (AE)，其输入和输出层维度相同。自编码器的主要目的是实现降维。在最近的文献中，研究人员发现使用多个隐藏层来学习原始数据的判别性和代表性特征。这种类型的网络称为DAE。与一般神经网络架构不同，后者被训练以学习预定义的输出变量，这些架构被训练以从输入中学习。因此，神经网络通过自身学习来重建输入数据。自编码器的架构与多层感知机 (MLP) 类似，即它有一个输入层、一个或多个隐藏层和一个输出层。如果自编码器有多个隐藏层，则从一层提取的特征会被进一步处理成能够重建数据的不同特征。在数据重建过程中，自编码器旨在最小化误差。因此，中间层的输出不过是其输入的编码版本，在特定条件下能够重建输入数据。

在常规变换中，会利用数据点选择一个特定的特征集，然后将这些特征输入到

在常规变换中，会利用数据点选择特定的特征集，然后将这些特征集输入到分类算法中。然而，自编码器（AE）采用无监督方法，从不同层提取不同特征，并将其传递给其他深度学习（DL）层，如卷积神经网络（CNN）、循环神经网络（RNN）以及包含CNN-RNN和CMLA的混合网络。自编码器有3种著名的变体：稀疏自编码器、降噪自编码器和收缩自编码器。在稀疏自编码器中，隐藏节点的数量多于输入和输出，以鼓励稀疏性。然而，在任意时刻只有部分隐藏单元被激活。这是通过惩罚额外节点的激活来实现的。降噪自编码器（DAE）从损坏的版本中恢复正确的输入，以提高模型的鲁棒性。收缩自编码器通过在重建误差函数中添加解析收缩惩罚来实现这一点。总体而言，DAE架构对噪声更鲁棒，而收缩自编码器可以捕捉数据所决定的局部变化方向。

3) 卷积神经网络 (CNN): CNN 是一种流行的最先进的方法，被应用于许多计算机视觉应用。CNN 由卷积层、池化层和全连接层 [21] 组成。卷积层使用内核或过滤器沿数据的各种维度（1D/2D/3D/4D）移动，以提取最佳特征，这些特征统称为特征图。然后，这些特征图被传递到池化层。卷积层和池化层都是平移不变的，因为它们考虑了相邻数据。最初，特征图被分成多个分区，并使用各种池化函数来降低特征图的维度，这实际上是一种非线性下采样操作。常见的池化操作包括最大值、最小值、平均值、随机值、空间金字塔和来自分区的变形值。随机池化与最大池化类似，但它通过用多项式分布确定的随机过程来替换传统的确定性池化操作，从而防止过拟合，该随机过程根据每个池化区域内的激活情况来确定。通常，CNN 网络只能处理固定长度的输入表示。为了处理可变长度的输入表示，可以使用空间金字塔池化，因为它可以处理不同尺度、大小和宽高比输入图像。与最大值和平均值池化相比，变形池化可以更有效地处理图像中的变形。通过结合不同的池化层来提升 CNN 架构的性能，可以探索 DL 架构中的新颖性。

基于卷积神经网络（CNN），研究人员提出了多种基准架构，并在ImageNet大规模视觉识别挑战赛（ILSVRC）上进行了评估。基于CNN的关键架构包括LeNet、AlexNet、ZFNet、GoogleNet/Inception、VGGNet、SPPNet、ResNet、DenseNet、squeezeNet、MobileNet和NASNet。这些架构都包含大量参数，通常应用于大型数据集。由于实时获取所有类别/任务的大型数据集非常困难，因此使用数据增强技术来增加数据样本，而不会引入额外的标注成本。CNN架构广为人知，可以在新任务中用于参数初始化，而不是使用随机参数值，这代表了预训练。这加速了学习过程，并提高了模型的泛化能力。

nation threshold that reflects the dependency of TPR on FPR, which is represented by the Receiver Operating Characteristics (ROC) curve. For the purpose of benchmarking, the area under the ROC curve (AUC) is estimated. AUC values typically lie between 0.5 to 1.0, and larger AUCs represent better performance.

B. Key Deep Learning Architectures

This section summarises the salient features of four key DL architectures using ANN (DNN) due to their wide applications in cyber security that are reported in recent literature

1) Deep belief network (DBN) and deep Boltzmann machine (DBM): DBN or deep networks is based on generative engineering related to the classical ANN. It contains an input layer, with at least one hidden layer and one output layer. It is to be noted that DBN with one layer is the same as feed forward network (FFN). Both the input and hidden layers should have at least one neuron, scientifically termed for a processing unit. The output layer has one unit for every class that is required for the inputs to be classified by the network. In addition, a network with more than one hidden layer may expend more opportunity for its assembly. For instance, an unsupervised learning component such as the restricted Boltzmann machine (RBM) could take in the minimized element vectors by passing an input vector through at least one of the RBM hidden layers within the preparation stage. DBN’s training phase has two steps namely pre-training and reconstruction. Given the training samples without class labels, the pre-training stage propagates the input stochastically across RBM layers. Each layer of RBM learns features which represent the data in the previous layer with associative memory present at the top layer. Conditional distribution is followed by each hidden layer unit to generate binary form feature vectors that are propagated in reverse direction to reconstruct the training samples. This procedure is followed iteratively for all the training samples.

2) Autoencoders (AE): A set of NN architectures made to learn alias representations of input data via linear or nonlinear operations are called AE which have identical input and output layer dimensions. The main purpose of AE is to achieve dimensionality reduction . In recent literature, researchers are found to utilize more than one hidden layer to learn discriminative and representative features of raw data. This type of network is called as DAE. Unlike the general NN architecture which is trained to learn predefined output variables, these are trained to learn from the input. As a result, the NN learns by itself to reconstruct the input data. The architecture of an AE is similar to the multi-layer perceptron (MLP), i.e., it has one input layer, one or more hidden layers, and one output layer. If AE has multiple hidden layers, then the features extracted from one layer are further processed to different features that are capable of reconstructing the data. During data reconstruction process, AE aims to minimize the error. Therefore, the outputs of intermediate layers are nothing but an encoded version of their inputs capable of reconstructing the input data under specific conditions.

In usual transformations, a particular set of features is selected utilizing the data points, which are then fed to

classification algorithms as input. However, AE follows unsupervised approach, where different features are extracted from different layers and passed on to the other DL layers such as CNN, RNN and hybrid networks that include CNN-RNN and CMLAs. AEs have 3 types of well-known variants: Sparse AE, Denoising AE, and Contractive AE. In sparse AE, there are more hidden nodes than the inputs and outputs in order to encourage sparsity. However, only a portion of the hidden units are activated at a given time. This is accounted for by penalizing the activation of additional nodes. DAE recovers the correct input from a corrupted version to increase the robustness of the model. Contractive AE achieves this by adding an analytic contractive penalty to the reconstruction error function. Overall, the DAE architecture is more robust for noise, while contractive AE can capture the local directions of variation dictated by the data.

3) Convolutional neural network (CNN): CNN is a popular state-of-the-art method adopted for many computer vision applications. CNN is composed of convolutional, pooling, and fully connected layers [21]. A convolutional layer uses kernels or filters to move along various dimensions (1D/2D/3D/4D) of data to extract optimal features, together called as feature maps. These feature maps are then passed into the pooling layer. Both convolutional and pooling layers are translational invariants because they consider the neighboring data into account. Initially, the feature maps are divided into partitions and various pooling functions are used to reduce the dimensionality of the feature maps which is nothing but a non-linear downsampling operation. The common pooling operations are maximum, minimum, average, stochastic, spatial pyramid, and deformation value from the partition. The stochastic pooling is similar to maximum pooling but it also prevents overfitting by replacing the conventional deterministic pooling operations with a stochastic procedure determined by the activation within each pooling region according to a multinomial distribution. Generally, CNN network can handle only the fixed length input representations. To handle variable length input representations, spatial pyramid pooling can be used as it can handle input images of variable scales, sizes and aspect ratios. A deformation pooling operation can handle deformation in image efficiently when compared to the max and average pooling. The novelty in the DL architecture could be explored by combining the different pooling layers to boost the performance of the CNN architecture.

Based on CNN, various benchmark architectures are proposed and evaluated on ImageNet Large Scale Visual Recognition Challenge (ILSVRC). Key architectures based on CNN are LeNet, AlexNet, ZFNet, GoogleNet/Inception, VGGNet, SPPNet, ResNet, DenseNet, squeezeNet, MobileNet, and NAS-Net. All these architectures contain large number of parameters and typically applied on large datasets. Since obtaining large datasets for all the classes/tasks in real-time is difficult, data augmentation is used to increase the data samples without introducing extra labeling costs. CNN architectures are well-known and can be employed for parameter initialization instead of random parameter value in newer tasks, which represents pretraining. This accelerates the learning process and improves the model generalization.

4) 循环神经网络 (RNN) : 循环结构主要应用于序列和时序数据建模任务。循环神经网络 (RNN) 是经典神经网络 (NN) 的一种高级模型, 其隐藏层具有自循环连接, 这有助于网络记住前一步的信息。在反向传播通过时间 (BPTT) 过程中处理长时序时, RNN会面临梯度消失和梯度爆炸问题。梯度裁剪是避免梯度爆炸问题的一种突出策略。为了缓解梯度消失问题, RNN的研究在三个重要方向上取得了进展: i) 使用无Hessian优化来改进优化方法, ii) 使用长短期记忆 (LSTM) 网络或参数集减少的LSTM变体, 以及门控循环单元 (GRU) 来为网络结构的循环隐藏层引入复杂组件, iii) 使用单位矩阵进行权重初始化的恒等循环神经网络 (IRNN) 。

C. 现有网络安全解决方案中的主要问题

深度学习因其BD特性, 以及多核CPU、GPU的可用性, 以及神经网络向训练多层隐藏层的演进, 为网络安全提供了最佳适配方案。然而, 由于机器学习和深度学习算法缺乏基准测试以及数据集, 其应用仍处于初级阶段。最近, 在网络安全中应用机器学习/深度学习技术的挑战和问题被讨论了 [24]。在表I中, 我们提供了所研究的威胁检测方法以及使用的基准数据集的总结。由于四个主要原因, 为网络安全用例找到一个令人满意的数据库往往很麻烦: 1) 大多数公开可用的数据集都已过时, 2) 它们不是真实的代理数据集, 3) 大多数研究人员采用不同的方法将数据划分为训练、验证和测试类别, 4) 由于安全和隐私原因, 它们没有广泛地被研究界获取。这导致了无法复现的实验结果。由于这些问题, 网络安全用例没有标准方法, 大多数企业避免使用机器学习/深度学习解决方案来改进其网络安全应用 [23]。

提出的解决方法: 最近通过将共享任务作为会议和研讨会的一部分来组织, 以提升系统性能的方法。共享任务是指研究人员或研究团队提交解决特定预定义挑战的系统的竞赛。共享任务的初始阶段是将训练数据集分发给参与者。利用测试数据集对训练好的模型进行评估。最后, 结果会公开并提供发表选项。共享任务在自然语言处理、计算机视觉和语音识别领域最为常见。最近, CDMC¹¹, IWSPA-API¹², DMD 2018¹³, 和 AICS 2019¹⁴ 是网络安全领域的4个共享任务。

III. 图像/信号处理、自然语言处理和大数据在网络安全中的应用

A. 图像和信号处理在网络安全中的应用

信号与图像处理领域中的几种特征工程机制已被成功应用于恶意软件分类。恶意软件特征以信号或灰度图像的形式表示, 作为十六进制或文本形式表示的恶意软件二进制代码的替代方案 [115]。每个信号的取值范围为 [0, 255] (0: 黑色, 255: 白色)。对于图像, 图像宽度是固定的, 而高度根据文件大小可变。基于信号和图像的恶意软件分析速度快, 无需反汇编、解包和执行二进制代码。最近, 提出了新的特征工程方法, 如谱平坦度、梅尔频率倒谱系数 (MFCC) 和音调特征, 以从信号和图像中准确提取重要特征。当前基于信号和图像的恶意软件检测方法存在两个主要问题: i) 使用基于信号和图像的特征对恶意软件进行表征并不能提供太多关于恶意软件实际行为的信息, ii) 由于该方法依赖于基于实例的学习, 其主要限制是它只能检测或分类与已观察到的恶意软件相似的恶意软件。然而, 零日或新的未见过恶意软件攻击无法被阻止。因此, 深度学习架构采用特征工程机制来提高恶意软件分析和检测的性能。

B. 自然语言处理在网络安全中的应用

自然语言处理 (NLP) 是分析并从自然语言中提取有用信息的方法, 以简化人机交互。NLP在网络安全的关键在于语言以数据形式的存在。网络安全领域的文本数据来源多样, 包括电子邮件、来自各种系统的交易日志和在线社交网络。利用NLP技术对各种网络事件日志和用户活动提供态势感知具有直接影响。文本以数字形式表示的几种方法, 如向量空间模型、分布式表示和分布式表示, 可用于将文本编码到词/字符级别。词/字符级别的文本编码包括预处理, 然后是作为初始步骤的词/字符级分词。这涉及清理和转换不必要的和未知的词/字符, 然后进行词/字符级分词。非序列和序列输入是文本表示的两种主要类型。词袋 (BoW)、项文档矩阵 (TDM) 和项频率-逆文档频率矩阵 (TF-IDF) 属于非序列表示。N-gram、Keras嵌入、Word2vec、神经词袋和FastText属于序列表示, 它具有提取词义相似性的能力。在网络安全的背景下, 捕获序列信息比词义相似性更重要, 因为大多数数据包含时间和空间信息。因此, 可以采用深度学习方法进行有效的恶意软件检测。

4) *Recurrent neural network (RNN)*: Recurrent structures are mainly used in sequential and temporal data modeling tasks. RNN is an advanced model of classical NNs which has a self-recurrent connection in the hidden layer that facilitates the network to remember the previous step information. It suffers from vanishing and exploding gradient issue when dealing with long time-steps during backpropagation through time (BPTT). Gradient clipping is one of the prominent strategies to avoid the exploding gradient issues. To alleviate the vanishing issue, research on RNN progressed on three significant directions: i) Hessian-free optimization for improving the optimization methods, ii) Long Short-term Memory (LSTM) or a variant of LSTM network with reduced parameters set, gated recurrent unit (GRU) for introducing complex components in recurrent hidden layer of network structure, and iii) Identity-recurrent neural network (IRNN) for weight initialization with an identity matrix.

C. Major issues in the existing Cyber security solutions

DL serves a best-fit for cyber security due to its BD characteristics, and the availability of multi-core CPUs, GPUs along with the evolution of NNs to train many hidden layers. However, the adoption is still in infancy due to lack of benchmarking in ML and DL algorithms as well as the datasets. Recently, the challenges and issues involved in employing ML/DL techniques in cyber security were discussed [24]. In Table I, we provide a summary of threat detection methods studied and benchmark datasets used. Finding a satisfactory dataset for cyber security use cases is often troublesome due to four main reasons: 1) the vast majority of publicly available datasets are outdated, 2) they are not genuine agent datasets, 3) most researchers follow different splitting methodologies to divide data into train, validate and test categories, and 4) they are not broadly accessible to the research community due to security and privacy reasons. This leads to experimental results that are not reproducible. Due to these issues, the use cases of cyber security do not have a standard approach and most enterprises avoid using ML/DL solutions for improving their cyber security applications [23].

Proposed approach to address the issue: Most recent way to enhance the performance of a system is by organizing the shared tasks as a part of a conference and workshop. Shared tasks are competitions to which researchers or teams of researchers submit systems that address specific, predefined challenges. Initial phase of the shared task is to distribute the training dataset among the participants. Evaluation of trained models is performed utilizing the testing dataset. Finally, the results are made publicly available with an option for publication. Shared tasks are most familiar in the field of NLP, computer vision and speech recognition. Recently, CDMC¹¹, IWSPA-API¹², DMD 2018¹³, and AICS 2019¹⁴ are 4 shared tasks in the field of cyber security.

III. IMAGE/SIGNAL PROCESSING, NATURAL LANGUAGE PROCESSING, AND BIG DATA FOR CYBER SECURITY

A. Image and Signal Processing for Cyber security

Several feature engineering mechanisms available in signal and image processing domain have been successfully employed for malware classification. The malware features are represented in the form of a signal or grayscale image as an alternative to malware binaries represented in Hexadecimal or text forms [115]. The range of each signal is [0, 255] (0: black, 255: white). In the case of an image, the width of the image is fixed, and the height is allowed to vary depending on the file size. Signal and image-based malware analysis is fast, and does not need disassembly, unpacking, and execution of binary code. Recently, novel feature engineering methods such as spectral flatness, mel-frequency cepstrum coefficients (MFCC), chroma features are proposed to accurately extract important features from signals and images. Current methods of signal and image-based malware detection exhibit two major problems: i) characterization of malware using signal and image-based features does not give much information about the actual behavior of the malware and ii) since the approach relies on instance-based learning, its main limitation is that it can only detect or classify malware similar to what has already been observed. However, zero-day or new unseen malware attacks cannot be prevented. Hence, feature engineering mechanisms are employed by DL architectures to enhance the performance of malware analysis and detection.

B. Natural Language Processing for Cyber security

NLP is the method of analyzing and extracting useful information from natural languages to make human-computer interaction simpler. The key to NLP success in cyber security is the availability of language in the form of data. The text data in cyber security domain have various sources such as emails, transaction logs from various systems, and online social networks. Leveraging NLP techniques have direct impact in providing situational awareness from various network event logs and user activities. Several methods of text representation in numeric form such as vector space models, distributional representation, and distributed representation can be used to encode text up to word/character level. Word/character level text encoding consists of preprocessing followed by tokenization as the initial step. This involves data cleaning and transformation of unnecessary and unknown words/characters, followed by word/character level tokenization. Non-sequential and sequential inputs are the two main types of text representation. Bag of words (BoW), term document matrices (TDM), and term frequency-inverse document frequency matrices (TF-IDF) belong to non-sequential representation. N-gram, Keras embedding, Word2vec, Neural-Bag-of-words, and FastText belong to sequential representation which has the capability to extract similarities in word meaning. In cyber security, capturing the sequential information is more important as compared to the similarities in word meaning due to the fact that most data contain time and spatial information. Hence, DL approaches could be adopted for an effective malware detection.

¹¹<http://www.csmining.org/cdmc2018/> ¹²<https://dasavisha.github.io/IWSPA-sharedtask/> ¹³<https://nlp.amrita.edu/DMD2018/> ¹⁴<http://www-personal.umich.edu/arunesh/AICS2019/index.html>

¹¹<http://www.csmining.org/cdmc2018/>

¹²<https://dasavisha.github.io/IWSPA-sharedtask/>

¹³<https://nlp.amrita.edu/DMD2018/>

¹⁴<http://www-personal.umich.edu/arunesh/AICS2019/index.html>

表 I 基准数据集详细信息。

Task	基准数据集	URL
入侵检测	KDDCup-99	http://kdd.ics.uci.edu/databases/kddcup99/kddcup99.html
	NSL-KDD	https://www.unb.ca/cic/datasets/nsl.html
	UNSW-NB15	https://www.unsw.adfa.edu.au/unsw-canberra-cyber/cybersecurity/ADFA-NB15数据集/
	ADFA-LD	https://www.unsw.adfa.edu.au/unsw-canberra-cyber/cybersecurity/ADFA-NB15数据集/
	UNM	https://www.cs.unm.edu/~immsec/系统调用.htm
	ISCX-IDS-2012	https://www.unb.ca/cic/数据集/ids.html
	CICIDS2017	https://www.unb.ca/cic/数据集/ids-2017.html
	京都	http://www.takakura.com/京都数据/
	UNIBS	http://netweb.ing.unibs.it/~ntw/工具/跟踪/
	CAIDA	https://www.caida.org/data/
僵尸网络与DGA分析	LBNL	https://commons.lbl.gov/display/cpp/100G+入侵+检测
	CIC拒绝服务攻击	https://www.unb.ca/cic/datasets/dos-dataset.html
	CSE-CIC-IDS2018	https://www.unb.ca/cic/datasets/ids-2018.html
	AWID	http://icsdweb.aegean.gr/awid/index.html
	WSN-DS	<https://www.hindawi.com/journals/js/2016/ 4731953/>
	UNB僵尸网络	https://www.unb.ca/cic/datasets/僵尸网络.html
	DGArchive	https://dgarchive.caad.fkie.fraunhofer.de/
	AmritaDGA	https://vinayakumarr.github.io/AmritaDGA/
	ISCX-URL-2016	https://www.unb.ca/cic/datasets/url-2016.html
	Sophos URL	https://github.com/Maddy12/SophosMachineLearningBuildingBlocksTutorial
垃圾邮件和网络钓鱼 电子邮件检测	CSDMC	http://csmining.org/垃圾邮件-电子邮件-数据集-.html/
	Enron	https://www.cs.cmu.edu/~.enron/
	TREC	<https://trec.nist.gov/data/spam.html>
	SpamAssassin	<https://垃圾邮件过滤器.apache.org/old/publiccorpus/>
恶意软件检测	EMBER	https://github.com/endgameinc/ember
	BIG 2015	<https://www.kaggle.com/c/malware-classification>
	Malrec	https://giantpanda.gtisc.gatech.edu/malrec/dataset/
	Microsoft 恶意软件预测	<style> 恶意软件预测竞赛: Microsoft恶意软件预测</style>
二进制分析	ByteWeight	http://security.ece.cmu.edu/byteweight/
图像垃圾邮件检测	图像垃圾邮件猎手	https://www.unb.ca/cic/datasets/android-adware.html
Android	AAGM	https://www.unb.ca/cic/datasets/android-adware.html
	CICAndMal2017	https://www.unb.ca/cic/datasets/andmal2017.html
	Drebin	https://www.sec.cs.tu-bs.de/~danarp/Drebin/
	Kharon	http://kharon.gforge.inria.fr/dataset/
流量分析	ISCXVPN2016	https://www.unb.ca/cic/datasets/vpn.html
	ISCXTor2016	https://www.unb.ca/cic/datasets/tor.html
物联网安全	N-BaIoT Bot-IoT	archive.ics.uci.edu/ml/datasets/检测物联网僵尸网络攻击 N BaIoT <https://ieee-dataport.org/documents/bot-物联网-dataset>
侧信道攻击	DPA竞赛	http://www.dpacontest.org/index.php
内部威胁检测	ASCAD	https://www.data.gouv.fr/zh-CN/datasets/ascad/
	CERT	https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=508099

C. 网络安全大数据分析

对CPS生成的BD进行实时分析对于各种网络安全应用非常重要，旨在保护数据、计算机系统、网络和物联网免受恶意活动侵害。随着越来越多的非结构化和噪声数据以前所未有的速度生成，需要先进的GPU技术、集群计算框架来高效地处理和海量数据。此类基础设施是BD技术的主要组成部分，用于使用机器学习技术存储、处理和分析数据。一般来说，BD技术分为两类，即批处理（如Hadoop）和流处理（如InfoSphere）。Hadoop框架包括Hadoop分布式文件系统用于存储大型文件和MapReduce编程模型用于处理大规模数据处理问题。采用机器学习框架的Hadoop工具包括Hive（一种SQL友好的查询语言）、Pig（一个用于复杂查询的平台和脚本语言）、Mahout和RHadoop。Spark 4等新框架旨在通过重复使用工作数据集来提高DM和机器学习算法的性能。因此，

专门为高效存储和查询大数据设计的数据库包括NoSQL数据库，如CouchDB、Cassandra、HBase、Greenplum数据库、Vertica和MongoDB。虽然批处理拥有像Hadoop这样的成熟技术，但流处理仍处于初期阶段，尚未广泛采用机器学习/深度学习方法。复杂事件处理（CEP）是流处理模型之一，通过聚合和组合信息流中的事件通知来生成高级事件。Storm、InfoSphere Streams和Jubatus是流技术的其他实现方案。

自编码器（AE）是一种生成模型，它以无监督的方式学习不同特征集的潜在表示，并在网络安全中因其显著的降维能力而被认为是网络流量分析的合适方法。这对于BD来说是有利的，因为大量数据需要在极短的时间内处理，而不会丢失任何信息。奇异值分解（SVD）和主成分分析（PCA）是常用的经典降维方法。限制人工智能（AI）和深度学习（DL）发展的主要因素是处理海量数据集的负担以及与其处理相关的繁重计算需求。传统上，大型数据集是由使用专用高端处理器的计算机集群来处理这些任务的。最近，GPU的强大功能被利用起来，并已被证明在大规模数据处理方面更快、更好，并且有多个平台专门用于开发机器学习（ML）和深度学习（DL）应用。实现深度学习（DL）架构的一些流行平台包括TensorFlow、Theano、Torch、Caffe和DeepDist。表II给出了流行深度学习（DL）框架的总结。大多数知名平台使用C[v1]，它支持后台的高级机器学习（ML）计算和Python前端。此外，Keras已成为支持TensorFlow、CNTK和Theano的常用高级API。

TABLE I
DETAILS OF BENCHMARK DATASETS.

Task	Benchmark Dataset	URL
Intrusion detection	KDDCup-99	http://kdd.ics.uci.edu/databases/kddcup99/kddcup99.html
	NSL-KDD	https://www.unb.ca/cic/datasets/nsl.html
	UNSW-NB15	https://www.unsw.adfa.edu.au/unsw-canberra-cyber/cybersecurity/ADFA-NB15-Datasets/
	ADFA-LD	https://www.unsw.adfa.edu.au/unsw-canberra-cyber/cybersecurity/ADFA-NB15-Datasets/
	UNM	https://www.cs.unm.edu/~immsec/systemcalls.htm
	ISCX-IDS-2012	https://www.unb.ca/cic/datasets/ids.html
	CICIDS2017	https://www.unb.ca/cic/datasets/ids-2017.html
	Kyoto	http://www.takakura.com/Kyoto data/
	UNIBS	http://netweb.ing.unibs.it/~ntw/tools/traces/
	CAIDA	https://www.caida.org/data/
Botnet and DGA Analysis	LBNL	https://commons.lbl.gov/display/cpp/100G+Intrusion+Detection
	CIC DoS	https://www.unb.ca/cic/datasets/dos-dataset.html
	CSE-CIC-IDS2018	https://www.unb.ca/cic/datasets/ids-2018.html
	AWID	http://icsdweb.aegean.gr/awid/index.html
	WSN-DS	https://www.hindawi.com/journals/js/2016/ 4731953/
	UNB Botnet	https://www.unb.ca/cic/datasets/botnet.html
	DGArchive	https://dgarchive.caad.fkie.fraunhofer.de/
	AmritaDGA	https://vinayakumarr.github.io/AmritaDGA/
	ISCX-URL-2016	https://www.unb.ca/cic/datasets/url-2016.html
	Sophos URL	https://github.com/Maddy12/SophosMachineLearningBuildingBlocksTutorial
Spam and phishing Email detection	CSDMC	http://csmining.org/spam-email-datasets-.html/
	Enron	https://www.cs.cmu.edu/~.enron/
	TREC	https://trec.nist.gov/data/spam.html
	SpamAssassin	https://spamassassin.apache.org/old/publiccorpus/
Malware Detection	EMBER	https://github.com/endgameinc/ember
	BIG 2015	https://www.kaggle.com/c/malware-classification
	Malrec	https://giantpanda.gtisc.gatech.edu/malrec/dataset/
	Microsoft Malware Prediction	https://www.kaggle.com/c/microsoft-malware-prediction
Binary Analysis	ByteWeight	http://security.ece.cmu.edu/byteweight/
Image spam detection	Image spam hunter	https://www.unb.ca/cic/datasets/android-adware.html
Android	AAGM	https://www.unb.ca/cic/datasets/android-adware.html
	CICAndMal2017	https:// www.unb.ca/cic/datasets/andmal2017.html
	Drebin	https://www.sec.cs.tu-bs.de/~danarp/drebin/
	Kharon	http://kharon.gforge.inria.fr/dataset/
Traffic Analysis	ISCXVPN2016	https://www.unb.ca/cic/datasets/vpn.html
	ISCXTor2016	https://www.unb.ca/cic/datasets/tor.html
Security in IoT	N-BaIoT Bot-IoT	archive.ics.uci.edu/ml/datasets/detection_of_IoT_botnet_attacks_N_BaIoT https://ieee-dataport.org/documents/bot-iot-dataset
Side channel attacks	DPA contest	http://www.dpacontest.org/index.php
Insider threat detection	ASCAD	https://www.data.gouv.fr/en/datasets/ascad/
	CERT	https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=508099

C. Big data Analytics for Cyber security

Real-time analysis of BD generated in CPS is important for various cyber security applications with the aim to protect data, computer systems, networks, and IoT from malicious activity. With more and more unstructured and noisy data being generated at an unprecedented rate, it requires advanced technologies in GPU, cluster computing frameworks to process and handle very large amount of data in an efficient way [22]. Such infrastructure is the primary component in BD technologies to store, process, and analyze data using ML techniques. In general, BD technologies are divided into 2 categories namely batch processing such as Hadoop, and stream processing such as InfoSphere. The Hadoop framework consists of Hadoop Distributed File System to store large files and MapReduce programming model to work on large-scale data processing problems. Hadoop tools that adopt ML frameworks include Hive (an SQL-friendly query language), Pig (a platform and a scripting language for complex queries), Mahout and RHadoop. New frameworks such as Spark 4 are designed to improve the performance of DM and ML algorithms by repeated reuse of the working dataset. Hence,

databases specifically designed for efficient storage and query of BD include NoSQL databases such as CouchDB, Cassandra, HBase, Greenplum Database, Vertica, and MongoDB. While batch processing has a dominant mature technology such as Hadoop, stream processing is still in its infancy to embrace ML/DL approaches. Complex Event Processing (CEP) is one of the models for stream processing where high-level events are produced by aggregating and combining notification of events which are considered from the information flow. Storm, InfoSphere Streams, and Jubatus are few other implementations of stream technologies.

Autoencoder (AE) is a generative model which learns the latent representation of different feature sets in an unsupervised manner and is considered suitable method for network traffic analysis due to its significant dimensionality reduction in cyber security. This is advantageous with BD as large amounts of data require to be processed in a fraction of time without any loss of information. Singular value decomposition (SVD) and principal component analysis (PCA) are commonly used classical methods for dimensionality reduction. The main factors limiting the growth of AI and DL are the burden

表 II 常见深度学习框架对比。

深度学习框架	创建者	许可证	核心语言	接口支持
Caffe ¹	伯克利中心	FreeBSD	C++	Python & MATLAB
Torch ²	Ronan Collobert 等	BSD	C, & Lua	C/C++, Lua, Python
Theano ³	蒙特利尔大学	BSD	Python	Python
Deeplearning4j ⁴	Skymind	Apache 2.0	Java	Java, Scala 和 Python
MXNet ⁵	Apache软件基金会	Apache 2.0	C++	C++, Python, R, Scala, Perl, Julia 等
TensorFlow ⁶	Google Brain团队	Apache 2.0	C++, & Python	Python, C/C++, Java, Go
H2O ⁷	H2O.ai	Apache 2.0	Java	R, Python, Scala, REST API
Chainer ⁸	日本ベンチャー企業 Preferred Networks 与 IBM 合作 与 Intel , Intel , Microsoft , Nvidia	MIT	Python	Python
CNTK ⁹	微软研究院	MIT	C++	Python, C++, BrainScript
Keras ¹⁰ (更高级别的库, 用于 TensorFlow, CNTK, Theano)	Franois Chollet	MIT	Python	Python

传统上，大型数据集是由使用专用高端处理器的计算机集群来处理这些任务的。最近，GPU的强大功能被利用起来，并已被证明在大规模数据处理方面更快、更好，并且有多个平台专门用于开发机器学习（ML）和深度学习（DL）应用。实现深度学习（DL）架构的一些流行平台包括 TensorFlow、Theano、Torch、Caffe和DeepDist。表II给出了流行深度学习（DL）框架的总结。大多数知名平台使用C++，它支持后台的高级机器学习（ML）计算和Python前端。此外，Keras已成为支持 TensorFlow、CNTK和Theano的常用高级API。

IV. 对抗深度学习在网络安全中的应用

随着机器学习被应用于各种关键系统，考虑其可靠性变得极为重要，因为这些算法容易受到攻击者的攻击。黑客利用机器学习框架的漏洞，使用对抗样本进行攻击，这与他们利用防火墙漏洞的方式非常相似。在部署前，通过在对抗环境中进行压力测试来识别机器学习漏洞，并考虑机器学习框架的不足之处，这是一项称为对抗性机器学习的研究。由于许多机器学习框架在关键系统中表现为黑盒，对抗性机器学习面临约束。在这样环境中，专家和客户难以理解模型结果，因为框架没有解释其决策。在没有对机器学习框架鲁棒性任何保证的情况下，它们在安全关键系统中的应用变得遥不可及。表III总结了网络安全文献中对抗深度学习和数据集的最新应用。

A. 领域生成算法

域名生成算法（DGA）已被多个恶意软件家族采用，用于建立命令与控制（C&C）连接，导致分布式拒绝服务（DDoS）攻击。

表III 对抗深度学习算法和数据集摘要

参考文献	数据集
[179]	Alexa & 私有
[180]	基因组、Contagio、VirusShare、Drebin
[181]	Drebin
[182]	BIG 2015
[184]	私有
[183]	恶意为 Windows 审计日志
[185]	VirusShare, Citadel, APT1
[186]	Spambase
[187]	野生数据集 & BIG 2015
[188]	GTSRB
[189]	Enron 垃圾邮件数据集
[190]	Mozilla Common Voice 数据集
[191]	NSL-KDD
[192]	AICS’ 2019 挑战数据集
[193]	HCRL 车辆入侵数据集
[194]	ADFA-LD
[195]	CIDDS-001
[196]	KDDCup-99
[197]	Ember
[198]	AmritaDGA
[199]	CycleGAN & StarGAN

最近，一种基于深度学习（DL）的架构——生成对抗网络（GAN）被训练用于生成对抗性域名，以有意绕过基于深度学习的检测器 [179]。另一项研究提出了一种黑盒对抗攻击方法，用于规避DGA，而无需任何关于分类器模型的先验知识。 [198]。该方法使用DMD-2018数据集进行测试，成功将F1分数从0.977降至0.495。

B. 恶意软件检测对抗

恶意软件重组变异（MRV），是一种基于对现有恶意软件进行语义分析以生成对抗性恶意软件样本的新方法，旨在规避恶意软件检测器并通过使用三种防御技术 [180] 来增强检测器的鲁棒性。另一项研究则利用现有的对抗性样本生成算法生成恶意软件，同时不损失其侵入性功能 [181]。研究人员提出了一种基于迁移式生成对抗网络（tGAN）的恶意软件检测框架，用于检测零数据攻击，并展示了良好的学习稳定性和96.39%的准确率 [182]。此外，为了

TABLE II
COMPARISON OF POPULAR DL FRAMEWORKS.

DL Frameworks	Creator	License	Core Languages	Interface Support
Caffe ¹	Berkeley Center	FreeBSD	C++	Python & MATLAB
Torch ²	Ronan Collobert et al.	BSD	C, & Lua	C/C++, Lua, Python
Theano ³	University of Montreal	BSD	Python	Python
Deeplearning4j ⁴	Skymind	Apache 2.0	Java	Java, Scala, & Python
MXNet ⁵	Apache Software Foundation	Apache 2.0	C++	C++, Python, R, Scala, Perl, Julia, etc.
TensorFlow ⁶	Google Brain Team	Apache 2.0	C++, & Python	Python, C/C++, Java, Go
H2O ⁷	H2O.ai	Apache 2.0	Java	R, Python, Scala, Rest API
Chainer ⁸	Japanese venture company Preferred Networks in partnership with IBM, Intel, Microsoft, Nvidia	MIT	Python	Python
CNTK ⁹	Microsoft Research	MIT	C++	Python, C++, BrainScript
Keras ¹⁰ (higher level library for TensorFlow, CNTK, Theano)	Franois Chollet	MIT	Python	Python

of handling massive datasets and the heavy computational requirements associated with their processing. Traditionally large datasets were handled by clusters of computers using dedicated high-end processors to perform these tasks. Recently, the power of GPU’s has been harnessed and has proven to be faster and better for large- scale data processing with a number of platforms tailored for developing ML and DL applications. Such popular platforms implementing DL architectures include TensorFlow, Theano, Torch, Caffe, and DeepDist. A summary of popular DL frameworks is given in Table II. Most of the well-known platforms use C++ that support high-level ML computations at the backend and a Python frontend. In addition, Keras has become the commonly used high-level API which supports TensorFlow, CNTK and Theano.

IV. ADVERSARIAL DEEP LEARNING FOR CYBER SECURITY

As ML is being applied for deployment in various critical systems, it is extremely imperative to consider the reliability of such algorithms as they are susceptible to attacks by adversaries. Hackers exploit the vulnerabilities of ML frameworks using adversarial samples much similar to how they are exploiting firewall vulnerabilities. It is important to consider the shortcomings of a ML framework by conducting stress tests in adversarial environments to identify ML vulnerabilities before deployment, and such a study is known as adversarial ML. As many ML frameworks behave as black boxes in critical systems, adversarial ML face constraints. It is exceptionally troublesome for experts and clients to comprehend the model results in such environments as there is no explanation about the decision made by the framework. In the absence of any assurance on the robustness of ML frameworks, their use in secure critical system becomes remote. In Table III, we summarise the state-of-the-art use of adversarial DL and datasets found in cyber security literature.

A. Domain Generation Algorithms

Domain generation algorithms (DGAs) have been adopted by several malware families utilize to establish command and control (C&C) connections resulting in distributed denial of

TABLE III
SUMMARY OF ADVERSARIAL DL AND DATASETS USED.

Reference	Dataset
[179]	Alexa & Private
[180]	Genome, Contagio, VirusShare, Drebin
[181]	Drebin
[182]	BIG 2015
[184]	Private
[183]	Malicious Behavior Windows Audit Logs
[185]	VirusShare, Citadel, APT1
[186]	Spambase
[187]	Wild dataset & BIG 2015
[188]	GTSRB
[189]	Enron Spam Dataset
[190]	Mozilla Common Voice Dataset
[191]	NSL-KDD
[192]	AICS’2019 challenge dataset
[193]	Car-Hacking Dataset from HCRL
[194]	ADFA-LD
[195]	CIDDS-001
[196]	KDDCup-99
[197]	Ember
[198]	AmritaDGA
[199]	CycleGAN & StarGAN

service (DDoS) attacks. Recently, a DL based architecture called generative adversarial network (GAN) was trained to generate adversarial domain names to intentionally bypass a DL based detector [179]. Another study proposed a black box adversarial attack approach for evading DGA without any prior knowledge of the classifier model. [198]. The approach was successfully tested using DMD-2018 dataset to degrade the F1-score from 0.977 to 0.495.

B. Malware Detection Adversary

Malware Re-composition Variation (MRV), a novel method generated adversarial malware examples based on semantic analysis of existing malwares to evade malware detector and to enhance the robustness of the detector by using three defense techniques [180]. Another study employed existing adversarial example generation algorithms to generate malwares without losing its intrusive functionality [181]. A malware detection framework based on transferred GAN (tGAN) was proposed to detect zero-data attacks and demonstrated good learning stability and accuracy of 96.39% [182]. Further, in order

提升深度学习模型（DL model）对抗对抗攻击的鲁棒性，研究人员提出了一种恶意软件检测模型，用于消除数据中的随机特征 [183]。基于六项原则对对抗攻击进行的改进尝试 [192], 包括可视化方法和基于API调用的对抗方法，以规避RNN、DNN和机器学习（ML）分类器等分类器。此类研究旨在训练GAN生成器，以增强深度学习模型，相较于CMLA。

C. 入侵检测系统（IDSs）对抗

一个基于深度自编码器（AE）的自适应IDS，用于测试其对抗样本的鲁棒性，与基于PCA的检测系统相比，准确率提高了15% [191]。一些研究提出了基于GAN的黑盒攻击，通过生成对抗性网络流量来逃避IDS的检测，成功率高，包括在智能车辆网络中[193]。基于GAN的主机端ID系统（HIDS）被提出，通过生成对抗性异常来训练人工神经网络（ANN）模型 [194]。类似地，研究了针对基于深度神经网络（DNN）的网络ID系统（NIDS）的三种黑盒对抗攻击，其中两个数据增强模块生成了对抗性数据，以解决NIDS中的数据不足问题 [196]。

D. 网络安全中的其他基于对抗的攻击和防御技术

集成防御能够提升检测模型对抗攻击的鲁棒性。研究人员已针对对抗攻击机制进行了研究，并提出了一种结合权重衰减防御的集成技术来提高其效能 [184], , 同时通过在基于异常的防御技术中增加少量隐藏层，进一步提升了线性分类器对中毒攻击的鲁棒性 [186]。然而，它们无法检测到攻击强度较弱的攻击，如标签翻转。另一项研究采用广义蒸馏学习方法，使用特权特征 [187]训练基于深度学习的检测模型。通过使用对抗样本将自动驾驶汽车中基于计算机视觉的模块中的广告和无害标志误分类，研究了现实世界应用中针对计算机视觉模块的攻击，成功率为95% [188]。另一项研究提出了一种针对黑盒环境中深度学习分类器的新型对抗攻击，通过修改数据中的微小文本扰动来生成对抗样本，导致IMDB和Enron数据集的准确率下降60% [189]。基于深度学习的自动语音识别系统被攻击，通过向输入音频波形添加微小扰动生成对抗样本，成功率为100%[190]。基于深度学习的模型使用生成对抗网络和三种预处理方法生成基于流的网络流量，并评估了生成流量数据的质量 [195]。另一项研究提出了检测由生成对抗网络生成的假图像的方法，通过在图像的RGB通道上计算共现矩阵，并使用这些矩阵训练卷积神经网络模型来检测假生成对抗网络图像 [199]。所提出的框架使用了cycleGAN和StarGAN等两个生成对抗网络数据集来测试模型，在两个数据集上均实现了99%的准确率。

表IV 网络安全文献中的强化学习（RL）研究和数据集。

参考文献	数据集
[153]	KDDCup-99
[154], [155], [156], [157], [160]	私有
[158]	网络钓鱼语料库、垃圾邮件过滤器、PhishTank
[159]	Ember

V. 强化学习在网络安全中的应用

强化学习(RL)是一种受巴甫洛夫经典条件反射技术和马尔可夫决策过程数学概念启发的进化式人工智能技术 [200]。强化学习算法由三个关键要素构成，即观察、奖励和行动。每次，算法被允许做出决策并观察场景的变化，从而获得相应的负奖励或正奖励。通常的目标是获得最大奖励。因此，算法借助奖励利用马尔可夫决策过程来获得最大奖励或达到特定目标。这个过程被称为经典强化学习，仅适用于较小的问题。对于较大问题，深度强化学习(DRL)利用神经网络或近似方法来寻找问题的最优值或解决方案。基于深度强化学习的网络安全应用解决方案仍处于起步阶段。该方法适用于网络安全应用，如僵尸网络检测和恶意软件检测。表IV提供了关于强化学习算法在网络安全中使用的数据库的研究总结。

A. 基于强化学习的入侵检测

近期文献综述报道了多项研究，探索了应用于公共数据集和私有数据集的各种强化学习框架。提出了一种基于机器学习的自适应入侵检测系统，采用多类支持向量机结合主成分分析进行特征降维，并使用强化学习方法进行预测[153]。该模型通过采用公开可用的基准数据集（如KDDCup-99数据集）进行训练，取得了令人满意的结果。随着我们步入工业4.0时代，网络中的DDoS攻击和物联网漏洞在信息物理系统（CPS）中愈发频繁。为应对这一问题，分布式入侵检测系统开始采用强化学习来增强安全性。提出了一种使用强化学习传感器代理来区分正常和异常网络状态，以及决策代理来学习传感器代理动作语义的分布式入侵检测系统 [154]。通过使用多代理路由限流，提出了一种更自主、可扩展且安全的分布式信息物理系统DDoS检测与响应系统[155]。另一项研究使用强化学习代理分析主机之间的数据流信息，检测了基于洪泛的拒绝服务攻击（DoS）和DDoS攻击 [157]。

B. 网络安全中的其他各种应用

一项最新研究探索了基于强化学习的框架，以深入了解恶意软件检测器的漏洞，从而

to boost the robustness of a DL model against adversarial attack, a malware detection model was proposed to nullify random features of data [183]. Enhancements to adversarial attacks based on six principles were attempted [192], including visualization methods and API call based adversarial to evade classifiers such as RNN, DNN, and ML classifiers. Such studies aim to train the GAN generator for enhancing DL models as compared to CMLA.

C. Intrusion Detection Systems (IDSs) Adversary

A deep AE based adaptive IDS to test its robustness against adversarial examples resulted in improving accuracy by 15% when compared to PCA based detection system [191]. Some studies have proposed black box attacks based on GAN against IDS by generating adversarial network traffic to evade detection with high success rate including in smart vehicle networks [193]. Host-based ID System (HIDS) based on GAN was proposed by producing adversarial anomalies to train an ANN model [194]. Similarly, three black box adversarial attacks against DNN based Network ID Systems (NIDS) were studied, and two data augmentation modules generated adversarial data for addressing the data insufficiency challenge in NIDS [196].

D. Other Adversarial based Attacks and Defense Techniques in Cyber security

Ensemble defenses are capable of improving the robustness of detection models against adversarial attacks. Such mechanisms against adversarial attack were studied and an ensemble technique was proposed with a weight decay defense to improve its efficacy [184], and by adding few more hidden layers in a outlier based defense technique, the robustness was improved further in linear classifiers against poisoning attacks [186]. However, they fail to detect attacks that are less aggressive such as label flipping. Another study employed generalized distillation learning approach to train the DL based detection model using privileged features [187]. Attacks against computer vision based modules of autonomous vehicles in real-world applications were studied by using adversarial examples to misclassify advertisements and innocuous signs with a success rate of 95% [188]. Another study proposed a novel adversarial attack against DL classifier in a black box environment by modifying the data with small text perturbation to produce adversarial examples that degraded the accuracy by 60% with IMDB and Enron datasets [189]. A DL based automatic speech recognition system was attacked by generating an adversarial example from an input audio waveform by adding a small perturbation with 100% success [190]. A DL based model generated flow-based network traffic by using GANs with three preprocessing methods and evaluated the quality of the generated traffic data [195]. Another study for the detection of fake images generated by GAN was proposed to compute the co-occurrence matrices on the RGB channels of the images by using those matrices to train the CNN model to detect fake GAN images [199]. The proposed framework used two GAN datasets such as cycleGAN and StarGAN to test the model resulting in 99% accuracy for both datasets.

TABLE IV
RL STUDIES AND DATASETS IN CYBER SECURITY LITERATURE.

Reference	Dataset
[153]	KDDCup-99
[154], [155], [156], [157], [160]	Private
[158]	PhishingCorpus, SpamAssassin, PhishTank
[159]	Ember

V. REINFORCEMENT LEARNING FOR CYBER SECURITY

Reinforcement learning (RL) is a revolutionary AI technique that is inspired by the psychological concept of Pavlov’s classical conditioning technique and the mathematical concept of Markov decision process [200]. There are three vital elements that construct an RL algorithm namely, observation, reward and action. Each time, the algorithm is allowed to take a decision and observes the changes in the scenario which in turn receives a corresponding negative or positive reward. Generally, the motive is to gain the maximum reward. Therefore, the algorithm aided by the reward uses the Markov decision process to either receive the maximum reward or reach a particular goal. This process is known as classical reinforcement learning which are only suitable for smaller problems. In case of larger problems, deep reinforcement learning (DRL) utilizes NNs or approximation methods for finding the optimal value or solution for the problem. DRL based solutions for the applications of cyber security are still in the beginning stage. This methodology can be suitable for cyber security applications such as botnet detection and malware detection. A summary of studies conducted on RL algorithms and the datasets used in cyber security is provided in Table IV.

A. Reinforcement Learning based Intrusion Detection

Recent literature surveys have reported several studies exploring various RL frameworks applied to both public and private datasets. An adaptive IDS based on ML was proposed using multi-class SVM with principal component analysis (PCA) for feature reduction and RL approach for prediction [153]. The model had achieved promising results by adopting training with benchmark datasets available publicly such as the KDDCup-99 dataset. As we embark into Industry 4.0, more DDoS over the network and IoT exploits happen in CPS. To address this, distributed IDS started adopting RL to enhance security. A distributed IDS was proposed using RL sensor agents to differentiate normal and abnormal network states and the decision agents to learn the semantics of the actions agents of sensors [154]. A more autonomous, scalable and secure decentralized DDoS detection and response system for CPS was proposed by using multi RL agent router throttling [155]. Another study detected flooding based DoS and DDoS attacks using RL agents to analyse the data flow information between hosts [157].

B. Other Various Applications in cyber security

A recent study explored the RL based framework to gain insights about the loopholes of the malware detector so as

预先阻止恶意攻击 [156]。提出了一种自适应神经网络和强化学习在线网络钓鱼电子邮件检测框架，以检测零日网络钓鱼攻击，准确率达98.63%，假阳性率极低为1.81% [158]。提出了一种新的深度强化学习（DRL）恶意软件执行控制框架，用于模拟恶意软件执行并在固定数量的系统调用后停止，从而将真阳性率大幅提高61.5%，同时与其他基线分类器相比保持极低的假阳性率1% [160]。

VI. 深度学习在网络安全中的应用

深度学习框架在网络安全中的应用正日趋成熟，该方向的研究也日益受到关注。然而，由于缺乏可用于适应工业4.0动态环境中高度复杂深度学习模型的标记数据集和基准数据集，该领域的优质研究成果受到阻碍。我们将应用于网络安全的深度学习方法按主要类别及其子类层次进行分类，如图3所示。在本节中，我们总结了文献中关于应用于网络安全的基于深度学习的威胁检测方法的关键成就和局限性。

A. 深度学习在入侵检测中的应用

在当今互联的CPS（信息物理系统）世界中，日常活动已变得自主化，更容易受到针对关键基础设施的攻击。例如，自动驾驶汽车可能被易受网络攻击的设备和智能手机控制。在表V中，我们提供了关于采用不同数据集进行入侵检测（ID）的深度学习架构研究综述。由于基准数据集在深度学习训练和测试过程中起着重要作用，我们简要回顾了12个不同基准数据集，并对采用深度学习架构与CMLA（类马尔可夫链自动机）的综述进行了比较研究。

1) *KDDCup-99* 数据集：一项研究使用长短期记忆网络来检测网络入侵，通过在 KDDCup-99 挑战中评估其在不同特征集上的性能，准确率达到 93.82%，优于其他方法 [201]。基于卷积神经网络的混合架构（如 CNN-LSTM、CNN-RNN 和 CNN-GRU）用于识别（IDS）的性能优于任何其他混合模型 [202]。在另一项研究中，基于深度神经网络的身份识别（IDS）系统与其他传统机器学习模型相比，达到了 99.9% 的最高准确率[222]。

2) *NSL-KDD*数据集：研究了一种用于二元和多元分类的RNN模型，表明具有80个隐藏神经元和学习率为0.1的模型在二元分类中取得了最佳结果，而在多元分类中，具有80个隐藏神经元和学习率为0.5的模型取得了最佳结果 [203]。此外，提出了一种基于自编码器（AE）的STL框架用于识别（ID），以高效地学习特征并降低特征的维度，以帮助基于SVM的检测器 [208]。一种基于CNN的字符级IDS通过将网络流量视为字符序列来预处理数据，其准确率优于传统的机器学习（ML）分类器，达到85.07% [210]。

在另一项研究中，研究了CNN、LSTM和自编码器（AE）模型在基于异常的识别（ID）系统中的有效性，CNN据报道比自编码器和其他传统机器学习（ML）分类器表现更好 [211]。在另一项研究中，一种基于堆叠稀疏自编码器（AE）的IDS框架通过使用高维稀疏特征加速了检测过程，用于分类正常和恶意流量 [221]。

3) 私有数据集：提出了一种基于卷积神经网络模型的低速端口扫描检测系统，用于过滤正常数据包，并使用其源和目的IP将剩余的可疑数据包分组。该卷积神经网络模型从输入中提取了区间和序列特征，以检测端口扫描，精确率为97.4%[215]。

4) *UNSW-NB15* 数据集：针对网络异常检测提出了一种使用卷积神经网络模型的编码方法，以提升其性能，相较于基于随机森林的检测器 [212]。所提出的编码方法在与其他灰度编码的比较中始终能获得更好的结果。

5) 京都数据集：提出了一种基于混合GRU和SVM的IDS，该IDS使用SVM代替softmax在最终输出层进行入侵检测，基于网络流量数据[205]。该模型准确率达到81.54%，而传统softmax方法的准确率仅为63.07。

6) *ISCX-IDS-2012*数据集：提出了一种基于CNN和随机森林算法的IDS，用于从原始网络流量中提取有效载荷特征。从网络流量中提取的统计特征和CNN的有效载荷特征用于训练随机森林分类器，准确率达到99.13%，误报率为1.18%。与其他SVM、NN、CNN和随机森林模型相比，该模型结果更好 [218]。所提出的模型准确率达到99.97%，并且误报率非常低的0.02%。

7) *CICIDS2017*数据集：在端口扫描异常检测实验中，深度学习模型准确率达到97.80%，而SVM准确率仅为69.79% [207]。

8) *AWID*数据集：使用自编码器和深度神经网络对WiFi网络入侵检测系统中的流量进行分类为正常、伪装攻击、洪泛攻击和注入攻击的基于深度学习的解决方案，分别达到了98.4%、98.3%、73.1%和99.9%的准确率，针对4种不同类别 [209]。

9) *HTTP DATASET CSIC 2010*数据集：基于字符级卷积神经网络的Web应用防火墙使用unicode编码的原始HTTP请求进行训练，达到了98.8%的准确率，平均处理时间为2.35毫秒 [213]。

10) *CIDDS-001*数据集：研究了长短期记忆网络在基于流量的NIDS中的有效性，使用不同的超参数组合，实验分析比较了LSTM模型与其他传统机器学习方法的表现。在另一项针对基于异常的IDS的研究中，机器学习和深度学习模型使用了各种技术来修复不平衡数据集，并训练了深度神经网络、变分自编码器、随机森林、投票和堆叠的机器学习模型 [220]。

11) *CTU-13*数据集：使用网络流量数据中的流特征，对5G网络进行了基于双层深度学习的自适应异常检测方法实验，并在第一层训练DBN或SAE模型，在第二层训练LSTM [223]。

to pre-empt the malicious attacks [156]. A self-adapting NN and RL based online phishing email detection framework was proposed to detect zero-day phishing attacks with 98.63% accuracy and a very low FPR of 1.81% [158]. A new deep RL (DRL) based malware execution control framework was proposed to emulate a malware execution and to stop after a fixed number of system calls, thereby improving TPR massively by 61.5% while keeping a very low FPR of 1% as compared to other baseline classifiers [160].

VI. DEEP LEARNING APPLICATIONS IN CYBER SECURITY

The application of DL frameworks in cyber security is maturing and research in this direction is of growing interest. However, quality research output in this domain is hampered by lack of labeled data and benchmark datasets that can suit highly complex DL models in the dynamic environment of Industry 4.0. We classify DL methods applied to cyber security falling under main categories and a hierarchy of sub-categories as shown in Fig. 3. In this section, we summarize the key achievements and limitations in literature related to DL-based threat detection methods applied in cyber security.

A. DL in Intrusion Detection

In today’s interconnected world of CPS, everyday activities have become autonomous and more vulnerable to attacks that are targeted against critical infrastructures. For instance, autonomous cars can be controlled by devices and smart phones that are prone to cyber-attacks. In Table V, we provide a summary of research studies conducted on DL architectures with different datasets adopted for ID. Since the benchmark dataset plays a major role in the DL training and testing processes, we provide short review of each of the 12 different benchmark datasets and conduct a comparative study on the surveys conducted with DL architectures versus CMLAs.

1) *KDDCup-99 Dataset*: A study used LSTM to detect network intrusion and outperformed other approaches with an accuracy of 93.82% by evaluating its performance with different feature sets in KDDCup-99 challenge [201]. CNN based hybrid architectures for IDS such as CNN-LSTM, CNN-RNN, and CNN-GRU had performed better than any other hybrid model [202]. In another study, a DNN based IDS achieved a maximum accuracy of 99.9% as compared to other traditional ML models.[222].

2) *NSL-KDD Dataset*: An RNN model for both binary and multi-class classification was studied showing that a model with 80 hidden neurons and a learning rate of 0.1 had achieved the best result in binary classification while in multi-class classification, the model with 80 hidden neurons and a learning rate of 0.5 had achieved the best results [203]. Further, an STL framework based on AEs for ID was proposed to learn efficiently from the features and reduces the dimensionality of the features to aid the SVM based detector [208]. A CNN based character level IDS preprocessed the data by considering the network traffic as sequences of character and performed better than traditional ML classifiers with an accuracy of 85.07% [210].

In another study, the effectiveness of CNN, LSTM, and AE models were studied for anomaly based ID system CNN and reported to perform better than AEs and other traditional ML classifiers [211]. In another study, a stacked sparse AE based IDS framework accelerated the detection process to classify normal and malicious traffic using high dimensional sparse features [221].

3) *Private Dataset*: A low speed port scan detection system based on CNN model was proposed to filter the normal packets and group the remaining suspicious packets using its source and destination IP. The CNN model had extracted the interval and sequential features from the input to detect port scan with a precision of 97.4%.[215].

4) *UNSW-NB15 Dataset*: A novel encoding approach using CNN model was proposed for network anomaly detection to enhance its performance as compared to a random forest based detector [212]. The proposed encoding approach gave consistently better results when compared to gray-scale encoding.

5) *Kyoto Dataset*: A hybrid GRU and SVM based IDS was proposed using SVM instead of softmax in the final output layer for detecting intrusion based on network traffic data [205]. The model achieved an accuracy of 81.54 while the accuracy of the traditional softmax approach was only 63.07.

6) *ISCX-IDS-2012 Dataset*: A IDS based on CNN and random forest algorithm were proposed to extract payload features from raw network traffic. The statistical features extracted from the network traffic and payload features from the CNN were used to train the random forest classifier and achieved an accuracy of 99.13% and FAR of 1.18%. The model results were better when compared to other SVM, NN, CNN and Random forest models [218]. The proposed model achieved 99.97% accuracy and a very impressive FAR of 0.02%.

7) *CICIDS2017 Dataset*: In a port scan anomaly detection experiment, a DL model achieved 97.80% accuracy while SVM achieved only 69.79% accuracy [207].

8) *AWID Dataset*: A DL based solution for WiFi NIDS using SAEs and DNN to classify the traffic into 4 classes such as normal, impersonation attack, flooding attack, and injection attack had achieved an accuracy of 98.4%, 98.3%, 73.1% and 99.9%, for the 4 different classes respectively [209].

9) *HTTP DATASET CSIC 2010 Dataset*: A character level CNN based web application firewall was trained using unicode encoded raw http requests and achieved an accuracy of 98.8% with average processing time of 2.35ms [213].

10) *CIDDS-001 Dataset*: The effectiveness of LSTM for flow-based NIDS was studied with different combination of hyperparameters and the experimental analysis compared the performance of the LSTM models with other traditional ML approaches [220]. In another study for anomaly-based IDS, ML and DL models used various techniques to fix the imbalanced dataset and trained DNN, VAE, random forest, voting, and stacking ML models.

11) *CTU-13 Dataset*: Applying a two-level DL based adaptive anomaly detection approach was experimented for 5G networks using flow-based features from the network traffic data and by training a DBN or a SAE model in the first level, and LSTM in the second level [223].

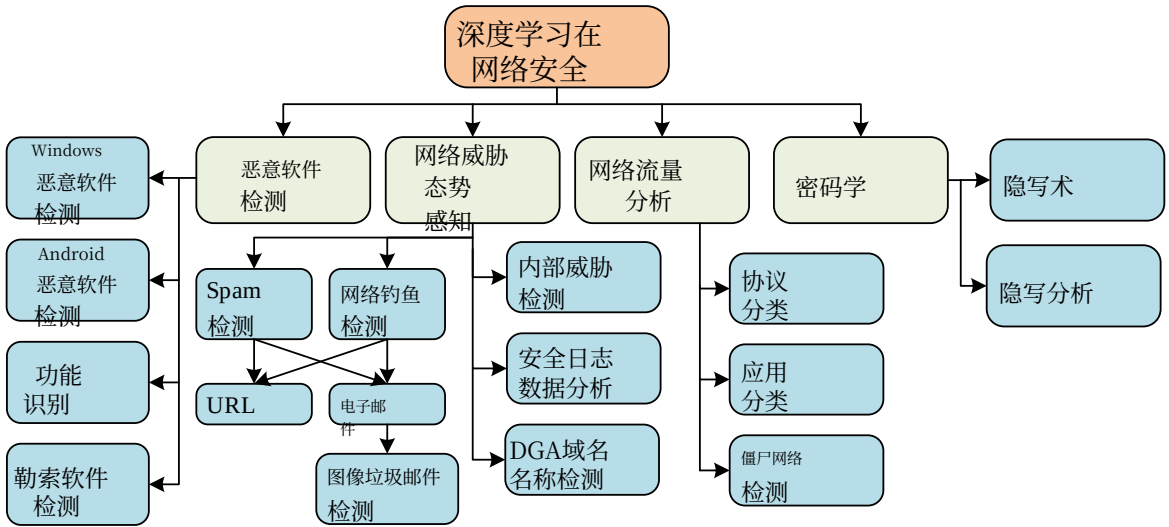


图3。网络安全中深度学习方法（DL）的层次分类。

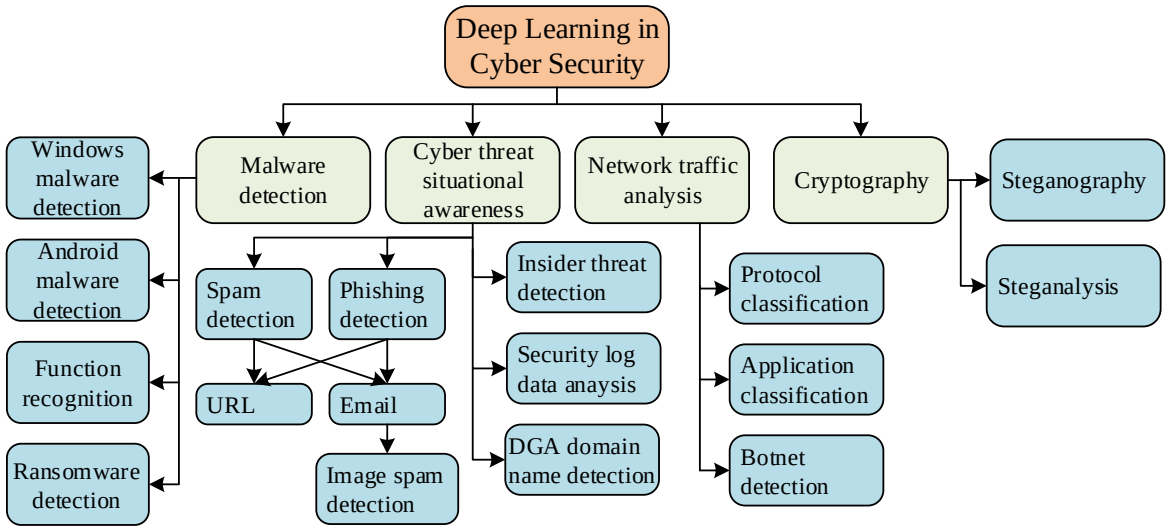


Fig. 3. A hierarchical classification of DL Methods in Cyber Security.

表V
关于深度学习架构和用于入侵检测的数据集的研究综述

参考文献	架构	数据集	与CML的比较
[201]	RS	KDDCup-99	Yes
[211]	DNN	NSL-KDD	No
[202]	卷积神经网络, CNN-RS	KDDCup-99	No
[210]	CNN	NSL-KDD	Yes
[208]	自编码器	NSL-KDD	Yes
[203]	RS	NSL-KDD	Yes
[222]	DNN	KDDCup-99	Yes
[227]	自编码器	KDDCup-99	No
[204]	卷积神经网络 & 反向传播算法	DARPA1998 & ISCX-IDS-2012	Yes
[205]	反向传播算法 & 支持向量机	京都	No
[206]	CNN	Balabit, & TWOS	No
[207]	DNN	CICIDS2017	Yes
[227]	自编码器	NSL-KDD	No
[212]	CNN	UNSW-NB15	No
[213]	CNN	HTTP数据集 CSIC 2010	Yes
[214]	RS	ISCX-IDS-2012 & AWID	Yes
[215]	CNN	私有	No
[216]	DBN	KDDCup-99, NSL-KDD, UNSW-NB15, CICIDS 2017	Yes
[217]	自编码器	NSL-KDD & UNSW-NB15	Yes
[218]	CNN	ISCX-IDS-2012	Yes
[219]	DNN	实时交通, KDDCup-99, UNSW-NB15	Yes
[220]	反向传播算法 & 深度神经网络	CIDDS-001	Yes
[221]	稀疏自动编码器	NSL-KDD	Yes
[223]	DBN, SAE, 反向传播算法	CTU-13	No
[225]	自编码器与深度神经网络	KDDCup-99, & UNSW-NB15	Yes
[226]	卷积神经网络 & 反向传播算法	CICIDS2017	No
[264]	DNN, RS, CNN	KDDCup 99 & 京都	Yes

12) 混合数据集：深度学习与机器学习研究对比：提出了一种基于CNN-LSTM的识别系统，其中CNN模型用于从网络流量数据中学习低级空间特征，LSTM模型用于学习高级时间特征 [204]。为雾计算环境中的网络攻击检测提出了分布式可扩展的LSTM架构 [214]。还提出了一种分布式DBN，结合基于集成SVM检测框架的大规模网络，其中分布式DBN用于非线性降维，基于Apache Spark的集成SVM模型用作检测分类器。该模型在

4个不同的数据集，并与其他相关模型进行了比较[216]。在另一项研究中，针对工业物联网专门设计了一种使用DAE和DNN的基于异常的IDS。DAE模型通过学习网络正常行为产生了最优参数，这些参数被用来有效地调整基于DNN分类器的参数[217]。提出了一种用于路由攻击检测的可扩展DNN框架，其中攻击数据集是从Cooja物联网模拟器中提取的 [219]。提出了一种基于堆叠AE的NIDS，通过使用UNSW-NB15和KDDCup-99数据集学习大量未标记数据中的重要特征，实现了89.134%和99.996%的准确率

TABLE V
SUMMARY OF STUDIES ON DL ARCHITECTURES AND DATASETS ADOPTED FOR INTRUSION DETECTION.

Reference	Architecture	Dataset	Compared CML
[201]	RS	KDDCup-99	Yes
[211]	DNN	NSL-KDD	No
[202]	CNN, CNN-RS	KDDCup-99	No
[210]	CNN	NSL-KDD	Yes
[208]	Autoencoder	NSL-KDD	Yes
[203]	RS	NSL-KDD	Yes
[222]	DNN	KDDCup-99	Yes
[227]	AutoEncoder	KDDCup-99	No
[204]	CNN & RS	DARPA1998 & ISCX-IDS-2012	Yes
[205]	RS & SVM	Kyoto	No
[206]	CNN	Balabit, & TWOS	No
[207]	DNN	CICIDS2017	Yes
[227]	Autoencoder	NSL-KDD	No
[212]	CNN	UNSW-NB15	No
[213]	CNN	HTTP DATASET CSIC 2010	Yes
[214]	RS	ISCX-IDS-2012 & AWID	Yes
[215]	CNN	Private	No
[216]	DBN	KDDCup-99, NSL-KDD, UNSW-NB15, CICIDS 2017	Yes
[217]	Autoencoder	NSL-KDD & UNSW-NB15	Yes
[218]	CNN	ISCX-IDS-2012	Yes
[219]	DNN	Real-time traffic, KDDCup-99, UNSW-NB15	Yes
[220]	RS & DNN	CIDDS-001	Yes
[221]	Sparse Autoencoder	NSL-KDD	Yes
[223]	DBN, SAE, RS	CTU-13	No
[225]	Autoencoder & DNN	KDDCup-99, & UNSW-NB15	Yes
[226]	CNN & RS	CICIDS2017	No
[264]	DNN, RS, CNN	KDDCup 99 & Kyoto	Yes

12) Mixed Dataset: DL studies compared with ML: A novel CNN-LSTM based ID system was proposed with the CNN model to learn the low level spatial features and the LSTM model to learn the high level temporal features from the network traffic data [204]. LSTM architectures for cyber attack detection in fog-of-things environment were proposed to be distributed and scalable to detect cyber-attacks [214]. Another distributed DBN with an ensemble SVM-based detection framework in a large scale network was proposed. The distributed DBN was used for non-linear dimensionality reduction and an apache spark based ensemble SVM model was used as a detection classifier. The model is trained on

4 different datasets and compared with other related models [216]. In another study, anomaly based IDS using DAE and DNN was specifically designed for industrial Internet of things. The DAE model produced optimal parameters by learning the normal behaviour of network which were used to effectively tune the parameters of DNN based classifier [217]. A scalable DNN based framework for routing attack detection was proposed where the attack dataset was extracted from the Cooja IoT simulator [219]. A stack AE based NIDS was proposed with learning of important features from a large quantity of unlabeled data using UNSW-NB15 and KDDCup-99 datasets achieved an accuracy of 89.134% and 99.996%

分别 [225]。使用NSL-KDD数据集评估了基于DNN的主机和网络层身份识别架构以及各种深度学习架构，如LSTM、GRU、RNN和IRNN [264]。

与机器学习进行比较的深度学习研究：提出并训练了一种基于二维（2D）卷积神经网络的鼠标行为认证系统，该系统使用Balabit和TWOS这两个公开数据集进行训练，平均AUC达到0.96，优于一维卷积神经网络和支持向量机模型 [206]。研究了几种深度学习模型在DDoS攻击检测中的有效性，其中多层感知机、卷积神经网络、长短期记忆网络和CNN-LSTM模型使用CICIDS2017数据集进行训练。所提出的CNN-LSTM模型通过达到97.16%的准确率，优于其他模型 [226]。在另一项研究中，基于深度学习的自适应和可扩展的误入入侵检测系统使用NSL-KDD和KDDCUUP 99数据集进行训练，并与静态入侵检测系统方案进行比较 [227]。

B. 基于深度学习开发网络威胁态势感知——使用DGA、URL、电子邮件和安全日志数据分析

随着CPS的不断发展，我们已识别出传统反恶意软件系统的局限性，这导致了对更高效、更具智能适应性的IDS的需求。通过分析与域名系统（DNS）、电子邮件、URL和社交媒体相关的网络威胁态势感知数据，这可能是可能的。从这些来源及时收集数据，并应用深度学习架构，将能够实现实时有效的恶意软件检测。在本节中，我们概述了可以成功采用的关键深度学习技术。

域名系统（DNS）：DNS是访问通过浏览器渲染的网页的主要互联网协议之一。DNS服务器分为两大类：递归服务器和非递归/迭代服务器。非递归DNS服务器作为权威起始点（SOA）工作，仅对其政府/本地域内的查询进行回复，而不管其他DNS服务器是否能满足查询。另一方面，递归DNS服务器通过将查询发送到其他服务器并将响应传递给用户，回复本地域以及所有类型域的查询。递归DNS服务器面临的最严重攻击包括根名服务器性能退化、DNS缓存中毒、分布式拒绝服务（DDoS）攻击和未经授权使用资源。由于DNS协议基本上不是为安全问题而创建的，并且存在漏洞，这些系统产生的大量事件数据可用于创建任何可能的网络威胁的态势感知。绕过DNS黑名单的一种方法包括攻击者通过使用变流技术嵌入具有固定域名和IP地址的恶意软件，以不断更改IP地址和域名。域名变流最常用的方法是使用域名生成算法（DGA）。

域名生成算法（DGA）：域名生成算法（DGA）利用种子值生成大量域名，而该种子值是攻击者已知的

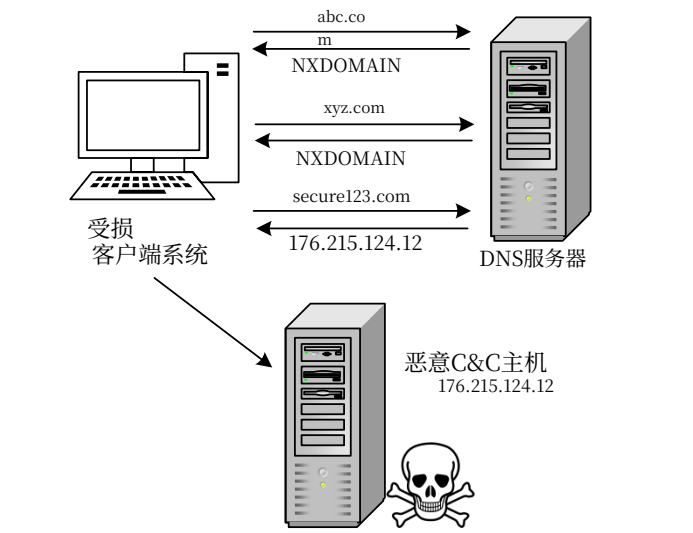


图4. 近期恶意软件采用的主动域名发现过程。

攻击者使用已知的种子值生成相同的域名集，并注册其中之一，部署命令与控制（C&C）服务器。DGA主要分为两种类型。一种是嵌入恶意软件二进制文件中的基于二进制的DGA，在恶意软件安装后触发。另一种是嵌入JavaScript中的基于脚本的DGA，在用户打开恶意网站时触发。域名流攻击的流程图如图4所示，其中受感染的系统尝试访问多个域名，试图联系C&C服务器，这已成为CPS领域日益增长的安全威胁。它尝试联系三个域名：abc.com、xyz.com和secure123.com。abc.com和xyz.com均未注册，受感染的系统从DNS服务器收到NXDOMAIN响应。第三个域名是一个活跃且已注册的域名。因此，DNS服务器使用该域名调用C&C服务器以发起攻击。僵尸网络是由被恶意软件感染的设备组成的网络。它可以被僵尸主通过命令与控制（C&C）通道 [117]远程控制。网络中的一个受损设备称为“僵尸”，僵尸主利用这些僵尸进行各种非法活动，如DDoS攻击 [117]、网络钓鱼、身份盗窃、恶意软件分发等。僵尸主使用C&C服务器向僵尸网络发布命令，根据这些命令，僵尸网络执行分配的任务并将结果反馈回来。根据C&C通信通道，僵尸网络分为互联网中继聊天（IRC）僵尸网络、超文本传输协议（HTTP）僵尸网络、点对点（P2P）僵尸网络和混合型僵尸网络，其中IRC僵尸网络采用集中式架构，P2P僵尸网络采用分布式架构。混合型僵尸网络结合了集中式和分布式架构，其检测通常比集中式和分布式架构更为复杂。

垃圾邮件和网络钓鱼电子邮件检测：垃圾邮件和网络钓鱼电子邮件攻击具有商业目的，以获取密码、信用卡号、银行账户详细信息和其他敏感信息，甚至感染电子邮件接收者的计算机上的恶意代码。这些可以大致分为两类

respectively [225]。A DNN based architecture for host and network level ID and various DL architectures such as LSTM, GRU, RNN, and IRNN were evaluated using NSL-KDD dataset [264].

DL studies without comparing with ML: A two-dimensional (2D) CNN based authentication system using mouse behaviour was proposed and trained on 2 publicly available datasets called Balabit and TWOS and it achieved an average AUC of 0.96 outperforming 1D-CNN and SVM models [206]. The effectiveness of several DL models was studied for DDoS attack detection where MLP, CNN, LSTM, CNN-LSTM models were trained using CICIDS2017 datasets. The proposed CNN-LSTM model outperformed other models by achieving 97.16% accuracy [226]. In another study, a DL based adaptive and scalable misuse IDS was trained using NSL-KDD and KDDCUUP 99 datasets and compared with a static IDS scheme [227].

B. DL in Developing Cyber Threat Situational Awareness using DGA, URL, Email and Security log Data analysis

With the growing developments in CPS, we have identified the limitations of traditional antimalware systems resulting in the requirement of a more efficient and intelligently adaptive IDS. This could be possible through the analysis of cyber threat situational awareness data related to domain name system (DNS), email, URL, and social media. Timely collection of data from such sources and the application of DL architectures would enable an effective malware detection in real-time. In this section, we provide an overview of key DL techniques that could be adopted successfully.

Domain Name System (DNS): DNS is one of the main Internet protocols for accessing web pages through a browser rendered via Internet. DNS servers have two broad categories: Recursive servers and Non-recursive/Iterative servers. Non-recursive DNS servers work as the Start of Authority (SOA), replying to the queries which are inside their government/local domain only, without considering other DNS servers regardless of whether or not they can cater to the query. On the other hand, Recursive DNS servers reply to the queries of not only local domain but also all types of domains by sending the queries to other servers and passing on the response to the user. Some of the most serious attacks on the Recursive DNS servers are root name server performance degradation, DNS cache poisoning, Distributed Denial of Service (DDoS) attacks, and unauthorized use of resources. As DNS protocol was not basically created with security issues in mind and has vulnerabilities, the large expanse of event data produced by these systems can be used to create situational awareness of any possible cyber threat. One of the methods to bypass DNS blacklisting includes an adversary embedding a malware with fixed domain name and IP address by using fluxing to constantly change the IP address and domain name. The most commonly used method for domain fluxing is achieved using a domain generation algorithm (DGA).

Domain generation algorithms (DGAs): The domain generation algorithm (DGA) facilitates the generation of large set of domain names using a seed value which is known

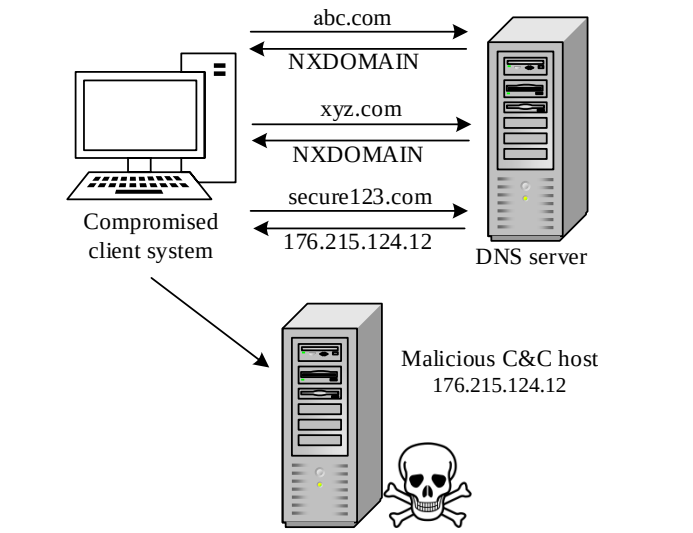


Fig. 4. Active domain name discovery process employed by recent malwares.

to the attacker. The attacker uses the known seed value to generate same set of domains and register one of the many generated domains and deploy the C&C server. The DGAs are broadly classified into 2 types. One is binary-based DGAs that are embedded in the malware binary and triggered after the installation of the malware. The second type is script-based DGAs that are embedded in the Javascript and triggered when the user opens a malicious website. The flow diagram of domain flux attack is illustrated in Fig. 4, where an infected system attempts to access many domains in an attempt to contact the C&C server, which is a growing threat in CPS. It contacts three domains, abc.com, xyz.com, and secure123.com. Both abc.com and xyz.com are not registered and an infected system receives an NXDOMAIN response from DNS server. The third domain is an active and registered domain. Hence the DNS server uses this domain to call C&C server to launch the attack. Botnets are networks formed by devices that are compromised by malware. It can be controlled remotely by the bot master using the command and control (C&C) channel [117]. A compromised device in a network called a bot and a bot master uses these bots to conduct various illegal activities such as DDoS [117], phishing, identity theft, malware distribution, etc. The C&C server is used by the bot master to issue commands to the botnets based on which the botnets perform their assigned tasks and send back the results. Based on the C&C communication channel, the botnet are grouped into Internet Relay Chat (IRC) botnet, Hyper Text Transfer Protocol (HTTP) botnet, Peer to Peer (P2P) botnet, and Hybrid botnet where an IRC botnet uses centralized architecture and P2P uses distributed architecture. A hybrid between centralized and distributed architectures for botnet detection is often more complex as compared to that of centralized and distributed architectures.

Spam and Phishing Email Detection: Spam and phishing email attacks have a commercial purpose to obtain passwords, credit card numbers, bank account details and other sensitive information or even to infect the email recipient’s computer with malicious code. These can be broadly classified under two

技术: i) 使用社会工程方案的网络钓鱼, ii) 基于恶意软件的网络钓鱼, 以欺骗受害者以获取个人和财务数据 [116]

统一资源定位符 (URL): URL是万维网 (World Wide Web) 上文档和其他资源的通用地址, 并在通过在线网络浏览器定位/访问文档和其他网络资源方面发挥着重要作用。随着互联网的增长, URL已成为攻击者托管恶意内容的最常用工具之一, 近年来需要采取保护措施。传统的黑名单和过滤虚假URL的方法虽然简单但不可扩展, 尽管存在一些使用模糊匹配技术的高级方法。其他方法尝试通过从URL字符串中提取特征来使用机器学习技术。

完全自动化的公共图灵测试来区分计算机和人类 (CAPTCHA): 这是一种用于识别用户是人类还是机器人的技术。用户必须通过CAPTCHA测试来证明自己是人类。常见的CAPTCHA类型基于文本、图像、视频、音频和拼图图像, 供用户识别, 是一种主动的安全技术, 用于保护DDoS攻击免受机器人侵害。它是普遍采用的第一道防线。

接下来, 我们提供一份关于使用DGA、URL、电子邮件和安全日志数据分析来分析态势感知的深度学习架构综述, 并总结在表VI中。

1) 电子邮件: 针对网络钓鱼电子邮件检测, 研究了CNN、RNN、LSTM和多层感知机等深度学习架构, 采用词嵌入和神经n-gram词袋, 结合语义和句法电子邮件相似度度量。实验结果表明, 使用LSTM方法的词嵌入在IWSPA-AP 2018¹语料库的两个不同反钓鱼共享任务中表现良好, 准确率分别为99.1%和97.1%。在另一项研究中, 提出了一种基于深度学习的反钓鱼系统, 其中分布式表示方法区分了钓鱼邮件和合法邮件, 在最佳情况下F1分数达到了99% [245]。

2) **DGA: 循环结构:** 基于LSTM的实时DGA域名检测模型在公开数据集上进行了评估, 实现了90%的检测率, 二元分类的AUC为0.9993 [235]。提出了一种基于深度学习的方法来解决时间序列解交错问题, 以生成合成数据集, 并使用增强隐马尔可夫模型 (AHMM) 和LSTM评估了该问题的各种推理策略。实验结果表明, LSTM方法优于基于AHMM的模型 [241]。研究了各种基于深度学习的方法, 用于可扩展的DGA检测, 使用DNS日志。RNN和LSTM等深度学习模型从DNS日志数据中提取了有用特征, 并优于最先进的机器学习模型 [248]。提出了一种基于深度学习的DGA检测引擎, 其中一维卷积神经网络模型从包含51个DGA恶意软件家族URL的大数据集中提取了重要特征, 实现了97%的准确率和0.7%的误报率 [250]。

卷积神经网络: 一个基于卷积神经网络的僵尸网络检测框架, 用于物联网和可穿戴设备, 其中网络流量数据被转换为图像格式, 并使用CTU-13数据集进行训练。它在最佳情况下达到了99.98%的准确率, 而支持向量机和逻辑回归分别达到了83.15%和78.56%的准确率。

racy的准确率达到99.98%, 而支持向量机和逻辑回归分别达到了83.15%和78.56%的准确率 [253]

混合方法: 一个基于深度学习的可扩展DGA检测框架, 在ISP级别运行, 使用DNS事件数据来检测DGA, 并提供态势感知[22]。另一种基于CNN-LSTM的新型DGA和恶意URL检测方法结合了自然语言处理技术, 并将性能与传统机器学习分类器 (使用二元特征表示和字符级卷积神经网络模型) 进行了比较。结果表明, 所提出的系统在恶意URL检测中达到了99%的准确率, 在DGA检测中达到了98.3%的准确率 [249]。

混合循环结构: 提出了一种使用 RNN 和 CNN 模型的 DGA 僵尸网络检测系统, 用于检测恶意域名, 并使用 DNS 流量数据提供关于感染主机和 C&C 域的信息。结果表明, CNN-LSTM 实现了 98.7% [251] 的最高准确率。使用各种数据集对 DGA 检测和分类进行了多种众所周知的基于字符的短文本分类建模, 以展示性能的相似性 [263]。

深度神经网络: 基于ADL的DGA分析方法在各种网络安全应用中进行了实验, 以泛化深度学习架构 [15]。在SDN环境中, 基于多层神经网络的僵尸网络检测系统使用HogZilla数据集进行特征选择和过滤训练, 使其更适用于SDN场景, 实现了96%的准确率 [254]。

双向循环结构: 提出了一种基于双向LSTM的僵尸网络检测引擎, 使用词嵌入将网络流量数据包转换为标记化的整数值, 对mirai、udp和dns攻击达到了良好的准确率 [252]。针对DGA检测和分类, 提出了多种深度学习架构, 并提供了名为AmritaDeepDGA和AmritaDGA的数据集用于DGA分析, 这些数据集已公开以供进一步研究 [262]。

3) **URL: 循环结构:** 一项研究基于LSTM的长可扩展模型, 通过从大量URL数据集中自动选择有用特征来检测网络钓鱼URL, 并与使用词法和统计分析提取的特征的随机森林进行比较。实验分析表明, LSTM和RF方法分别达到了98.7%和93.5%的准确率 [247]。

卷积神经网络: 提出了一种卷积神经网络架构, 并使用来自短字符串 (如URL、注册表键、文件路径、命名互斥体和命名管道) [236]的特征进行训练。一个基于卷积神经网络的系统使用代理日志的事件去噪来提取序列以检测URL重定向 [239]。

混合方法: 一种最近开发的卷积门控循环单元模型, 使用从212个URL中提取的特征进行训练, 以检测恶意URL, 准确率达到99.6% [257]。

混合循环结构: 通过采用Keras嵌入, 对用于恶意URL检测的LSTM、RNN、GRU和CNN等多种深度学习架构进行了性能评估, 并与RF、DT、MT、AB和NB等多种CMLA进行了比较 [258]。

techniques: i) deceptive phishing that uses social engineering schemes and ii) malware-based phishing to trick the victim for capturing personal and financial data [116]。

Uniform resource locator (URL): URL is an universal address of documents and other resources on the World Wide Web and plays the important role of locating/accessing documents and other web resources via a web browser online. With Internet growth, the URL has become one of the most commonly used tool to host malicious contents by an adversary and a protection is warranted in recent years. The traditional methods of blacklisting and filtering fake URLs are simple but not scalable, even though some advanced methods using fuzzy matching techniques exist. Other approaches try to use ML techniques by extracting features from URL strings.

Completely Automated Public Turing Test to Tell Computers and Humans Apart (CAPTCHA): It is a technique used to identify whether the user is a human or a bot. The user has to pass the CAPTCHA test to prove that the user is a human. The common types of CAPTCHA are based on text, image, video, audio, and puzzle image for users to recognize and is an active security technique to protect DDoS attacks from bots. It is a commonly adopted first line of defense.

Next, we provide a survey on DL architectures for analyzing situational awareness using DGA, URL, Email and security log data analysis and is summared in Table VI.

1) *Email:* DL architectures such as CNN, RNN, LSTM, and MLP were studied for phishing email detection using word embedding and neural bag of n-grams with semantic and syntactic email similarity measures. The experiment results showed that word embedding with LSTM approach performed well with accuracy of 99.1 and 97.1 in two different anti-phishing shared tasks from IWSPA-AP 2018¹ corpus [244]. In another study, a DL based anti-phishing system was proposed where a distributed representation method had differentiated phishing and legitimate emails with F1 score of 99% achieved for the best case [245].

2) **DGA: Recurrent Structures:** An LSTM based real-time DGA domain detection model was evaluated on open datasets and achieved 90% detection rate with AUC of 0.9993 for binary classification [235]. A DL based approach to address the problem of time series deinterleaving was proposed to generate synthetic dataset and evaluated various inference strategies for the problem using Augmented Hidden Markov Model (AHMM) and LSTM. The experimental results showed that the LSTM method outperformed the model based on AHMM [241]. The effectiveness of various DL based approaches were studied for scalable DGA detection using DNS logs. DL models such as RNN and LSTM had extracted useful features from DNS log data and outperformed state-of-the-art ML models [248]. A DL based DGA detection engine was proposed where a 1D-CNN model extracted important features from a large dataset containing URLs from 51 DGA malware families, achieving 97% accuracy and 0.7% FAR [250].

Convolutional Neural Network: A CNN based botnet detection framework for IoT and wearable devices where the network traffic data was converted into image format was trained using CTU-13 Dataset. It achieved the best case accu-

racy of 99.98% while SVM and logistic regression achieved accuracies of 83.15% and 78.56% respectively [253].

Mixed Approaches: A DL based scalable DGA detection framework which works at ISP level used event data from the DNS to detect DGA and provided situational awareness [22]. Another novel CNN-LSTM based approach for DGA and malicious URL detection incorporated NLP technique and the performance was compared with traditional ML classifiers using bi-gram feature representation and character-level CNN model. The results showed that the proposed system achieved an accuracy of 99% for malicious URL detection and 98.3% for DGA detection [249].

Hybrid Recurrent Structures: A DGA botnet detection system using RNN and CNN models was proposed for detecting the malicious domains using DNS traffic data to provide information about the infected host and C&C domain. The results showed that CNN-LSTM achieved the highest accuracy of 98.7% [251]. Various well-known character based short-text classification were modeled for DGA detection and classification using various datasets to show the similarity of performance [263].

Deep Neural Network: A DL based method for DGA analysis was experimented on various cyber security applications to generalize the DL architecture [15]. A multi-layered neural network based botnet detection system in SDN environment was trained with feature selection and filtering using HogZilla dataset to make it realistic for SDN scenario, achieving 96% accuracy [254].

Bi-directional Recurrent Structures: A bi-directional LSTM based botnet detection engine using word embedding for the conversion of network traffic packets into tokenized integer values was proposed, achieving good accuracy for mirai, udp, and dns attacks [252]. Various DL architectures were proposed for DGA detection and classification, with datasets termed as AmritaDeepDGA, and AmritaDGA for DGA analysis that have been made publicly available for further research [262].

3) **URL: Recurrent Structures:** A study on LSTM based scalable model for detecting phishing URL by selecting useful features automatically from a large dataset of URLs compared random forest using features extracted from lexical and statistical analysis. The experimental analysis showed that the LSTM and RF approach achieved an accuracy of 98.7% and 93.5% respectively [247].

Convolutional Neural Network: A CNN architecture was proposed and trained using features from a short character strings such as URLs, registry keys, file paths, named mutexes, and named pipes [236]. A CNN based system used event denoising from proxy logs to extract the sequences to detect URL redirection [239].

Mixed Approach: A recently developed convolutional GRU model used feature extraction from 212 URLs for training to detect malicious URLs with an accuracy of 99.6% [257].

Hybrid Recurrent Structures: A performance evaluation of various DL architectures such as LSTM, RNN, GRU, and CNN for malicious URL detection was conducted by employing Keras embedding and compared with various CMLAs such as RF, DT, MT, AB, and NB [258].

表 VI 深度学习架构用于态势感知分析所使用数据集的研究综述

参考文献	架构	Task	数据集	Text 表示	对比 CML
[235]	RS	域名	公开可用的资源 (PASs)	Keras嵌入	Yes
[236]	CNN	URL	私有	独热编码	Yes
[237]	AlexNet, VGGNet, GoogleNet	CAPTCHA	私有	-	No
[238]	DNN & 卷积神经网络	CAPTCHA	公开可用的资源	-	No
[239]	CNN	URL	公开可用的资源	独热编码	Yes
[240]	CNN	CAPTCHA	公开可用的资源	-	Yes
[241]	RS	域名	私有	独热编码	No
[250]	卷积神经网络, 反向传播算法	域名	公开可用的资源	Keras嵌入	No
[258]	反向传播算法, 卷积神经网络, CNN-RS	URL	公开可用的资源	Keras嵌入	Yes
[242]	VAE	YouTube视频地址	私有	Sent2vec嵌入	Yes
[243]	DNN	网页	私有	词袋模型	Yes
[244]	卷积神经网络 & 反向传播算法	电子邮件	IWSPA-2018	Word2vec & 神经 n-gram词袋	No
[245]	FastText	电子邮件	IWSPA-2018	FastText	No
[246]	DBN	URL	公开可用的资源	手动特征工程	Yes
[247]	RS	URL	公开可用的资源	独热编码	Yes
[248], [249], [251]	反向传播算法, 卷积神经网络, CNN-RS, 深度神经网络,	域名	公开可用的资源	Keras嵌入	Yes
[22]	卷积神经网络, 反向传播算法	域名	PASs & 实时数据集	Keras嵌入	Yes
[15]	DNN	域名 & URL	公开可用的资源	n-gram & feature 工程	Yes
[252]	双向长短期记忆网络	网络流	私有	Keras嵌入	No
[253]	CNN	网络流	CTU-13数据集 实时数据集	交通到图像表示	Yes
[254]	DNN	网络流	HogZilla数据集, CTU-13数据集, ISCX-IDS-2012数据集	-	No
[255]	DNN	URL	私有	词袋模型	No
[256]	DNN	URL	公开可用的资源	手动特征工程	No
[257]	卷积神经网络 & 反向传播算法	URL	私有	Keras嵌入	Yes
[262]	卷积神经网络, 反向传播算法, CNN-RS	域名	AmritaDGA	Keras嵌入	No
[263]	最终目标, 纽约大学, MIT, 卡内基梅隆大学, Invincea , DNN, CNN-RS	域名	PASs & 实时数据集	Keras嵌入	Yes

深度神经网络：基于深度学习的恶意URL检测框架从静态HTML文件中提取特征，并利用空间信息实现了97.5%的准确率和低假阳性率 [243]。一个深度神经网络框架使用真实URL数据集进行训练，检测准确率达到94.18%[255]。在另一项研究中，提出了基于人工神经网络和深度神经网络的钓鱼URL检测系统，使用73575个URL进行训练，实验分析显示人工神经网络的准确率为92%，深度神经网络的准确率为96%，均优于机器学习分类器[256]。

自编码器和DBN：针对YouTube视频中的点击诱饵问题，提出了一种基于变分自编码器的方法 [242]。一种基于深度学习的恶意URL检测系统采用贪婪的多层DBN自动提取有用特征，并使用27,700个URL进行训练以实现极低的假阳性率 [246]。

4) **CAPTCHA：**一种基于文本的CAPTCHA技术，采用非模态补全方法，与基于深度学习的CAPTCHA解器进行了比较。由于模拟非模态补全的难度较大，基于深度学习的解器需要更多时间来解题 [237]。此外，提出了一种基于卷积神经网络的CAPTCHA解题技术，能够破解11种CAPTCHA方案，准确率超过50%，并且性能优于多层感知机、单层感知机以及基于卷积神经网络的解器 [238]。进一步地，提出了一种基于深度卷积神经网络的CAPTCHA破解器，用于解决基于字母的CAPTCHA，并且对于单字母分类效果良好

表 VII 关键深度学习模型网络威胁检测

Ref.	架构	数据集	对比 CML
[228]	自编码器	私有	No
[229]	CNN	私有	No
[230]	卷积神经网络, & 反向传播算法	TIM 大数据挑战 2015	Yes
[231]	RS	GEANT	No
[232]	卷积神经网络, & 反向传播算法	摩尔	Yes
[233]	DNN	私有	Yes
[234]	RS	私有	No
[259]	RS	DARPA 1999, AMP, MAWI, NIMS	Yes

[240]。

C. 深度学习在网络流量分析中的应用

区分网络中每条信息流是一个随着BD不断增长的重要问题。基于端口（例如HTTP端口80和SSL端口443）、特征（例如基于字符串或十六进制格式的有效载荷）和事实性亮点（例如传输间隔时间、数据包时间和重哈希活动）的模式是用于网络流量分析的标准方法。本节简要回顾了用于基于网络流量分析的识别的深度学习架构，并总结在表VII中。

TABLE VI
A SUMMARY OF STUDIES ON DL ARCHITECTURES DATASETS USED FOR SITUATIONAL AWARENESS ANALYSIS

References	Architecture	Task	Data set	Text representation	Compared CML
[235]	RS	Domain name	Publically available sources (PASs)	Keras embedding	Yes
[236]	CNN	URL	Private	One-hot	Yes
[237]	AlexNet, VGGNet, GoogleNet	CAPTCHA	Private	-	No
[238]	DNN & CNN	CAPTCHA	Publically available sources	-	No
[239]	CNN	URL	Publically available sources	One-hot	Yes
[240]	CNN	CAPTCHA	Publically available sources	-	Yes
[241]	RS	Domain name	Private	One-hot	No
[250]	CNN, RS	Domain name	Publically available sources	Keras embedding	No
[258]	RS, CNN, CNN-RS	URL	Publically available sources	Keras embedding	Yes
[242]	VAE	YouTube video address	Private	Sent2vec embedding	Yes
[243]	DNN	Web page	Private	bag-of-words	Yes
[244]	CNN & RS	Email	IWSPA-2018	Word2vec & Neural Bag-of-ngrams	No
[245]	FastText	Email	IWSPA-2018	FastText	No
[246]	DBN	URL	Publically available sources	Manual feature engineering	Yes
[247]	RS	URL	Publically available sources	One-hot	Yes
[248], [249], [251]	RS, CNN, CNN-RS, DNN,	Domain name	Publically available sources	Keras embedding	Yes
[22]	CNN, RS	Domain name	PASs & real-time data set	Keras embedding	Yes
[15]	DNN	Domain name & URL	Publically available sources	n-gram & feature engineering	Yes
[252]	BLSTM	Network flow	Private	Keras embedding	No
[253]	CNN	Network flow	CTU-13 Dataset real-time data set	Traffic to image representation	Yes
[254]	DNN	Network flow	HogZilla dataset, CTU-13 dataset, ISCX-IDS-2012 dataset	-	No
[255]	DNN	URL	Private	Bag-of-words	No
[256]	DNN	URL	Publically available sources	Manual feature engineering	No
[257]	CNN & RS	URL	Private	Keras embedding	Yes
[262]	CNN, RS, CNN-RS	Domain name	AmritaDGA	Keras embedding	No
[263]	Endgame, NYU, MIT, CMU, Invincea, DNN, CNN-RS	Domain name	PASs & real-time dataset	Keras embedding	Yes

Deep Neural Network: A DL based malicious URL detection framework extracted features from static HTML files and used spatial information to yield 97.5% accuracy and low FPR [243]. A DNN framework was trained using real-life datasets of URLs and achieved a detection accuracy of 94.18% [255]. In another study, ANN and DNN based phishing URL detection system was proposed and trained using 73575 URLs with experimental analysis showing 92% accuracy for ANN and 96% accuracy for DNN outperforming ML classifiers [256].

Autoencoder and DBN: A VAE based method was proposed for clickbait problem in Youtube videos [242]. A DL based malicious URL detection system employed greedy multi-layered DBN to extract useful features automatically and was trained using 27,700 URLs to achieve very low FPR [246].

4) **CAPTCHA:** A text-based CAPTCHA technique with amodal completion was compared with a DL based CAPTCHA solver with DL based solver taking more time to solve due to difficulty in emulating the amodal completion [237]. A CNN based CAPTCHA solving technique was proposed and was capable of breaking 11 CAPTCHA schemes with more than 50% accuracy and performed better than MLP, SLP and CNN based solvers [238]. Further, a deep CNN based CAPTCHA breaker was proposed to solve letter-based CAPTCHAs and worked well for single-letter classification

TABLE VII
KEY DL MODELS NETWORK THREAT DETECTION.

Ref.	Architecture	Dataset	Compared CML
[228]	Autoencoder	Private	No
[229]	CNN	Private	No
[230]	CNN, & RS	TIM Big Data Challenge 2015	Yes
[231]	RS	GEANT	No
[232]	CNN, & RS	Moore	Yes
[233]	DNN	Private	Yes
[234]	RS	Private	No
[259]	RS	DARPA 1999, AMP, MAWI, NIMS	Yes

[240].

C. DL in Network Traffic Analysis

Differentiating each stream of information in a network is an important issue with the ever-growing BD. Patterns based on port (e.g. HTTP port 80 and SSL port 443), signature (e.g. strings or hex based payload) and factual highlight (e.g., transmission interim time, parcel time, and rehashing activity) are standard approaches used for network traffic analysis. This section provides a short review of DL architectures for ID using network traffic analysis and is summarized in Table VII.

深度学习研究与机器学习对比: 提出了一种基于深度学习的多任务架构用于预测移动互联网流量 [230]。CNN-RNN架构提取了地理和时间流量特征, 而采用基于深度学习的SDN架构对网络应用进行分类 [232]。softmax回归层和SAE的混合使用SDN控制器数据进行了训练, 与SVM分类器相比实现了高准确率。另一项研究比较了不同的深度学习技术用于对移动网络加密流量进行分类[233]。循环结构(如RNN和LSTM)使用从实时人类活动中收集的三个不同数据集进行评估, 用于识别SSH流量和非SSH流量 [259]。深度学习架构使用4种不同类型的公开可用网络流量数据集, 其性能优于其他经典分类器, 如RF、AB、DT、KNN、NB和SVM。然而, 通过将整个有效载荷和输入传递给深度学习架构, 可以避免特征工程。

与机器学习不进行比较的深度学习研究: 为协议分类、未知协议识别和异常协议检测等应用提出了一个用于网络流量分析的深度学习框架, 使用实时数据[228]。此外, 提出了一个结合循环神经网络和卷积神经网络的深度学习模型, 用于对物联网网络流量进行分类, 以优于机器学习模型 [229]。还采用了各种循环神经网络架构, 如长短期记忆网络、门控循环单元和改进循环神经网络, 使用来自GEANT骨干网络的实时数据对网络流量进行预测, 其中长短期记忆网络优于其他循环神经网络架构 [231]。此外, 还采用了字节片段神经网络和基于循环神经网络的编码器来对网络流量进行分类 [234]。

D. 深度学习在Windows恶意软件分析中的应用

随着基于Windows的应用的日益普及, 它们成为各种类型的恶意软件(包括病毒、木马、蠕虫、后门、Rootkit、间谍软件、勒索软件和恐慌软件)进行网络犯罪、间谍活动和非法活动的平台。为了克服这一问题, 深度学习架构最近被用于Windows恶意软件分析。在勒索软件中, 受害者的PC信息被窃取, 通常在恢复信息被解码并归还给受害者之前会提出加密的财务要求 [269]。表VIII总结了深度学习在Windows恶意软件分析中的应用。

1) 深度神经网络 (DNN): 提出了一种用于分类二进制恶意软件的多任务深度学习框架, 使用450万文件进行训练, 200万文件进行测试深度学习模型 [126]。此外, 使用dropout对深度和浅层神经网络框架进行错误率显著降低, 并且通过使用修正线性激活函数减少了训练模型的轮数。使用BD训练了一个深度学习模型, 并使用复杂数据集进行评估, 结果准确率为97%, ROC为0.99 [138]进一步地, 在EMBER基准数据集上 [133]将用于恶意软件分类的DNN架构与其他浅层模型(如LR、NB、KNN、DT、AB和SVM)进行了比较。然而, 这种方法的主要限制是所提出的DNN架构依赖于特征工程。

2) 卷积神经网络 (CNN): 一种卷积前馈神经网络, 采用分层特征提取机制, 利用 PE 文件的元数据 [129]检测和分类恶意软件。将表示为图像的恶意程序输入 CNN 进行分类, 并使用不同的内核和数据大小对深度学习算法进行训练, 最终在恶意软件检测方面实现了 0.9973 的高 AUC [131]。提出了一种使用名为 Malimg 和 Microsoft 恶意软件的两个基准数据集 [135]对恶意软件进行分类的深度学习方法。灰度图像特征被输入 CNN, 分别使用 Malimg 和 Microsoft 数据集实现了 98.52% 和 99.97% 的准确率 [135]。此外, 使用 Microsoft 数据集分析了 GoogleNet 和 ResNet 模型在恶意软件检测方面的性能 [136]。

3) 循环结构 (RS): 一个结合了 Echostate 网络和 RNN 的混合模型使用无监督数据进行训练, 投影阶段采用了 Half-Frame 模型和最大池化, 最终得到 98.3% 的 TPR 和 0.1% 的 FPR [122]。此外, LSTM 和 GRU 模型使用半监督学习, 结合注意力机制和时序最大池化来检测勒索软件 [130]。

4) 自编码器 (AE) 和深度信念网络 (DBN): 恶意软件特征生成和分类是使用 DBN 进行的, 而深度堆叠的 DAEs 被用于恶意软件行为的紧凑表示, 该模型达到了 98.6% 的准确率 [123]。一个深度学习框架采用了 SAE 模型, 通过无监督特征学习 Windows API 调用, 并借助监督参数进行微调, 以高效检测恶意软件 [124]。在另一项研究中, 恶意软件被表示为指令序列, 并输入到 DBN 中进行恶意软件分类 [127]。

5) 混合深度学习架构: 提出了一种基于进程行为的两阶段深度神经网络模型用于恶意软件检测以检查终端是否感染了恶意软件 [125]。该模型使用长短期记忆网络从表示进程行为的API调用序列中构建特征, 并使用循环神经网络提取图像特征, 这些图像被输入到卷积神经网络中进行分类。在另一项研究中, 提出了一种混合神经网络, 具有两层卷积层用于提取层次化特征, 结合了全序列建模和n-gram特征的卷积, 用于恶意软件分类, 并优于许多机器学习方法, 如支持向量机和隐马尔可夫模型 [128]。MalNet, 一个使用卷积神经网络和长短期记忆网络的自动特征学习模型, 用于从 40000 个样本的原始数据中检测恶意软件, 这些样本被转换为灰度图像, 并在恶意软件分类中达到了99.88%的准确率 [132]。最近提出了一种异构深度学习架构, 由自编码器、多层玻尔兹曼机和关联记忆层组成, 用于Windows API调用提取、预训练和微调以检测恶意软件 [137]。

6) 混合循环结构 (HRS): 一个结合了卷积神经网络 (CNN) 和双向长短期记忆网络 (BLSTM) 的深度学习框架, 采用数据驱动方法来识别用于分类九种不同类型恶意软件的复杂特征 [134]。提出了一种使用 API 调用特征构建的神经序列分类恶意软件模型, 其假阳性率 (FPR) 低于其他神经分类模型 [139]。

DL studies compared with ML: A DL based multitask architecture for forecasting mobile Internet traffic was proposed [230]. A CNN-RNN architecture extracted geographical and temporal traffic features, while a DL based SDN architecture was adopted to classify network applications [232]. The hybrid of softmax regression layer and SAE was trained using data from SDN controller achieving high accuracy as compared to SVM classifier. Another study compared different DL techniques to classify mobile network encrypted traffic [233]. Recurrent structures such as RNN and LSTM were evaluated using three different datasets collected from real-time human activities for identifying SSH traffic and Non-SSH traffic [259]. The DL architectures outperformed other classical classifiers such as RF, AB, DT, KNN, NB, and SVM using 4 different types of publicly available network traffic datasets. However, feature engineering could be avoided by passing the entire payload and input to the DL architectures.

DL studies without comparing with ML: A DL framework for network traffic analysis was proposed for applications such as protocol classification, unknown protocol identification, and anomalous protocol detection using real-time data [228]. Further, a DL based model combining RNN and CNN for classifying IoT network traffic was proposed to outperform ML models [229]. Various RNN architectures such as LSTM, GRU, and IRNN were employed for network traffic prediction using real-time data from GEANT backbone networks with LSTM outperforming other RNN architectures [231]. Further, byte segment neural network (BSNN) and RNN based encoders were adopted to classify network traffic [234].

D. DL in Windows Malware Analysis

With the growing prevalence of Windows based applications, they become platforms for cybercrime, espionages and other illegal activities by various types of malware including viruses, trojans, worms, backdoors, rootkits, spyware, ransomware and panic software. To overcome this, DL architectures are being recently adopted for Windows malware analysis. In ransomware, the information on a casualty's PC is bolted, and normally an encrypted financial request is made before the recovered information is decoded and gets back to the casualty [269]. Table VIII summarises on the application of DL in windows malware analysis.

1) Deep Neural Network (DNN): A multi-task DL framework for classifying binary malware was proposed with 4.5 million files for training and 2 million files for testing the DL model [126]. Furthermore, the error rate was reduced significantly using dropout for both deep and shallow neural frameworks, and the number of epochs to train the model were also reduced by using rectified linear activation function. A DL model was trained using BD and was evaluated using a complex dataset resulting in an accuracy of 97% and ROC of 0.99 [138] Further, a DNN architecture for malware classification was compared to other shallow models such as LR, NB, KNN, DT, AB, and SVM on EMBER benchmark data set [133]. However, the main limitation of this method is that the proposed DNN architecture relies on feature engineering.

2) Convolutional Neural Network (CNN): A convolutional FFN employed hierarchical feature extraction mechanism to detect and classify malwares using metadata of PE files [129]. Malware programs represented as an image were fed into CNN for classification and training of the DL algorithm was performed with different kernel and data size resulting in a high AUC of 0.9973 for malware detection [131]. A DL approach was proposed for classifying malware using two benchmark datasets called Malimg and Microsoft malware. Grayscale image features were fed into a CNN and accuracies of 98.52% and 99.97% were achieved with Malimg and Microsoft datasets respectively [135]. Further, GoogleNet and ResNet models were analysed for malware detection using a Microsoft dataset [136].

3) Recurrent Structures (RS): A hybrid model of Echo state networks and RNN was trained using unsupervised data and the projection stage employed Half-Frame models and Max Pooling resulting in 98.3% TPR and 0.1% FPR [122]. Further, LSTM and GRU models used semi-supervised learning with attention mechanism and temporal maximum pooling to detect ransomware [130].

4) Autoencoder (AE) and Deep belief network (DBN): Malware signature generation and classification were performed using DBN and deep stack of DAEs were utilized for compact representation of the malware behavior and the model achieved an accuracy of 98.6% [123]. A DL framework employed SAE models with unsupervised feature learning of 4)Windows API calls and fine tuning with the help of supervised parameters to detect malwares efficiently [124]. In another study, malwares were represented as opcode sequences and fed to DBNs for malware classification [127].

5) Mixed DL architectures: A two-stage DNN model was proposed for malware detection based on process behavior to check if a terminal is infected or not [125]. It used LSTM to construct the features from API call sequences that represented process behavior and an RNN for extraction of features as images that were fed into a CNN for classification. In another study, a hybrid NN with two convolution layers for extracting hierarchical features having both full sequential modeling and convolution of n-gram features was proposed for classification of malware and outperformed many ML methods such as SVMs and hidden markov models [128]. MalNet, an automatic feature learning model using CNN and LSTM was proposed for detecting malware from raw data of 40000 samples that were converted to grayscale images and achieved 99.88% accuracy for malware classification [132]. Heterogeneous DL architecture was recently proposed consisting of AE, multilayer boltzmann and layers of associate memory for windows API call extraction, pretraining and finetuning to detect malware [137].

6) Hybrid Recurrent Structures (HRS): A DL framework with CNN and BLSTM network utilized data driven approach to identify complex features for classifying nine different types of malware [134]. A neural sequential classification malware model was proposed using API call features to achieve FPR lower than other neural classification models [139].

表 VIII 恶意软件分析中深度学习应用研究的综述。

参考文献	架构	数据集	对比CML
[122]	RS	私有	Yes
[123]	DBN	私有	No
[124]	自编码器	私有	Yes
[125]	卷积神经网络, 反向传播算法	私有	
[126]	DNN	私有	No
[127]	DBN	私有	Yes
[128]	卷积神经网络, 反向传播算法	VirusShare, Malttrieve, 私有	Yes
[129]	CNN	VirusShare, Malttrieve, 私有	Yes
[130]	RS	私有	No
[131]	CNN	VirusShare	No
[135]	CNN	BIG 2015	Yes
[132]	卷积神经网络, 反向传播算法	BIG 2015	Yes
[133]	DNN	Ember	Yes
[134]	卷积神经网络, CNN-RS	BIG 2015	No
[136]	CNN	BIG 2015 与私有收集样本	No
[137]	DBN、自编码器、堆叠自编码器	私有	Yes
[138]	DNN	MalwareTrainingSets	Yes
[139]	长短期记忆网络、CNN-RS	私有	No
[135]	CNN	Maling	Yes

E. 深度学习在Android恶意软件检测中的应用

Android 操作系统 (OS) 是一个开源系统，具有多种重要的金融和个人应用，容易受到恶意攻击。黑客利用恶意软件窃取私有敏感数据，或删除/修改现有数据，以获取经济利益。Android 应用程序托管在多个第三方商店中，这些商店允许用户无意中重新打包包含恶意代码的Android 应用程序。Android OS 在安装阶段自动分配一个唯一的 Linux 用户 ID，以知道每个应用程序都运行其自己的虚拟机实例。这促进了沙盒的创建，沙盒将应用程序彼此隔离。它通过 Android 权限提供授权机制。通过已root或未root的设备收集 Android 功能，这些设备作为输入传递给机器学习 (ML) 模型，以学习区分良性应用程序和恶意应用程序的特征。然而，来自托管应用程序的授权市场商店（如 Google Play）的伪装恶意应用程序以及权限系统可能会欺骗移动用户以获取安装权限。用户在安装应用程序过程中可能会盲目地授予权限，而最终用户对这种影响知之甚少。因此，基于 Android 的权限系统进行风险评估非常重要。

随着技术的演进，针对智能设备操作系统（如Android操作系统）的攻击将持续增长，因为基于签名的检测方法和基于启发式的方法在零日恶意软件检测中完全失效。针对不断增长的Android恶意软件的检测技术正越来越多地被探索。由动态分析、机器学习和深度学习算法组成的自学习系统可以为Android恶意软件检测提供新的感知能力，这些能力可以被扩展以实现规模化应用。此外，这些方法还具备检测现有恶意软件变种或全新恶意软件的能力。研究人员遵循两种基本的技术分类方法，从Android操作系统中收集特征，这与个人计算机环境非常相似，即静态分析和动态分析[v1}。静态分析通过解包或反汇编应用程序来收集一组特征，而无需运行时执行；动态分析则检查应用程序的运行时执行行为，例如系统调用、网络连接、内存利用率、功耗和用户交互。混合分析是一个两步过程，首先执行静态分析，然后执行动态分析，这导致计算成本较低、资源利用率低、轻量级且耗时少。混合分析方法越来越多地被智能手机安全厂商采用，因为它提供了更高的检测率。关于深度学习在Android恶意软件分析中应用的综述总结报告在表IX中。

静态分析通过解包或反汇编应用程序来收集一组特征，而无需运行时执行；动态分析则检查应用程序的运行时执行行为，例如系统调用、网络连接、内存利用率、功耗和用户交互。混合分析是一个两步过程，首先执行静态分析，然后执行动态分析，这导致计算成本较低、资源利用率低、轻量级且耗时少。混合分析方法越来越多地被智能手机安全厂商采用，因为它提供了更高的检测率。关于深度学习在Android恶意软件分析中应用的综述总结报告在表IX中。

1) 动态分析： 一个用于Android恶意软件检测的深度学习框架通过系统调用图进行动态分析得到训练，在用真实世界恶意软件样本进行测试时，其性能优于其他传统检测器 [164]。一种基于卷积神经网络的新Android恶意软件检测方法将API调用执行路径作为图使用，达到了98.86%的准确率[178]。

2) 静态和动态分析： 一个基于深度信念网络的Android恶意软件检测框架采用静态和动态分析来获取200多个特征，在准确率为96% [161]的情况下，其性能优于NB、DT、SVM、MLP和LR等其他模型。一种新的混合深度神经网络Android恶意软件分类器采用分层多核学习算法，在组合特征集上提升了检测准确率至94.7% [162]。

总体而言，与动态分析相比，静态分析在准确率上更高，而动态分析对于在线检测恶意软件更为重要。在一项比较RNN和LSTM模型检测Android恶意软件的研究中，LSTM的表现优于RNN，在使用动态分析和静态分析时分别达到了93.9%和97.5%的检测准确率 [174]。研究人员提出了一种基于CNN的Android恶意软件识别框架，该框架使用API调用序列和保护级别作为特征 [177]。

TABLE VIII
A SUMMARY OF STUDIES ON DL APPLICATIONS FOR MALWARE ANALYSIS.

Reference	Architecture	Dataset	Compared CML
[122]	RS	Private	Yes
[123]	DBN	Private	No
[124]	Autoencoder	Private	Yes
[125]	CNN, RS	Private	
[126]	DNN	Private	No
[127]	DBN	Private	Yes
[128]	CNN, RS	VirusShare, Malttrieve, Private	Yes
[129]	CNN	VirusShare, Malttrieve, Private	Yes
[130]	RS	Private	No
[131]	CNN	VirusShare	No
[135]	CNN	BIG 2015	Yes
[132]	CNN, RS	BIG 2015	Yes
[133]	DNN	Ember	Yes
[134]	CNN, CNN-RS	BIG 2015	No
[136]	CNN	BIG 2015 & privately collected samples	No
[137]	DBN, Autoencoder, Stacked Autoencoder	Private	Yes
[138]	DNN	MalwareTrainingSets	Yes
[139]	LSTM, CNN-RS	Private	No
[135]	CNN	Maling	Yes

E. DL in Android Malware Detection

Android operating system (OS), an open source with several important financial and personal applications is prone to malicious attacks. Hackers make use of malware to steal the private sensitive data or delete/ alter the existing data with the aim to gain financial benefits. Android applications are hosted in various third-party stores which allow the user to inadvertently repackage Android applications along with malicious code. Android OS automatically assigns a unique Linux user ID during the installation phase to know that each app runs its own instance of virtual machine. This facilitates the creation of a sandbox which isolates the apps from each other. It provides authorization mechanism using Android permissions. Android features are collected via either rooted or unrooted devices that are passed as an input to ML models to learn the characteristics for distinguishing between the benign and malicious apps. However, malicious Apps in disguise from authorized market store that hosts the apps such as Google play and the permission systems could trick the mobile user to receive permissions for installation. Naive users may follow the blind approach in granting permissions during the installation procedure of apps and the impact is less known to the end user. Hence, it is important for Android based permission systems to undergo risk assessment.

Attacks on smart device OS such as Android OS will continue to grow as the technology evolves since signature based and heuristics-based methods are completely failing in zero-day malware detection. Detection techniques of the growing Android malware are increasingly being explored. Self-learning systems composed of DM, ML and DL algorithms could provide new sensing capabilities for Android malware detection which could be enhanced to work to scale. Moreover these approaches have the capability to detect the variants of already existing malware or entirely new malware itself. There are two fundamental taxonomies of techniques followed by researchers for collecting features from Android OS much similar to personal computer environments, namely static analysis

and dynamic analysis [270]. While static analysis collects a set of features from apps by unpacking or disassembling them without the runtime execution, dynamic analysis examines the runtime execution behavior of apps such as system calls, network connections, memory utilization, power consumption, and user interactions. The hybrid analysis is a two-step process where initially static analysis is performed before the dynamic one which results in less computational cost, low resource utilization, lightweight, and less time-consuming in nature. Hybrid analysis approach is increasingly being used by antivirus providers for the smartphones as it provides higher detection rates. A summary of review on DL applications in Android malware analysis is reported in Table IX.

1) *Dynamic Analysis:* A DL framework for Android malware detection was trained using dynamic analysis of system call graphs and outperformed other traditional detectors when tested with real-world malware samples [164]. A new CNN based android malware detection approach used API call execution paths as graphs and achieved an accuracy of 98.86% [178].

2) *Static and Dynamic Analysis:* A DBN based malware detection framework for Android employed static and dynamic analysis to obtain more than 200 features and outperformed other models such as NB, DT, SVM, MLP, and LR with an accuracy of 96% [161]. A new hybrid DNN Android malware classifier employed a hierarchical multiple kernel learning on combined feature set that improved its detection accuracy to 94.7% [162].

Overall, static analysis resulted in a higher accuracy when compared to dynamic analysis which is more required for online detection of malware. In a comparative study between RNN and LSTM models for detection of Android malware, the LSTM performed better than RNN, and achieved detection accuracy of 93.9% and 97.5% using dynamic and static analysis respectively [174]. A CNN based Android malware identification framework using API call sequence and protection levels as features was proposed [177].

表IX Android恶意软件检测中深度学习应用研究的综述。

参考文献	分析	特征	架构	数据集	与CML
[161]	动态、静态分析	敏感 API 调用	DBN	公开可用的资源	Yes
[162]	静态、动态分析	10 个静态、动态特征集	自编码器	公开可用的资源	No
[163]	静态分析	API调用块	DBN	公开可用的资源	Yes
[164]	动态分析	系统调用	SAE	公开可用的资源	Yes
[165]	静态分析	超过1,000个特征	DNN	Drebin, 收集自PASs	Yes
[166]	指令序列	学习检测指令序列 表明恶意软件	CNN	公开可用的资源	No
[167]	静态分析	API调用序列	CNN	公开可用的资源	Yes
[168]	传输类。dex 转换为RGB彩色图像	从传输的图像中提取特征	CNN	由研究人员收集	No
[169]	静态分析	字节序列	CNN	公开可用的资源	Yes
[170]	静态分析	危险的API调用, 高风险权限	DBN	Drebin, 从PASs收集的应用	No
[171]	静态分析	请求的权限, 权限 过滤的意图限制API调用 硬件特征、可疑API调用、 代码相关特征	CNN	公开可用的资源	Yes
[172]	静态分析	API序列调用	CNN	Drebin, 从PASs收集的应用	No
[173]	静态分析	Android字节码的语义结构	CNN-RS	公开可用的资源	No
[174]	动态, 静态分析	权限事件生成 由 Monkey Tool 提供	RS	公开可用的资源	Yes
[175]	静态代码分析	从smali获取token语义特征	DNN	私有	Yes
[176]	静态分析	指令序列	CNN	Drebin, 收集自PASs	Yes
[177]	静态, 动态分析	API序列到图像	CNN	公开可用的资源	No
[178]	动态分析	API调用图	CNN	AMD, AndroZoo , Drebin恶意软件库, ISCX安卓僵尸网络	Yes

3) 图像处理： 最近，研究人员开始采用静态图像分析进行Android恶意软件检测。一个基于CNN的模型将classes.dex的字节码转换为图像，以有效检测Android恶意软件 [168]。我们综述了采用图像处理技术进行Android恶意软件静态分析的DL算法的应用。

深度学习模型与机器学习的比较： 一个基于DBN的安卓系统恶意软件检测器使用了API调用块作为特征，而不是简单的API调用，并且优于CMLA[163]。一个基于DNN的安卓恶意软件检测系统使用遗传算法来修改深度学习模型的参数和配置，并与基于SVM的模型进行了比较，准确率达到91% [165]。在另一项研究中，一个CNN模型与LSTM模型在基于API调用序列的安卓恶意软件检测中进行了比较，结果表明虽然深度学习模型比基于n-gram的检测模型实现了更高的准确率，但CNN模型出人意料地比LSTM模型表现更好 [167]。

提出了新的用户行为模式感知模型，用于检测 Android 恶意软件。一个基于卷积神经网络的通用 Android 恶意软件检测系统评估了用户注意力图，其检测准确率优于传统方法[169]。一个基于卷积神经网络和自编码器的混合大规模 Android 恶意软件检测系统被提出，并使用 13000 个恶意应用和 10000 个良性应用进行测试，与简单的卷积神经网络模型相比，其训练周期缩短了 83% [171]。一个基于深度神经网络的 Android 恶意软件检测器使用了 smali 文件的标记和语义特征，并使用来自 50 个 apks 的 smali 文件进行训练，以优于 CMLAs，在 WPDP 和 CPDP 模式下分别达到了 85.98% 的 AUC 和 70%，并优于 CMLAs [175]。一个基于卷积神经网络的 Android 恶意软件检测框架使用了反编译 APK 文件中的指令序列进行训练，实现了 99% 的检测准确率，且误报率非常低[169]。

一个基于卷积神经网络的 Android 恶意软件检测框架使用了反编译 APK 文件中的指令序列进行训练，实现了 99% 的检测准确率，且误报率非常低 [176]。

与机器学习模型相比的深度学习模型： 一个基于CNN的android恶意软件检测框架，使用原始指令序列的静态分析特征进行训练，在与其他基于n-gram的检测模型相比时，计算效率更高 [170]。在另一项研究中，一个基于DBN的android恶意软件检测系统从API调用和权限中提取特征来构建检测模型。虽然DREBIN使用545000个特征达到了90%的准确率，但DBN模型仅使用237个特征就达到了相同的准确率[166]。

提出了一种基于深度学习的自动恶意软件检测框架，用于在服务器、移动设备和物联网设备上部署，通过探索应用程序的原始API调用序列，实现了高F1分数（96%-99%）[172]。另一个基于多层感知机和长短期记忆网络的多检测层用于Android恶意软件检测，使用第一层的xml文件和第二层的字节码语义进行训练，达到了97.74%的准确率 [173]。

F. 深度学习在侧信道攻击检测中的应用

科赫在1996年引入了侧信道攻击，黑客曾利用这种攻击手段入侵密码设备[121]。侧信道信息硬件，如时序信息、电磁辐射、功耗以及加密设备的统计信息，都可能被黑客用来发起侧信道攻击。这类攻击速度极快且易于实施，对安全构成重大威胁。任何设备，从小型嵌入式设备如RFID到笔记本电脑，都可能受到侧信道攻击的影响。我们回顾了针对侧信道攻击的有效检测的深度学习框架研究，并在表X中进行了总结。

TABLE IX
A SUMMARY OF STUDIES ON DL APPLICATIONS IN ANDROID MALWARE DETECTION.

Reference	Analysis	Features	Architecture	Dataset	Compared CML
[161]	Dynamic, Static analysis	Sensitive API calls	DBN	Publically available sources	Yes
[162]	Static, Dynamic Analysis	10 static, dynamic feature sets	Autoencoder	Publically available sources	No
[163]	Static analysis	API call blocks	DBN	Publically available sources	Yes
[164]	Dynamic analysis	System calls	SAE	Publically available sources	Yes
[165]	Static analysis	More than 1,000 features	DNN	Drebin, collected from PASs	Yes
[166]	Opcode Sequence	Learn to detect sequences of opcode that indicates malware	CNN	Publically available sources	No
[167]	Static analysis	API call sequences	CNN	Publically available sources	Yes
[168]	Transfer classes. dex into RGB color images	Extract features from transferred images	CNN	Collected by the researchers	No
[169]	Static analysis	Byte sequences	CNN	Publically available sources	Yes
[170]	Static analysis	Dangerous API calls, risky permissions	DBN	Drebin, apps collected from PASs	No
[171]	Static analysis	Permissions requested, permissions filtered intents restricted API calls, hardware features, suspicious API calls, code related features	CNN	Publically available sources	Yes
[172]	Static analysis	API sequence calls	CNN	Drebin, apps collected from PASs	No
[173]	Static analysis	Semantic structure of Android bytecode	CNN-RS	Publically available sources	No
[174]	Dynamic, Static analysis	Permissions events generated by Monkey Tool	RS	Publically available sources	Yes
[175]	Static code analysis	Token semantic features from smali	DNN	Private	Yes
[176]	Static analysis	Opcode sequences	CNN	Drebin, collected from PASs	Yes
[177]	Static, Dynamic analysis	API sequences to image	CNN	Publically available sources	No
[178]	Dynamic Analysis	API calls graph	CNN	AMD, AndroZoo, Drebin Malware Collection, ISCX Android Botnet	Yes

3) *Image Processing:* Static image analysis were adopted recently by researchers for Android malware detection. A CNN based model converted the bytecode of classes.dex into images for an effective detection of Android malware [168]. We present a survey of literature on the application od DL algorithms that have adopted image processing techniques for static analysis of Android malware detection.

DL models compared with ML: A DBN based malware detector for Android systems used blocks of API calls as features instead of simple API calls and outperformed CMLA [163]. A DNN based android malware detection system used a genetic algorithm to modify the parameters and configurations of the DL model and compared with SVM based models with an accuracy of 91%. [165]. In another study, a CNN model was compared with LSTM model for the android malware detection based on API call sequences and showed that while DL models achieved better accuracy than n-gram based detection models, CNN model surprisingly performed better than LSTM model [167].

New models of user pattern sensing were proposed for detecting Android malware. A CNN based generalized Android malware detection system evaluated user attention map and achieved better detection accuracy than conventional methods [169]. A hybrid large-scale android malware detection system based on CNN and DAE was proposed and was tested with 13000 malicious and 10000 benign Apps and showed considerable reduction in the training period by 83% when compared to simple CNN model [171]. A DNN based android malware detector used token and semantic features of smali files and was trained using smali files from 50 apks to outperform CMLAs with an AUC of 85.98% and 70% in both WPDP and CPDP mode respectively [175]. A CNN based android

malware detection framework used opcode sequences from decompiled APK files for training and achieved a detection accuracy of 99% with very low false positives [176].

DL models without comparing with ML: A CNN based android malware detection framework trained using features from static analysis of raw opcode sequences was computationally more efficient when compared to n-gram based detection models [170]. In another study, a DBN-based android malware detection system extracted features from API calls and permissions to build the detection model. While DREBIN attained accuracy of 90% with 545000 features, the DBN model achieved the same accuracy with just 237 features [166].

A DL based automatic malware detection frame-work for android systems was proposed for deployment on servers, mobile and IoT devices by exploring raw sequences of API calls from Apps and achieved high F1 score (96%-99%) [172]. Another multi-detection layer based on MLP and LSTM for Android malware detection was trained using xml files in the first layer and bytecode semantics in the second layer and achieved an accuracy of 97.74% [173].

F. Deep Learning in Side Channel Attacks Detection

Kocher introduced Side-Channel attacks in 1996 that were employed by hackers to break into cryptographic devices [121]. Side channel information hardware such as timing information, electromagnetic radiation, power consumption, and statistics of encryption devices could be utilized by hackers to launch a side channel attack. These attacks are very fast and can be implemented easily, posing a great threat to security. Any device from small embedded devices such as RFID to

表 X: 用于侧信道攻击检测的深度学习模型。

参考文献	方法	数据集	与 CML 相比
[140]	自归一化神经网络 & 卷积神经网络	ASCAD数据库	No
[141]	深度神经网络及其变体	ASCAD数据库	No
[142]	CNN	DPA竞赛v4	Yes
[143]	CNN	DPA竞赛v2 & DPA竞赛v4	No
[144]	CNN	MNIST	No
[145]	DNN	私有	No

笔记本电脑都容易受到侧信道攻击。我们回顾了针对侧信道攻击的有效检测的深度学习框架研究，并在表X中进行了总结。

将深度学习应用于侧信道攻击分析，并结合多个超参数，并与基准模型进行比较，表明VGG-16模型优于许多基线模型 [196]。此外，为基于卷积神经网络（CNN）的模型提供了深度学习在侧信道攻击检测方面的概述 [197]。此外，提出了两种深度学习方法来提高侧信道攻击的有效性。第一种方法通过在神经网络中使用新的扩散层来减少训练和攻击轨迹以检索密钥。第二种方法基于混淆矩阵有效地纠正模型预测 [198]。研究了多种机器学习和深度学习模型用于侧信道攻击，发现当噪声水平低且特征数量高时，CNN表现更好。在其他场景中，随机森林（RF）和XGBoost比CNN具有更低的计算成本且表现更优 [199]。此外，一项关于CNN与CMLA在四种侧信道攻击上的比较研究表明，CNN模型达到了99.3%的更高准确率，而在DPA竞赛v2数据集上，支持向量机（SVM）则优于CNN [200]。训练了一种基于CNN的新型模型用于检测侧信道攻击，使用测量的功率轨迹实现了在MNIST数据集上的高准确率[201]。基于深度学习的侧信道攻击被用于检索AES加密电路的密钥。通过分析捕获的电磁（EM）辐射和功耗，使用深度神经网络（DNN）建立了EM噪声与功率噪声之间的关系，并通过分析一组32,500个明文检索出了密钥 [202]。

G. 深度学习用于功能识别

功能识别是一个识别二进制码中功能的过程，这在恶意软件检测中很有用。我们回顾了深度学习架构在功能识别中的应用。

提出了一种深度学习方法，用于识别各种应用（如语言建模和语音识别）二进制文件中的函数，并与机器学习方法进行了比较，显示出良好的结果 [50]。EKLAVYA 是一个基于循环神经网络的系统，用于解决函数类型签名恢复问题，通过学习与给定领域知识和调用约定匹配的习语 [51]。Gemini 是一种基于深度神经网络的方法，用于为二进制函数生成嵌入，能够显著识别易受攻击的固件图像 [52]。MobileFindr 是一种动态策略，用于映射函数相似性，成功识别移动应用中的细粒度函数相似性 [50]。Diff 是一种基于深度神经网络的增强解决方案，应用了三个语义特征，即模块间、函数间和函数内特征，以解决跨版本二进制代码相似度检测问题 [51]。一种深度学习方法采用了词嵌入和图嵌入来从两个二进制文件中提取特征，以找到它们之间的相似性，并且比机器学习方法表现更好 [52]。

成功识别移动应用中的细粒度函数相似性 [53]。Diff 是一种基于深度神经网络的增强解决方案，应用了三个语义特征，即模块间、函数间和函数内特征，以解决跨版本二进制代码相似度检测问题 [54]。一种深度学习方法采用了词嵌入和图嵌入来从两个二进制文件中提取特征，以找到它们之间的相似性，并且比机器学习方法表现更好 [55]。

自注意力功能嵌入（SAFE）被引入，基于自注意力神经网络进行功能嵌入，以在不使用CFG [56]的情况下提高速度。SySeVR是一种深度学习架构，通过识别基于语法、语义和向量的功能来检测多个软件漏洞 [57]。INNEREYE-BB是一种基于神经网络和词嵌入的相似性比较工具，使用LSTM [58]解决了跨架构代码包含的问题。一种深度学习方法被应用于可视化二进制码图像并将其分类，以解决二进制码相似性问题 [59]。此外，使用RNN架构实现了一种通用的反编译方法，从二进制机器代码中恢复源代码结构 [60]。

H. 深度学习用于隐写分析和隐写术

在网络通信中，隐写术技术是一种在隐藏通信存在的同时发送消息的艺术，用于发送隐藏在普通信息中的秘密消息，如安全密钥，使秘密消息不可见。隐写分析，即检测包含隐藏消息的通信存在的过程，最近已成为网络安全领域的一个重要研究方向，用于识别公共网络中的隐蔽攻击。隐写术和隐写分析技术可以应用于不同类型的数据，如文本、图像和视频 [61]。实验利用人工神经网络进行，展示了基于机器学习的隐写术的能力 [62]。

深度隐写术是利用深度神经网络提出的一种技术，它不仅用于隐藏信息，还能揭示隐藏的消息。实验是在ImageNet数据库上进行的 [63]。研究人员提出了一种两阶段的信息隐藏过程，应用深度学习架构进行信息隐藏，并使用多种隐写算法研究了其性能 [64]。此外，将深度神经网络的数据隐藏能力与经典数据隐藏技术进行了比较，结果表明深度神经网络模型更高效、更鲁棒 [65]。在另一项研究中，计算机视觉中著名的纹理合成方法被用于图像隐藏，以实现隐写术和数字水印 [66]。在另一项研究中，研究人员没有使用经典的数学函数和特征进行纹理合成，而是采用生成对抗网络来增强卷积神经网络模型在图像隐写术中的鲁棒性 [67]。

无监督生成对抗网络被引入以避免隐写术和有效载荷容量所需的专家知识及复杂的人工规则，通过从秘密消息生成隐写图像而不需要载体图像 [68]。此外，基于生成对抗网络的方法被提出以成功地在图像中隐藏二进制数据 [69]。此外，基于深度学习隐写分析进行实验分析以

TABLE X
KEY DL MODELS FOR SIDE CHANNEL ATTACK DETECTION.

Reference	Method	Dataset	Compared CML
[140]	Self-Normalizing Neural Networks, & CNN	ASCAD Database	No
[141]	DNN & its variants	ASCAD Database	No
[142]	CNN	DPA Contest v4	Yes
[143]	CNN	DPA Contest v2 & DPA Contest v4	No
[144]	CNN	MNIST	No
[145]	DNN	Private	No

laptops are vulnerable to side channel attacks. We review studies on DL frameworks for an effective detection of side channel attacks and provide a summary in Table X.

The application of DL for the analysis of side-channel attacks with several hyperparameters, and comparison with benchmark models showed that VGG-16 model outperformed many baseline models [196]. Further, an overview of DL for side channel attacks detection was provided for CNN based models [197]. Further, two DL approaches to enhance the effectiveness of side-channel attacks were proposed. The first approach decreased the training and attack traces to retrieve the key by using new spread layer in NNs. The second approach efficiently corrected the model predictions based on confusion matrix [198]. Several ML and DL models for side-channel attacks were studied to find that CNN performed better when the noise level was low and the number of features were high. RF and XGBoost performed better than CNN with low computational cost in other scenarios. [199]. Further, a comparative study of CNN versus CMLAs on four side channel attacks showed that CNN model achieved better accuracy of 99.3%, while with the DPA contest v2 dataset, SVM outperformed CNN [200]. A novel CNN based model for detecting side-channel attack was trained using the measured power traces achieving high accuracy with MNIST datasets [201]. A DL based side-channel attack was used to retrieve the secret key of AES cryptographic circuit. The relationship between EM noise and power noise was modeled using DNN by analysing the captured EM emission and power dissipation and the secret key was retrieved by analysing a set of 32,500 number of plain texts [202].

G. Deep Learning for Function Recognition

Function recognition is a process of recognizing functions in a binary code that is useful in malware detection. We review the application of DL architectures for function recognition.

A DL approach to recognize functions in binary files of various applications such as language modeling and speech recognition was proposed and compared with MLA showing promising results [50]. EKLAVYA, a RNN based system was introduced to address the problem of function type signature recovery by learning using idioms that match the given domain knowledge and their calling conventions [51]. Gemini, a DNN based approach to generate embedding for binary function was able to identify vulnerable firmware images significantly [52]. MobileFindr, a dynamic strategy for mapping function similarities was developed identify the fine-grained function similari-

ties successfully in mobile Apps [53]. Diff, a DNN augmented solution applied three semantic features namely, inter-module, inter-function and intra-function features to solve cross-version BCSD problem [54]. A DL method employed word embedding along with graph embedding for extracting features from two binary files to find the similarities between them and performed better than ML approaches [55].

Self-attentive function embeddings (SAFE) was introduced for embedding of functions based on self-attentive NN to enhance speed without using CFG [56]. SySeVR, a DL architecture detected several software vulnerabilities by recognising functions based on syntax, semantics, and vectors [57]. INNEREYE-BB, a similarity comparison tool based on NN and word embedding solved the problem of cross-architecture code containment using LSTM [58]. A DL approach was applied to visualize images for binary codes and classify them to solve the problem of binary code similarities [59]. Further, a generic method of decompilation to recover the structure of source code from binary machine code was implemented using RNN architecture [60].

H. Deep Learning for Steganalysis and Steganography

In network communications, steganography technique, an art of sending messages while hiding the existence of the communication have been used to send secret messages such as security keys that are concealed inside ordinary information, making the secret message invisible. Steganalysis, the process of detecting the presence of communication that contains the concealed messages, has been a significant area of research in cyber security recently to recognize covert attacks in public network. Steganography and steganalysis techniques can be applied on different kinds of data such as texts, images and videos [61]. Experiments were performed utilizing ANN to show the capability of ML based steganography [62].

Deep Steganography was proposed using DNN to work as a pair to not only hide but also uncover the concealed messages and experiments were performed using Imagenet database [63]. A two-stage process for hiding information was proposed by applying a DL architecture for information hiding and its performance was studied using multiple steganographic algorithms [64]. Further, the capability of DNN for data hiding was compared with the classical data hiding techniques showing that DNN models were more efficient and robust [65]. In another study, texture synthesizing, a well-known method in computer vision was used for image concealing for achieving steganography and watermarking [66]. In another study, instead of using the classical method of using mathematical functions and features for texture synthesizing, GANs were employed to enhance the robustness of the CNN model in image steganography [67].

Unsupervised GANs were introduced to avoid expert knowledge and complex artificial rules required for steganography and payload capacity by generating the steganographic image from the secret message without the cover image [68]. Further, GANs based method was proposed for hiding the binary data inside an image successfully [69]. Further, an experimental analysis based on DL based steganalysis was studied for to

表 XI 内部威胁检测的关键深度学习应用。

参考文献	方法	数据集	对比 CML
[146]	深度神经网络全反向传播算法	CERT 内部威胁	Yes
[147]	CNN & 反向传播算法	CERT 内部威胁	No
[148]	RS	洛斯阿拉莫斯国家实验室 (LANL) 网络安全	Yes
[149], [151]	RS	CERT 内部威胁	No
[150]	DBN	CERT 内部威胁	No
[152]	CNN	Balabit 鼠标动态挑战	Yes

挫败基于最低有效位（LSB）的隐写术 [70]。在另一项研究中，深度卷积生成对抗网络（DCGAN）被采用以避免在隐写术中嵌入信息[71]。使用DCGAN，提出了一种安全的隐写术方法，可自动生成用于图像的容器，与原始图像派生的容器相比，其安全性更高 [72]。深度残差架构被引入作为先前基于卷积神经网络（CNN）模型的改进，由于其需要良好的启发式方法来识别其各种参数值[73]。此外，深度残差多尺度卷积网络被提出用于隐写分析，其性能优于现有的基于CNN和其他经典隐写术方法。在另一项研究中，采用了一种TL方法进行图像隐写分析，并使用深度残差神经网络进行了详细的实验分析 [74]。提出了一种使用长短期记忆网络（LSTM）编码器和解码器模型的文本隐写术方法，用于生成中文律诗 [75]。为了在VoIP流中隐藏信息，采用了量化索引调制（QIM），并结合基于循环神经网络（RNN）的语言学隐写术和基于CNN的文本隐写分析进行语义分析 [76]。此外，提出了一种基于深度神经网络（DNN）的语音信号隐写术方法[77]。

I. 深度学习在内部威胁检测中的应用

此前已有多项研究针对外部恶意软件攻击系统进行检测。然而，直到最近研究人员才开始考虑系统内部的潜在安全威胁。为应对所谓的内部威胁，基于各种机器学习架构的检测系统已被探索。此类系统的目标是从系统内部数据的行為中识别敌对活动。随着网络威胁随着物联网和工业4.0的发展而演变，内部威胁的识别变得更加困难。在入侵检测系统（IDS）中，传统的内部威胁检测系统依赖于过去攻击的知识，但这种方法被认为效率低下。我们总结了采用深度学习进行内部威胁检测的关键研究综述，如表XI所示。

一个在线无监督深度学习框架，由深度神经网络和长短期记忆网络模型组成，通过系统日志实时开发单个用户行为的异常分数，以有效地识别内部威胁，并优于现有的异常检测基线，如隔离森林、PCA 和支持向量机[146]。另一项研究提出了一种新颖的内部威胁检测

一个基于长短期记忆网络的用户行为特征提取系统和卷积神经网络分类系统，实现了0.9449的AUC[147]。在另一项研究中，一种用于检测异常活动的灵活无监督技术基于双向长短期记忆网络，在计算机安全日志数据上进行训练，性能明显优于标准主成分分析和基于隔离森林的检测模型，实现了0.98的AUC[148]。此外，一个基于长短期记忆网络-循环神经网络和主成分分析的内部门户威胁检测框架评估了行为异常性，并优于支持向量机、主成分分析和隔离森林 [149]。一个基于自适应优化DBN的多层隐藏层内部门户威胁检测器通过分析用户日志提取行为模式，实现了97.872%的检测准确率 [150]。此外，一个基于长短期记忆网络的内部门户威胁系统使用系统日志训练模型，以区分异常行为和正常用户行为 [151]。在另一项研究中，一种基于卷积神经网络的用户认证技术通过每7秒检查鼠标的动态行为来认证用户，实现了仅2.94%的假阳性率[152]。

J. 社交媒体数据用于网络安全

诸如Google Talk、Orkut、Facebook、Twitter和WhatsApp等社交媒体平台正以指数级速度获得普及。由于社交媒体在两个远程地点之间实时通信方面的诸多优势，也存在一些需要解决的挑战，例如网络跟踪、网络欺凌、黑客攻击、传播反社会言论和意识形态的反社会分子，甚至虚假新闻的传播。由于社交媒体平台上的虚假身份画像日益加剧，监控相关文本流并预测DDoS攻击的可能性 [118]是一项巨大挑战。一项将NLP模型应用于仅使用社交媒体作为来源检测DDoS攻击的新颖应用，在FFN和部分标记的LDA模型等先前最先进技术 [119]方面表现更优。另一项罕见的研究研究了自2010年以来发布勒索软件、病毒和其他恶意软件的Twitter用户集群，并利用这些信息对新攻击进行自动分类[120]。一项研究提出了DNN架构，并采用网络威胁态势感知技术对Twitter推文中的加密文本进行分类，以检测和分类勒索软件事件 [265]。此外，使用循环结构将加密数据转换为数值表示，并使用Keras嵌入在字符和单词级别的文本表示上评估模型[260]。

第七节 深度学习：保障下一代通信网络安全

A. 深度学习在智慧城市物联网应用中的部署

物联网和CPS的近期发展促使世界不同国家的政府将智慧城市项目作为工业4.0的未来来推广。许多用于实现智慧城市的创新应用，如电网、供水、道路交通和智能社区服务，正在迅速增长。各种CPS的物联网网络将各种设备连接起来，这些设备的存储空间有限

TABLE XI
KEY DL APPLICATIONS FOR INSIDER THREAT DETECTION.

Reference	Method	Dataset	Compared CML
[146]	DNN & RS	CERT Insider Threat	Yes
[147]	CNN & RS	CERT Insider Threat	No
[148]	RS	Los Alamos National Laboratory (LANL) cyber security	Yes
[149], [151]	RS	CERT Insider Threat	No
[150]	DBN	CERT Insider Threat	No
[152]	CNN	Balabit Mouse Dynamics Challenge	Yes

defeat LSB-based steganography [70]. In another study, deep convolutional generative adversarial networks (DCGANs) was employed to avoid embedding information in steganography [71]. Using DCGAN, a secure steganography method was proposed to automatically generate container for images that enhanced the security against steganalysis method as compared to container derived from original images [72]. A deep residual architecture was introduced as an improvement to previous CNN based models for steganography due to its requirement of good heuristics to identify the value for its various parameters [73]. Further, deep residual multi-scale convolutional network was proposed for steganalysis which outperformed the existing methods based on CNN and other classical steganography methods. In another study, a TL approach was employed for image steganalysis with a detailed experimental analysis using a deep residual NN [74]. Text steganography method using LSTM encoder and de-coder models was proposed to generate Chinese quatrains [75]. To hide information in VoIP streams, Quantization index modulation (QIM) was employed with RNN based linguistic steganography and CNN based text steganalysis for semantic analysis [76]. Further, a DNN based method for steganography in speech signals was proposed [77].

I. DL in Insider Threat Detection

There have been several research studies for the detection of external malware attacking a system. However, only recently researchers have started to consider the possibility of security threats within the system. To address this so-called insider threat, detection systems based on various ML architectures have been explored. The aim of such systems is to identify hostile activities from the behavior of data inside the system. As the network threat evolves with IoT and Industry 4.0, identification of internal threat has become more difficult. In IDS, classical insider threat detection systems have been functioning on the acquired knowledge of past attacks which has been deemed to be inefficient. We present a review of key research studies that have employed DL for insider threat detection is summarised in Table XI.

An online unsupervised DL framework consisting of DNN and LSTM models developed anomaly scores from individual user behavior in real-time using system logs to efficiently identify the insider threat and outperformed the existing anomaly detection baselines such as Isolation Forest, PCA, and SVMs [146]. Another study presented a novel insider threat detection

system based on LSTM for user behaviour feature extraction and CNN for classification and achieved an AUC of 0.9449 [147]. In another study, a flexible unsupervised technique for the detection of anomalous activities using BLSTM was trained on computer security log data and performed significantly better than standard PCA and isolation forest based detection models achieving an AUC of 0.98.[148]. Further, an insider threat detection framework based on LSTM-RNN and PCA evaluated the behaviour abnormality and outperformed SVM, PCA, and Isolation Forest [149]. A novel insider threat detector based on adaptive optimization DBN with multiple hidden layers extracted the behaviour patterns by analysing user logs and achieved detection accuracy of 97.872% [150]-Further, an LSTM based insider threat system used system log to train the model to differentiate anomalous behaviour from normal user behaviour [151]. In another study, a CNN based user authentication technique used the dynamic behaviour of the mouse to authenticate users by checking every 7 seconds and achieved FPR of only 2.94%.[152].

J. Social Media Data for Cyber Security

Social media platforms such as Google Talk, Orkut, Facebook, Twitter and WhatsApp have gained popularity at an exponential rate. With several advantages of social media for real-time communication between two remote locations, there are also challenges to be addressed such as cyber stalking, cyber bullying, hacking, anti-social elements spreading their propaganda and ideologies, and even the spread of fake news. Due to the escalating fake identity profiling on social media platforms, there lies a huge challenge in monitoring relevant text streams in social media and predicting the likelihood of DDoS attacks [118]. A novel application of NLP models to detect DDoS attacks using only social media as a source outperformed the previous state-of-the-art techniques such as FFN and a partially labelled LDA models [119]. Another rare research studied the clusters of Twitter users tweeting about ransomware, virus and other malware since 2010 and used the information for automatic classification of new attacks [120]. A study proposed DNN architectures and employed cyber threat situational awareness techniques for encrypted text classification from Twitter tweets to detect and classify the ransomware events [265]. Further, recurrent structures were employed to convert the data in encrypted form to numeric representation, and using Keras embedding the model was evaluated on both character and word level text representation [260].

VII. DEEP LEARNING FOR SECURING NEXT-GENERATION COMMUNICATION NETWORKS

A. DL in IoT Applications of Smart cities

Developments in IoT and CPS have recently prompted governments of different countries around the world to promote smart city projects as the future of Industry 4.0. Many innovative lot applications for realizing smart cities such as power grid, water supply, road traffic, and intelligent community services are rapidly growing. IoT networks of various CPS connect various devices with limited storage

物联网的主要目标是为建筑物、智能家居、智能车辆、智能电网、智慧城市、智能医疗、智能农业等安全、可靠且完全自动化的互联智能环境提供支持，这会产生BD。由于这些异构物联网设备本身具有较低的安全特性，它们对智慧城市构成了巨大威胁。当攻击者获得对这些物联网 [34]的访问权限时，可能会发起DDoS攻击和僵尸网络等多种类型的攻击。受损的物联网设备不仅可能入侵企业网络，还可能危害智慧城市环境中的生命安全。

除了在网络和主机级系统中开发具有安全防火墙的安全策略外，还可以采用基于启发式的方法，使用机器学习和深度学习来保护物联网环境免受恶意活动侵害。最近的研究成功地将深度学习方法应用于物联网僵尸网络检测 [35], [36]。此外，还提出了一种基于网络的深度学习方法，用于检测从受损物联网设备发起的攻击 [35]。通过为每个设备的物联网流量行为模式训练自编码器，进行异常检测，以获得熟练度。当自编码器无法重新创建快照时，物联网设备被检测为受损。此外，还开发了一个名为Bot-IoT的测试环境，用于创建一个包含各种类型攻击、合法流量和模拟物联网流量的数据集，这将有助于未来的研究[36]。

B. 深度学习在区块链技术中的网络安全应用

最先进的深度学习算法已被应用于广泛领域的各种应用，如自然语言处理、语音识别和计算机视觉，并取得了令人满意的结果。然而，由于对实际可行性要求非常低的假阳性率给予了高度重视，深度学习在网络安全领域的应用仍处于起步阶段。此外，深度学习框架需要在对抗环境中运行，这意味着任何黑客都不应该有能够规避这些安全措施。

隐私保护深度学习是一种策略，其中模型和训练数据都不应暴露给外界。DeepChain是一个用于安全深度训练协作的鲁棒且公平的去中心化平台，其利用了四个不同方面，特别是吞吐量、密文大小、训练时间和训练准确率 [39]。如果分布式计算机节点受损，它将暴露算法和数据于各种网络安全威胁。交替方向乘法（ADMM）作为一种分布式优化问题的解决方案，被用于攻击检测 [40]。另一项研究对使用ADMM解决的广义分布式优化问题的潜在攻击向量进行了详细分析 [41]。

区块链技术可以被定义为一种解决网络犯罪的现实方案，因为它能够在多台计算机上实现去中心化和安全的公共账本。区块链技术受欢迎的原因在于其固有的韧性，能够应对身份盗窃、数据泄露和犯罪攻击。

最近，针对各种与安全相关的服务，开展了一项关于使用区块链技术的综合调查 [37]。此外，还探讨了区块链解决网络安全挑战的能力，并通过在不同数据服务器上使用大数据集训练深度学习模型，对各种基于区块链的方法进行了比较，以提供安全服务。区块链技术提供的分布式解决方案，用于维护防篡改系统，以及其解决安全问题各种方案的能力得到了讨论[38]。区块链技术可以利用其去中心化和协调的平台作为计算能力来管理其BD。它也将使深度学习决策或结果更加可信、透明和可解释。此外，它还提供了安全的数据共享环境。在最近的一项研究中，提出了一个用于深度学习应用的去中心化区块链架构 [42]。研究人员已经采用协作ID网络来提高IDS的性能。此类IDS的主要显著问题是数据的可用性和信任管理，这会影响IDS模型的有效性。为了避免这种情况，另一项研究展示了区块链技术在IDS中的应用 [43]。

C. 深度学习用于密码学

密码学是一种用于安全通信的技术，通过这种技术，只有发送者和接收者才能查看消息内容。在互联网传输中，密码学被广泛用于将数据转换为不可读格式（加密），在数据解密为原始形式后，该格式只有授权人员才能理解。现代密码学在网络安全背景下用于确保数据完整性、身份验证、数据机密性和不可否认性。它可以大致分为两种类型：对称密钥（单钥）和非对称密钥（公钥和私钥）。

深度学习算法已被应用于密码学，以实现更安全的数据加密，难以被轻易破解[78]。尽管深度神经网络计算成本较高，但将其应用于智慧城市已成为全球研究人员极大关注的应用领域。随着未来量子计算的帮助，这一计算成本将非常微小，破解加密将变得不可能。在最近的研究中，神经网络被用于通过对抗方法进行加密和解密，其应用于密码分析的应用被考虑用于多种操作，如隐写术、伪随机数生成和完整性检查[79] [81]。另一项研究讨论了将神经网络应用于加密数据的理论方面 [80]。各种实验研究表明，人工智能代理可以学习安全的加密方法，以及基于循环神经网络和卷积神经网络的方法在S-box和密码设计中的应用及其分类 [82][83] [84] [85] [86]。此外，研究人员提出了一种名为“密码网络”的神经网络集合，结合深度学习，允许用户将加密数据发送到云端，模型通过多种实验进行训练，以在不解密数据的情况下预测风险，从而提升性能 [87] [88]。

capacity and processing power to collaborate, associate, and exchange information in a peer-to-peer manner. The principle objective of IoT is to make secure, reliable, and fully automated interconnected smart environments such as buildings, smart homes, smart vehicles, smart grids, smart cities, smart healthcare, smart agriculture, and so on and this generates BD. As these heterogeneous IoT devices inherently possess low security features, they pose a great threat to smart cities. Different types of attacks including DDoS and botnet could be launched when attackers gain access to these IoT [34]. Compromised IoT devices could then be used to not only invade corporate networks, but also to endanger lives in a smart city environment.

Apart from developing security policies in network and host level systems with secure firewalls, heuristics based approaches using ML and DL can be employed to secure the IoT environment from malicious activities. Recent studies applied DL approaches successfully for IoT botnet detection [35], [36]. Further, an empirically evaluated network-based DL approach was proposed for detecting attacks launched from compromised IoT devices [35]. Anomaly detection was performed by training a DAE with behavioral snapshots of the IoT traffic for each device behavior patterns to gain proficiency. When an AE fails to recreate a snapshot, the IoT device was detected to be compromised. Further, Bot-IoT, a testbed environment, was developed to create a dataset with a combination of various types of attacks with legitimate and simulated IoT traffic that would be useful for future research [36].

B. DL in Blockchain Technology for Cyber Security

State of art DL algorithms have been employed in various applications in wide range of areas such as NLP, speech recognition and computer vision achieving promising results. However, DL is still in its infancy with its applications to cyber security domain due to the heavy importance given to have a very low FPR for practical viability. Further, the DL framework requires to be in adversarial environment which implies that any hacker should not have the capacity to circumvent the security.

Privacy preserved DL is a strategy in which neither the model nor the training data ought to be exposed to the outside world. DeepChain, a robust and fair decentralized platform for secure collaboration of deep training was proposed utilizing four different aspects in particular, throughput, ciphertext size, training time and training accuracy [39]. If the distributed computer nodes are compromised, it will expose the algorithm and data to various cyber security threats. Alternating Direction Method of Multipliers (ADMM), a solution for the distributed optimization problem was used for the detection of attacks [40]. Another study performed a detailed analysis on potential attack vectors for the generalized distributed optimization problems addressed using ADMM [41].

Blockchain technology can be defined as a realistic solution for cybercrime as it enables decentralized and secure public ledger on multiple computers. The reason behind the popularity of block technology is that its inherent resilience to identity

theft, data breach, and criminal attacks. Recently, a comprehensive survey on using blockchain technology was conducted for various security related services [37]. Furthermore, the ability of blockchain to resolve cyber security challenges was explored and various blockchain based methodologies were compared for providing security services by training a DL model with big datasets on different data servers. The distributed solution offered by the blockchain technology to maintain tamper-resistant system and its ability to provide various solutions to tackle security problems were discussed [38]. Blockchain technology can utilize its decentralized and coordinated platform as the computing power for managing its BD. It will likewise make the DL decisions or outcomes to be more trustworthy, transparent, and explainable. Additionally, it provides secure data sharing environment. In a recent study, a decentralized blockchain-based architecture for DL applications was proposed [42]. Collaborative ID networks have been employed by researchers to enhance the performance of IDS. The main significant issues with such IDS is the availability of data and trust management which impact on the effectiveness of the IDS model. To avoid this, another research study showed the application of blockchain technology in IDS [43].

C. DL for Cryptography

Cryptography is a technique used for secure communication by which only the sender and the receiver can view the contents of the message. In Internet transmission, cryptography is extensively used to convert data into unreadable format (encrypted) that is not understandable by anyone except for the authorized individuals after data is decrypted to the original form. Modern cryptography is used in cyber security to ensure data integrity, authentication, data confidentiality, and non-repudiation. It can be broadly classified into two types: symmetric-key (single key) and asymmetric-key (public and private key).

DL algorithms have been applied to cryptography to achieve a more secure data encryption that cannot be easily broken [78]. Even though DNNs have high computation cost, their application to smart cities is of immense interest among researchers worldwide. With the help of Quantum computing in future, this computation cost will be very minimal and breaking the encryption would become impossible. In recent studies, NNs were trained for encryption and decryption using adversarial methods and their applications to cryptanalysis were considered for a variety of operations such as steganography, pseudorandom-number generation and integrity checks [79] [81]. Another study discussed the theoretical aspects of applying NNs to encrypted data [80]. Various experimental studies showed how an artificial agent can learn a secure encryption method and the use of RNN and CNN based methods for S-box and cipher design as well as their classification [82] [83] [84] [85] [86]. Further, cryptonets, a set of NNs with DL was proposed to allow the user to send the encrypted data to the cloud and the models were trained to predict risks without decrypting the data with various experiments to enhance their performance [87] [88].

D. 深度学习与云安全

互联网正由云计算改革，因为它提供了一种便捷的按需网络接入能力，可访问大量共享、可配置的计算机系统资源，这些资源主要存在于三种主要的云服务模型中，即软件即服务（SaaS）、平台即服务（PaaS）和基础设施即服务（IaaS） [89]。云计算在企业中很受欢迎，因为它不仅节省成本，还节省了在主动管理和监控计算机系统资源及数据安全威胁方面所涉及的时间和精力。假设云资源处于一个安全的环境中，配备必要的防火墙和入侵检测系统，以防止未经授权用户访问。然而，云渗透事件正在上升，恶意软件攻击者正变得越来越容易绕过防火墙。随着机器学习和深度学习技术在许多领域的应用，它们在云计算中与信任、隐私和安全相关的问题上的应用得到了研究 [90]。Google和其他云平台提供者已将安全置于最高优先级，并采用机器学习和人工智能技术，不仅阻止恶意活动，还在其云环境中出现任何可疑行为模式时发出警告。

随着云计算进入云端存储和云服务器的时代，各种深度学习架构可以通过云端数据训练来提升性能。云服务提供商正在引入新的服务模式，如机器学习即服务（MLaaS）和深度学习即服务（DLaaS）成为可能。DLaaS使用户能够更快地设计、开发和训练深度学习应用，并通过商业服务（如Microsoft Azure ML、Google Cloud ML Platform、Amazon ML和IBM Watson Analytics）提供 [91]。作为IaaS中云基础设施的重要组成部分，一些实验性研究已经探索了启发式方法和深度学习的应用，例如使用长短期记忆网络（LSTM）检测虚拟机网络流量的异常行为 [92], [93]。此外，在云端数据背景下，基于循环神经网络（RNN）的识别和其他深度学习方法（如卷积神经网络）被提出用于移动云计算环境中的网络攻击检测 [94]。

E. 基于深度学习的边缘计算网络安全

边缘计算和雾计算是作为补充云计算的新兴概念，将计算更靠近数据位置，通过优化物联网设备和网络应用来提升响应时间。雾计算是边缘计算的标准，其中终端用户之间交换的信息通过雾层处理，这些层通过改变数据的结构、大小和有效性来处理输入信息，数据可能非常庞大。这些层包含来自不同提供商的雾节点，这导致了更多信任和网络安全问题，需要高度关注，因为它们对智慧城市、电子医疗、智能家居、移动应用等产生重大影响。最近的研究探索了雾计算中的新隐私和安全问题，使用深度学习架构对雾安全进行云端数据分析 [107], [100]。

与雾计算类似，多接入边缘计算（MEC）的概念由欧洲电信标准化协会（ETSI）移动边缘计算行业规范组（MEC ISG）于2014年末引入。

ETSI将MEC定义为一项新技术，它能在无线接入网络中以及靠近终端用户的位置提供云计算能力和功能 [101], [102]。MEC与云计算的区别在于，它具备本地数据、低延迟、位置感知和网络上下文文化等优势特性，但也存在计算能力有限、缓存和存储能力不足，以及个人用户数据安全和隐私等问题[89], [102]。由于MEC的典型部署位于云和终端用户之间，因此容易受到严重恶意攻击。为此，研究人员在多种基准方案（包括安全部分卸载、安全完全卸载、本地计算和无窃听的部分卸载）的基础上，对MEC物理层的安全性进行了研究评估，重点考察了能量消耗问题 [103]。

分析了可能影响边缘范式的安全威胁和挑战 [104], [105]。此外，随着MEC与即将到来的5G 6G、非正交多址接入（NOMA）、云无线接入网络（C-RAN）以及无人机通信等技术的集成，安全与隐私挑战日益加剧。MEC系统的挑战被进一步放大。针对MEC威胁态势、安全漏洞以及潜在解决方案的综述，为深度学习架构在安全边缘计算中的应用开辟了未来研究方向 [89], [106]。

F. 深度学习与网络安全在自动驾驶技术中

AI在智慧城市未来应用中最重要之一是自动驾驶技术。由于特斯拉、Waymo等汽车公司及其他初创企业的近期创新，街道上的自动驾驶汽车正成为可预见的未来。由于自动驾驶汽车与传统内燃机汽车相比，更像现代智能手机，这引发了关于控制这些自动驾驶汽车系统的网络安全、安全鲁棒性以及可被攻击性的问题。在最近的研究中，识别了CPS中汽车控制器局域网（CAN）的网络安全犯罪以及自动驾驶汽车的可能攻击、漏洞和利用方式，并测试了用于检测CAN流量和数据广播中异常的统计方法 [261], [266]。典型现代自动驾驶汽车系统中内置的深度神经网络（DNN）的任何错误都可能导致潜在的致命后果。提出了DeepTest这一工具，用于对DNN驱动的自动驾驶汽车进行自动化测试，以系统地探索DNN逻辑的各个部分，构建更鲁棒的系统，这些系统可以基于DNN对抗性深度强化学习（DRL）算法进行训练，以提高系统的鲁棒性 [267]。此外，另一项研究讨论了在车载网络中对自动驾驶套件进行安全测试的重要性，并使用深度神经网络实现了一个入侵检测系统（IDS） [268]。

G. 深度学习与普适计算中的网络安全

普适计算，也称为泛在计算，是随着CPS发展而出现的一种新技术。

D. DL for Cloud Security

Internet is being reformed by cloud computing since it offers a convenient, on-request network accessibility to huge amount of shared, configurable computer system resources available in three main cloud services models, namely Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS) [89]. Cloud computing is popular among enterprises as it not only saves cost but also the time and effort involved in active administration and monitoring of the computer system resources and data from security threats. The assumption is that the cloud resources are in a secure environment with the required firewall and IDS to prevent any access by unauthorized users. However, cloud infiltrations have been on the rise and malware attackers are becoming more capable of bypassing the firewall easily. With the recent use of ML and DL techniques in many domains, their application to issues related to trust, privacy, and security in cloud computing were studied [90]. Google and other cloud platform providers have given top priority for security and employ ML and AI techniques to not only block malicious activities but also show warning when there is any suspicious behaviour patterns in their cloud environment.

With cloud computing seeing the era of BD stored in cloud and cloud servers, various DL architectures could be trained with BD to enhance their performance. Cloud providers are introducing new service models such as ML as-a-service (MLaaS) and DL as-a-service (DLaaS) possible. DLaaS facilitates users to design, develop and train DL applications faster and available via commercial services such as Microsoft Azure ML, Google Cloud ML Platform, Amazon ML, and IBM Watson Analytics [91]. Virtual machines being an important component of cloud infrastructure in IaaS, some experimental studies have explored heuristics and the application of DL such as LSTM for VMs network traffic anomalous behavior detection [92], [93]. Further, RNN based ID and other DL approaches such as CNN were proposed for cyberattack detection in mobile cloud computing environment in the context of BD [94].

E. DL based Cyber Security in Edge Computing

Edge and fog computing are new emerging concepts introduced to complement cloud computing to bring computing closer to the data location for enhancing response time by optimizing IoT devices and Web applications. Fog computing is the standard adopted for edge computing where the information exchanged between the end users is through the fog layers which process the input information by changing structure, size and validity of the data which can be very large. These layers have fog nodes which are from various providers which leads to more issues of trust and cyber security that need much attention due to their impact on smart city, e-Health, smart homes, mobile applications, etc. Recent studies have explored new privacy and security problems in fog computing, analysis of BD for fog security using DL architectures [107], [100].

Similar to fog computing, the concept of multi-access edge computing (MEC), has been introduced by the European Telecommunications Standards Institute (ETSI) Mobile Edge

Computing Industry Specification Group (MEC ISG) in late 2014, the ETSI defines MEC as a new technology that provide the cloud-computing capabilities and functionalities within radio access networks and in close proximity to end users [101], [102]. MEC differs from cloud computing as it possesses advantageous features such as on-premise data, low latency, location awareness, and network contextualization but suffers from limited computing power, caching, and storage capabilities, and security and privacy of personal user data [89], [102]. Since the typical deployment of MEC lies between the cloud and end users, it is vulnerable to severe malicious attacks. Hence, the security of MEC physical layer was studied and evaluated in terms of energy consumption over several baseline schemes, including secure partial offloading, secure full offloading, local computing, and partial offloading without eavesdropping [103].

Security threats and challenges that can affect edge paradigms were analyzed and reviewed [104], [105]. Further, as MEC is becoming integrated with forthcoming technologies such as 5G 6G, non-orthogonal multiple access (NOMA), cloud radio access network (C-RAN), and unmanned aerial vehicle (UAV) communications security and privacy. challenges of MEC systems are much exacerbated. Reviews on MEC threat landscapes, security vulnerabilities, and potential solutions were conducted that opened up future research directions in the application of DL architectures for secured edge computing [89], [106].

F. DL and Cyber Security in Autonomous Vehicle Technology

One of the most important of applications of AI for smart cities of the future is autonomous vehicle technology. Due to the recent innovation by automobile companies such as Tesla, Waymo and other startups companies, autonomous driving cars on streets are becoming the foreseeable future. Since autonomous cars have more similarities with a modern smartphone than a traditional combustion engine car, it raises the question of cyber safety, security robustness, and hackability of the system that controls these autonomous cars. In recent research studies, cybercrime in CPS of automotive Controller Area Network (CAN) and the possible attacks, vulnerabilities and exploitations of autonomous vehicles were identified and statistical methods to detect the anomalies in the CAN traffic and data broadcasts were tested [261], [266]. Any error in the built-in DNNs of a typical modern-day autonomous vehicle system would result in potentially fatal outcome. DeepTest, a tool for automated testing of DNN-driven autonomous cars was proposed to systematically explore various parts of the logic of DNN for building robust systems that could be trained based on DNNs adversarial DRL algorithm in order to make the system more robust [267]. Further, another study discussed the importance of testing the safety of autonomous driving suites within in-vehicular networks and implemented an intrusion detection system (IDS) using deep neural networks [268].

G. DL and Cyber Security in Pervasive computing

Pervasive computing, also called ubiquitous computing, is another new technology developing with CPS advancement for

普适计算，也称为泛在计算，是随着CPS发展而出现的一种新技术，旨在让嵌入式系统随时随地可用。虽然它试图通过在日常活动中增加计算能力来互联、通信和更高效地执行，但普适安全问题已成为一个至关重要的关注点。关于这项新技术的安全性和普适安全领域当前开放问题的综合研究已被报道 [108][111]。通过使用反驳和单例不变量进行的实验，以识别当门关闭时冰箱内的灯是否关闭，以确定监控、证据收集和普适计算中的安全协调方法是否是更好的安全方式 [109][110]。实验已进行。另一项研究提出了一种结合去中心化信任和声誉管理、网络级观察和语义网语言声明策略的方法，以应对普适安全面临的挑战 [112]。在一个深度学习框架中，提出了一个在分布式传感器网络上的CNN——MicroDeep，用于预测来自物联网设备多个传感器的数据，并证明MicroDeep的性能优于简单的CNN [113]。此外，深度学习框架被嵌入到物联网设备中，去噪自编码器被用于检测受损物联网设备的可疑网络流量，以识别物联网僵尸网络攻击 [35], [114]。

H. 深度学习用于生物识别安全

生物识别安全涉及通过个人的独特生理特征和行为特征来识别一个人。生理特征包括面部、指纹、掌纹、虹膜等，而行为特征则涉及声音、签名、步态、按键等。通过使用这些生物测量数据进行个人识别，入侵者很难突破任何系统。因此，当需要访问机密数据时，可以实施生物识别安全。由于深度学习算法能够学习层次化特征，它们在语音、自然语言处理和计算机视觉等生物识别相关领域变得非常流行。最近一项研究综述了用于生物识别安全的深度学习模型 [95]。

第八章 机器学习与深度学习在网络安全杂项问题中的应用

A. 机器学习在网络安全应用中的迁移学习

迁移学习（TL）是一种利用现有模型为特定任务构建另一个相关任务的方法。这种方法在深度学习领域非常流行，特别是在与自然语言处理和计算机视觉相关的问题中。这是通过用新的输出层替换分类输出层来实现的。

TL 的主要优势在于节省时间和提升性能，因为无需重新创建模型。如图 5 所示，TL 可应用于在网络安全日志数据集上训练的深度学习模型，以检测恶意活动并识别僵尸网络，利用 DNS 查询信息。在这种情况下，无需从头开始训练，而是可以通过使用已训练的模型来提升深度学习模型在僵尸网络检测方面的性能。提出了一种新的聚类增强型层次 TL 技术，该技术发现已知和新之间的关系。

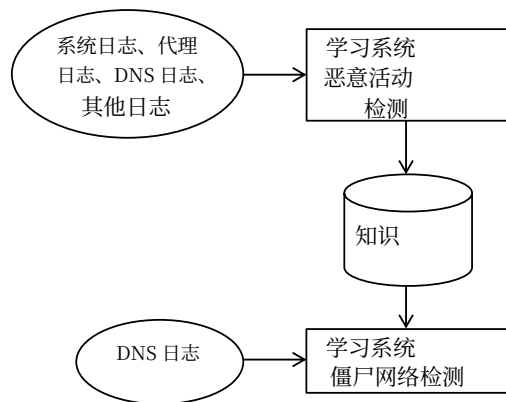


图5. 迁移学习。

提出了一种针对攻击的新 TL 技术，并使用 DT、随机森林、KNN、SVM 和 NB 等多种传统机器学习分类器的组合，以及各种现有的 TL 方法 [224] 对该 TL 技术进行了评估。

B. 无监督机器学习助力构建更优的网络安全系统

无监督学习是一种机器学习方法，用于推断描述隐藏结构的函数。近年来，人们越来越关注无监督学习方法，以理解和学习词语的表示，流行的方法如word2vec嵌入模型，它学习词语的句法和语义表示。与公开可用的数据集不同，大多数现实生活中的数据集通常是未标记的或标记质量差的。在这种情况下，监督学习和CMLA不可靠。为了解决这种缺点，采用了无监督机器学习，并提出了一个使用无监督学习的网络安全领域新型入侵检测系统（IDS）的例子，以展示其在数据几乎总是未标记时的优势 [25]。还提出了一种使用无监督学习的神经语言模型，用于特征提取，这是法医日志分析的关键部分，并且它优于其他特征提取技术 [26]。在另一项研究中，一个企业级框架采用分而治之的策略，结合行为分析和时间序列建模，AUC达到0.943 [27]。此外，开发并测试了一个基于神经网络学习的实时集体异常检测架构，使用KDDCup-99数据集的时间序列版本 [28]。

在机器学习和深度学习中，张量被用于泛化多维数组，其中包含数值，这些数组可以表示超过一维的矩阵。张量分解是一种将复杂张量表示为一个或多个更简单的张量的方法，以便更容易地进行操作和理解。为了提取网络流量中随时间变化且跨多视角变化的共同行为以检测不一致性 [29]，开发了一种联合概率张量分解方法来推导潜在张量子空间。当张量具有高阶维度且较为复杂时，用于查找密集块的分解技术在准确率、速度和灵活性方面并不令人满意。

embedded systems to be available anywhere at any time. While it is trying to increase embedding in IoT with computational capability in everyday activities to interconnect, communicate and perform more efficiently, pervasive security becomes a vital concern. Comprehensive studies on the security of this new technology and the current stage of open problems in pervasive security were reported [108][111]. Experiments using falsification and singleton invariant were conducted to identify if the light in refrigerator was off when the door was closed to ascertain methods of monitoring, evidence gathering and reconciliation for security in pervasive computing are a better way for security [109][110]. experiments were. Another study proposed a method to combine decentralized trust and reputation management, network-level observations and Semantic Web languages declarative policies to address the challenges faced by pervasive security [112]. MicroDeep, a CNN over a distributed sensor network was proposed in a DL framework to predict the data coming from multiple sensors of IoT devices and demonstrated that MicroDeep performed better than simple CNN [113]. Further, DL frameworks were embedded in IoT devices, and DAEs were utilized to detect suspicious network traffic of compromised IoT devices to identify IoT botnet attacks [35], [114].

H. DL for Biometric Security

Biometrics security deals with identifying a person by their unique physiological characteristics and behavioral characteristics. Physiological characteristics include face, fingerprints, palmprints, iris etc., whereas behaviors characteristics involve voice, signature, gait, keystroke, etc. By using such biometric measurements for personal identification, it becomes extremely difficult for an intruder to break into any system. Hence, biometric security could be enforced when access to confidential data is required. Since DL algorithms can learn hierarchical features, they have become highly popular in biometric related fields such as speech, natural language processing, and computer vision. A recent study reviewed DL models for biometric security [95].

VIII. ML AND DL FOR MISCELLANEOUS CYBER SECURITY ISSUES

A. Transfer Learning in Cyber Security Applications

Transfer learning (TL) is a method of making use of an existing model for a particular task for another related task. This method is very popular in DL particularly in various problems related to natural language processing and computer vision. This is achieved by replacing the output layer for classification by a new output layer.

The key advantages of TL are saving of time and enhanced performance as the model need not be created again. As illustrated in Fig. 5, TL could be employed on a DL model trained on network log dataset to detect malicious activities to detect botnets using DNS query information. In this case instead of training from scratch, the performance of the DL model can be enhanced for botnet detection by using the already trained model. A new clustering-enhanced hierarchical TL technique which finds the relation between known and new

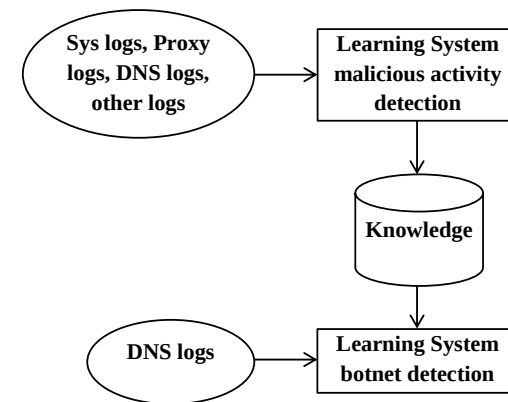


Fig. 5. Transfer Learning.

attack was proposed and the TL technique was evaluated using combination of different traditional ML classifiers such as DT, random forest, KNN, SVM, and NB with various existing TL approaches [224].

B. Unsupervised ML leads to build better Cyber Security system for an organization

Unsupervised learning is a ML approach for inferring a function to describe the hidden structure. Recently, there has been a lot of interest in unsupervised learning methods to understand and learn the representation of words, popular methods like word2vec embedding model, which learns the syntactic and semantic representation of a word. Unlike the openly available datasets, most of the real-life datasets are often unlabeled or poorly labeled. In such circumstances, supervised learning and CMLAs are not dependable. To address this disadvantage, unsupervised ML is employed and a classical example of unsupervised learning is ANN. A novel approach for IDS using unsupervised learning in the field of cyber security was proposed to demonstrate its advantage when the data is almost always unlabeled [25]. A neural language model using unsupervised learning was proposed for signature extraction which is the key part in forensic log analysis and it outperformed other signature extraction techniques [26]. In another study, an enterprise-grade framework used a divide and conquer strategy by combining the analytics of behavior and modeling of time series with an AUC of 0.943 [27]. Further, a real-time collective anomaly detection architecture based on NN learning was developed and tested using time series version of KDDCup-99 dataset [28].

In ML and DL, tensors which are multi-dimensional arrays that contain numerical values are employed to generalize matrices of more than one dimension. Tensor decomposition is a method of representing a complex tensor in the form of one or more simpler tensors for easier manipulation and understanding. A joint probabilistic tensor factorization method to derive the latent tensor subspace was developed to extract common behaviors in network traffic that vary in time across multi-views to detect inconsistency [29]. When the tensor is complex with high order dimension, the decomposition techniques used for finding dense blocks were not satisfactory

针对这一问题，M-ZOOM 被开发出来，在可扩展性、准确率、灵活性和有效性方面取得了令人鼓舞的结果，其 AUC 分数为 0.98 [30]。TensorDet 被开发出来，通过利用具有新颖方法（如顺序张量截断和两阶段异常检测 [31]）的分解结构来提高张量分解的实时计算效率。

C. 网络安全应用中的离线与实时部署

深度学习架构在网络安全中的离线与实时部署应用，需要在设计底层模型时考虑重要因素。理解深度学习架构、解释训练的机器学习模型学到了什么，是鲁棒验证程序的重要因素。可解释性是网络安全相关应用中的关键因素，其中必须建立模型对正确特征的依赖关系。通常，简单模型（线性模型）比复杂模型（非线性模型）更容易解释。可解释性是指理解特征预测的能力，例如文本或图像的特征，而非可解释性则指隐藏层特征、由文本表示和词嵌入产生的向量空间。热力图是最常用的理解分类决策的方法之一。热力图图像的像素提供了对分类的贡献。

目前还没有关于深度学习架构解释和透明性的明确数学证明和理论。因此，很难找到一个具体的原因来识别深度学习架构模型为何会将数据样本误分类。识别哪种深度学习架构更合适、为网络结构和网络参数确定最佳参数，是一项艰巨的任务。此外，需要更多实践知识来为学习率、正则化等参数确定合理的值。目前，这些参数是按经验确定的。提出了一种识别最佳特征图数量的方法，该方法对极小的感受野 [32]效果很好。提出了一种可视化方法，促进了中间特征的可视化 [33]。

缺乏知名标记基准数据集：由于隐私和安全原因，标记数据集不会公开用于研究目的。通过手动方法标记数据样本是一项艰巨的任务。标记数据样本最常用的解决方案是基于供应商提供的黑名单和白名单。基本上，机器学习中使用了三中不同类型的数据集。它们被称为训练、验证和测试数据集，并且这些数据集是互不相交的。这意味着当我们收集数据样本以开发网络流量分析系统时，我们必须从包含不同用户以及不同应用程序访问的不同网络收集这些数据集。这三个数据集还应包括时间信息。也就是说，训练数据应来自 t_1 ，测试数据集应来自 $t_1 + 2$ ，验证数据集应来自 $t_1 +$

1. 异常检测在许多领域更受欢迎，但在网络安全领域不太受欢迎。这是因为实现低误报率是网络安全异常检测中的一项最大任务。在单个错误分类的情况下，可能会给公司造成数百万美元的损失。半监督和无监督学习方法在网络安全领域是首选的方法。在数据集收集过程中应考虑的主要因素有：1) 不同的测量质量，2) 不同的主题，3) 随着时间的推移技术演变，4) 不同的示例标记方式，5) 不同的集中程度，6) 不同的环境，7) 不同的协议，以及 8) 一天中的时间。

攻击者-防御者方法与漂移概念：网络安全是一个不断发展的领域，为了适应对手使用的新类型模式；基于机器学习的系统必须不断进行训练。由于各种信息通信系统生成的数据集规模庞大，特征工程是一项艰巨的任务，因此在这种情况下可以使用深度学习架构。这有助于简单地按照相关方法学习对手使用的新类型模式。

不平衡的数据样本：数据不平衡是网络安全中最常见的问题之一。大多数情况下，恶意软件的样本是稀有的，特别是在网络安全的 多类分类领域中，几乎所有的数据都是不平衡的。

领域自适应：领域自适应是一种衡量训练数据集和测试数据集之间差异的方法。这两个数据集的分布应该是完全不同的。网络安全的领域包含许多形式的数据集；这包括网络流量、垃圾邮件、网络钓鱼等。这些高度相关，并且可以帮助有效检测恶意软件。一个主要的挑战是从一个领域采用有效的防御方法到另一个领域。

实时系统中部署机器学习模型需考虑的重要因素：尽管这些机器学习算法和深度学习架构具有区分新型恶意模式的能力，但在企业安全系统中采用它们仍处于早期阶段。最近，一个通常称为可解释人工智能的新研究方向可以更好地为错误决策提供理由。网络安全系统中的错误决策可能造成数百万美元的损失。例如，如果在一个企业系统中，合法应用程序被标记为恶意，并且在工作时间无法被任何一方接受，那么它将造成巨大的损害。可解释人工智能可以更好地理解复杂问题。可解释性对于CMLA和深度学习架构至关重要，因为单个错误决策可能极其昂贵。通常，深度神经网络学习分层特征表示。每一层都有多个具有相似结构但权重参数不同的神经元。在网络安全系统中存在数据异质性的情况下，确保分类器使用正确的特征可能很棘手。可解释的机器学习模型可用于验证训练好的模型，或从模型中学习。通过敏感性分析可以学习预测的变化。它还讨论了可解释深度神经网络建模以解释预测的重要性。

with respect to accuracy, speed, and flexibility. To address this, M-ZOOM was developed providing promising results in terms of scalability, accuracy, flexibility, and effectiveness with an AUC score of 0.98 [30]. TensorDet, was developed to enhance real-time computational efficiency of tensor decomposition by exploiting the factorization structures with novel methodologies such as sequential tensor truncation and two-phase anomaly detection [31].

C. Cyber Security Applications in Off-line and Real-time Deployment

The application of DL architectures in cyber security for both off-line and real-time deployment requires important factors to be considered during designing the underlying models. An understanding of DL architectures, interpreting what the trained ML model has learnt is an important factor of a robust validation procedure. Interpretability is an essential factor in applications related to cyber security where the reliance of the model on the correct features must be established. Generally, the simple models (linear models) are easier to interpret than the complex models (non-linear models). Interpretability is the ability to understand what the predictions of features are, such as features of texts or images, while non-interpretability refers to hidden layer features, vectors spaces produced by say text representations and word embedding. Heat map is one of the most commonly used approaches to understand the classification decision. The pixel of a heat map image provides the contribution towards the classification.

There is no clear mathematical proof as well as theory to DL architectures interpretation and transparency. Thus, it is very difficult to arrive at a specific reason to identify why DL architecture model misclassifies a data sample. Identifying which DL architecture is more suitable, identifying optimal parameters for network structure and network parameters is one of the daunting tasks. Additionally, more practical knowledge is required to identify sensible values for parameters such as learning rate, regularizer, etc. Currently, these are determined on an ad-hoc basis. A method to identify the optimal number of feature maps was proposed and it worked well for extremely small receptive fields [32]. A visualization approach was proposed that facilitated for intermediate feature visualization [33].

Unavailability of well-known labeled benchmark datasets: Due to privacy and security reasons, the labeled datasets are not publicly available for research purpose. Labeling data samples by using manual approach is one of the daunting tasks. Most commonly used solution to label data sample is based on vendor provided blacklist and whitelist. Basically, there are three different types of datasets are used in ML. They are called as train, valid and test datasets and these datasets are disjoint to each other. It means when we are collecting a data sample to develop network traffic analysis system, we have to collect these datasets from different networks which include different users as well as different applications access. These three datasets should also include time information. It means the train data should be from t_1 , test dataset should be from $t_1 + 2$ and valid dataset from $t_1 +$

1. Anomaly detection is more popular in many domains and less preferred in the area of cyber security. This is due to the reason that achieving low false positive is one of the biggest tasks in cyber security anomaly detection. There are chances where a single misclassification can cause millions of dollars damage to the company. The semi-supervised and mostly unsupervised learning methodology is the preferred method in the domain of cyber security. The main important factors to be considered during dataset collection are 1) different qualities of measurement, 2) different subjects, 3) evolution of technology over time, 4) different ways of labeling examples, 5) different level of concentration, 6) different environments, 7) different protocols, and 8) time of the day.

Attacker-Defender Approach and Concept of Drift: cyber security is an evolving area, to adapt to new types of patterns used by adversary; the ML based system has to be continuously trained. Since the datasets generated by various ICT systems is huge, feature engineering is a difficult task, thus in this case application of DL architectures can be used. This helps to lean the different types of new patterns used by an adversary simply following pertaining method.

Imbalanced data samples: Data imbalance is one of the most common problems in cyber security. Most of the time the samples of malware are rare and particularly almost all the data's are imbalanced in multiclass classification in the field of cyber security.

Domain adaptation: Domain adaptation is a method to measure the difference between train and test datasets. Both of these datasets distribution should be completely different. The domain of cyber security contains many forms of datasets; this includes network traffic, spam, phishing, etc. These are highly correlated and can help to detect malware effectively. A major challenge is to adopt an effective defense method from one domain to another.

Important factors to be considered in deployment of ML models in real-time systems: Though as these ML algorithms and DL architectures have the capability to discriminate the new types of malicious patterns, there are still is in early stage in adopting in enterprise security systems. Recently, a new research direction typically called as explainable AI can give better reasons for incorrect decision. The incorrect decision in cyber security system can cause dollars of damage. For example, If a legitimate application is flagged as malicious and the application is not acceptable by any of them in a working hours in an enterprise system, then it is going to cause a lot of damages. The explainable AI can better understand the complex problems. Interpretability is crucial for CMLAs and DL architecture because a single wrong decision can be extremely costly. Generally, DNNs learn hierarchical feature representations. Each layer has multiple neurons with similar structure but with different weight parameters. In the presence of the data heterogeneity in cyber security systems, it can be tricky to ensure that the classifier uses the right features. Interpretable ML model can be used to validate a trained model, or to learn something from the models. Variation in the prediction can be learned by using sensitivity analysis. It also discusses the importance of interpretable DNNs modeling explaining the predictions.

D. 可解释人工智能在网络安全中的作用

如今，基于人工智能的系统在执行特定任务时，其速度和效果已超越人类智能，这促使人类依赖这些系统在实时环境中做出复杂决策。然而，虽然人类决策可以通过解释来证明其合理性，但基于人工智能的推理在每一步中的基本原理通常无法解释模型的具体操作、选择过程等。由于这一局限性，基于人工智能的决策在决策者中缺乏信任和信心。当安全因素被考虑时，信任是主要的接受因素之一。为了克服这一局限性，通过可解释人工智能（XAI）[48]可以逐步提升对基于人工智能模型所做决策的信任。可解释人工智能是一个可行且理想的概念，它允许人类了解人工智能的决策过程，因为该技术通过算法为所有发生的事情（包括结果背后的原因）提供理由和解释。这里最大的挑战是创建人类可以信任的基于人工智能的系统。一项近期研究对可解释人工智能进行了详细调查 [49]。

深度学习是一种最先进的[[技术]]，其模型由多个层组成，与人类大脑的构成有相似之处。虽然深度学习在网络安全中的实时部署仍在发展中，但可解释人工智能被用于对抗性方法，以解决身份识别系统中的错误分类，并通过图形显示直观地解释了错误分类的原因 [44]。在另一项研究中，通过正则化技术对基于深度学习的控制系统（CPS）进行了一项调查，以提升其泛化能力 [16]。此外，还讨论了关键基础设施安全与韧性（CISR） [45]。提出了LEMNA方法，该方法将深度学习模型视为黑盒，并为网络安全中的每个分类结果提供解释 [46]。另一项研究通过将结果映射到3个不同的任务，并使用各种解释和可视化方法进行了详细分析，展示了可解释人工智能 [47]。

E. 基于深度学习的因果关系理论在网络安全中的应用

当前最先进的深度学习算法已被应用于网络安全，并提升了准确率。然而，由于缺乏明确的推理机制来理解这些模型所使用的推理过程，它们并未得到广泛商业应用。因果推理是用于此目的的一种方法，因为它能够回答与数据分布和干预变化相关的问题。因果理论可应用于网络安全，以理解深度学习模型，并回答涉及现有框架变化的“假设”类型问题。研究人员提出了一种使用因果推理来理解深度学习架构的框架，以展示该模型的有效性 [96]。另一项研究在网络流量中应用因果理论，以确认TCP-SYN洪水DDoS攻击 [97]。此外，因果关系防御措施被用于攻击检测 [98]。PRIOTRACKER被提出用于通过优先调查异常因果依赖关系来跟踪进程 [99]。

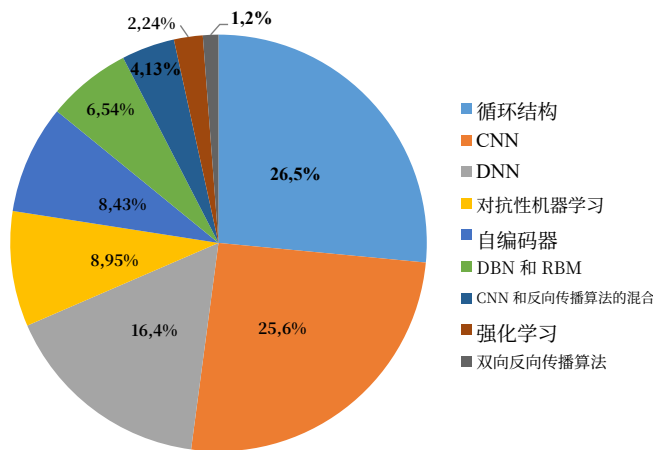


图6. 网络安全中深度学习架构的统计数据。

IX. 网络安全领域中深度学习应用的统计分析

近年来，将新型深度学习方法应用于网络安全，并评估其性能以构建最优深度学习框架，已成为安全研究人员的关键研究方向。图6提供了流行深度学习架构对网络安全贡献百分比的统计总结。由于深度学习架构众多，我们将相似的架构分组，详细信息如下。

- 1) DBN、RBM（所有其他DBN和RBM相关架构）。
- 2) DNN - 具有多个全连接层的网络，包括 $ReLU$ （包括Invincea、Endgme、NYU、CMU、MIT）。
- 3) 自编码器和其他自编码器变体。
- 4) CNN及其变体架构，如AlexNet、VGG16、SqueezeNet、Inception-BN-21k、Inception-BN-1k、Inception V4和ResidualNet152。
- 5) 循环结构 - RNN、LSTM、GRU和IRNN。
- 6) CNN与循环结构的混合 - CNN-RNN、CNN-LSTM、CNN-GRU、CNN-IRNN。
- 7) 双向循环结构 - 双向循环神经网络、双向LSTM、双向门控循环单元和其他变体。
- 8) 强化学习和其变体。
- 9) 对抗性机器学习。

从图6可以看出，基于循环结构的深度学习架构、卷积神经网络和深度神经网络被广泛使用，其使用率均超过了卷积神经网络和深度神经网络。这可能是因为大多数网络安全数据集都涉及序列和时间序列信息，而这类信息与深度学习架构中的循环结构非常契合。

图7展示了采用深度学习方法的各种网络安全应用研究的统计总结。我们考虑了以下13种网络安全应用：1) Windows恶意软件检测，2) Android恶意软件检测，3) 入侵检测，4) 网络流量分析，5) DGA、电子邮件、URL和安全日志数据分析，6) 侧信道攻击检测，7) 内部威胁检测，8) 功能识别，9) 隐写分析和隐写术，10)

D. Role of Explainable Artificial Intelligence in Cyber Security

Today, AI-based systems can perform specific tasks quicker and better than human intelligence, which motivate humans to rely on these systems for making complex decisions in real-time. However, while human based decisions are justified through explanations, the fundamental principles of AI based reasoning in each step are usually not explainable in terms of what the model did, how the choice was made, etc. Due to this limitation, AI based decisions carry low trust and confidence among decision makers. Trust is one of the major acceptance factors when safety comes into picture. To overcome this limitation, trust on decisions made by AI based models could be improved over time by Explainable AI (XAI) [48]. Explainable AI is a possible and desirable concept where humans can access the decision-making procedure of the AI as it gives reasons and explanations for everything that is happening by the algorithms including the reason behind the outcome. The biggest challenge here is to create AI based systems which humans can trust. A recent study undertook a detailed survey on explainable AI [49].

DL is a state-of-the-art technique where models are composed of multiple layers that has similarities with the human brain composition. While real-time deployment of DL in cyber security is still growing, XAI was applied using an adversarial approach for the incorrect classification in an ID system and the reason behind the incorrect classification was explained visually with graphical displays [44]. In another study, a survey to improve the generalization capability of DL based CPS using regularization techniques was conducted [16]. Further, Critical Infrastructure Security and Resilience (CISR) were discussed [45]. LEMNA, a method to treat a DL model as a black box and derive explanations for each and every classification outcome in cyber security was proposed [46]. Another study demonstrated XAI by the mapping outcomes to 3 different tasks with detailed analysis using various interpretation and visualization methods [47].

E. Casual Theory with DL for Cyber Security

State of art DL algorithms have been applied to cyber security with enhanced accuracies. However, they are not widely used commercially due to the fact that there is no explicit reasoning to understand the inference mechanism used by these models. Causal inference is one of the methods used for this purpose as it has the ability to answer questions related to data distribution and intervention changes. Causal theory could be applied to cyber security for understanding DL models and in answering what-if type of questions that involves changes to the existing framework. A framework to understand a DL architecture using causal inference was proposed to show the effectiveness of this model [96]. Another study used causal theory in network traffic to confirm the TCP-SYN flooding DDoS attacks [97]. Further, causality countermeasures were utilized for detection of attacks [98]. PRIOTRACKER was proposed for tracking processes by prioritizing the investigation of abnormal causal dependencies. [99].

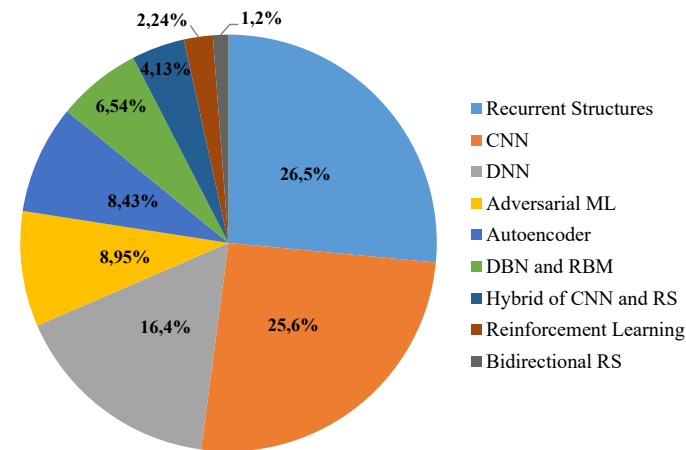


Fig. 6. Statistics of the DL Architectures in cyber security.

IX. A STATISTICAL SUMMARY OF DL APPLICATIONS IN CYBER SECURITY

In recent years, applying novel DL methods to cyber security as well as evaluating their performance to arrive at an optimal DL framework have become key research directions for security researchers. Fig. 6 provides a statistical summary of the percentage of contribution by popular DL architectures for cyber security. Since there are many DL architectures, we have grouped similar architectures to a group and the details are given below.

- 1) DBN, RBM (all other variants of DBN and RBM related architectures).
- 2) DNN - a network with many fully connected layers with $ReLU$ (includes Invincea, Endgme, NYU, CMU, MIT).
- 3) Autoencoder and other variants of Autoencoder.
- 4) CNN and variant architectures such as AlexNet, VGG16, SqueezeNet, Inception-BN-21k, Inception-BN-1k, Inception V4, and ResidualNet152.
- 5) Recurrent structures - RNN, LSTM, GRU, and IRNN.
- 6) Hybrid of CNN and Recurrent structures - CNN-RNN, CNN-LSTM, CNN-GRU, CNN-IRNN.
- 7) Bidirectional recurrent structures - Bidirectional recurrent neural network, Bidirectional LSTM, Bidirectional gated recurrent unit, and other variants.
- 8) Reinforcement learning and variants.
- 9) Adversarial ML.

From Fig. 6, it is evident that DL architectures based on recurrent structures, CNN and DNN are largely used with surpassing both CNN and DNN. This may be due to the reason that most of the cyber security datasets involves the sequencing and time series information which forms a good fit with recurrent structures of DL architectures.

In Fig. 7 we present the summary of statistics of various studies of cyber security applications that have adopted DL approaches. We have considered the following 13 cyber security applications: 1) Windows malware detection, 2) Android malware detection, 3) intrusion detection, 4) network traffic analysis, 5) DGA, Email, URL, and security log data analysis, 6) side channel attack detection, 7) insider threat detection, 8) function recognition, 9) steganalysis and steganography, 10)

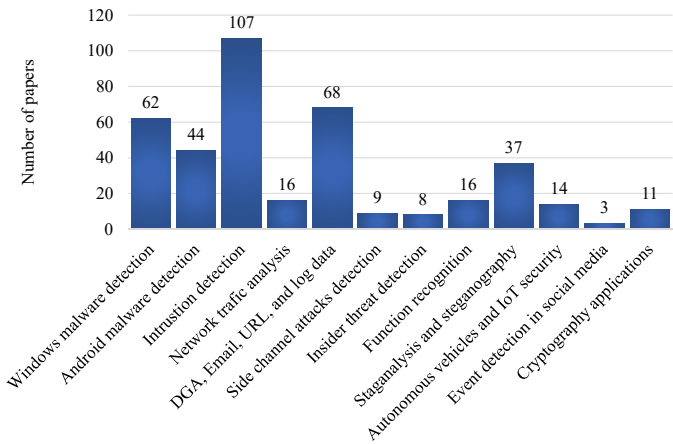


图7. 基于深度学习的网络安全应用统计。

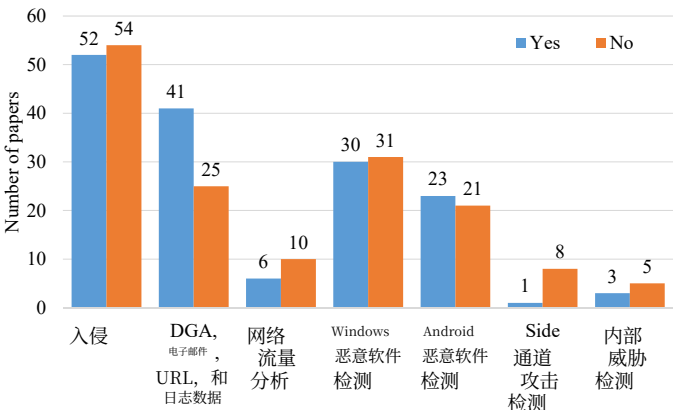


图8. 基于与经典机器学习算法的比较，网络安全应用深度学习方法统计。

内部威胁检测, 11) 自主车辆中的攻击检测, 12) 社交媒体中的事件检测, 13) 密码学应用。在这些应用中，我们发现深度学习架构主要应用于入侵检测，文献中报告的研究数量最多。其次是Windows恶意软件检测、Android恶意软件检测和DGA、电子邮件、URL及安全日志数据分析。

在基于深度学习的网络安全应用的研究工作中，主要的已发表研究尚未比较深度学习架构与现有CMLA的性能。这非常必要，因为在某些网络安全应用中，CMLA比深度学习架构更充分。为了识别这一点，我们在图8中展示了基于深度学习的各种网络安全应用的统计数据。我们考虑了以下网络安全应用：1) 入侵检测，2) DGA、电子邮件、URL及安全日志数据分析，3) 网络流量分析，4) Windows恶意软件检测，5) Android恶意软件检测，6) 侧信道攻击检测，以及7) 内部威胁检测。该图表明，基于深度学习架构的已发表研究工作较少与CMLA比较结果。最重要的是，在大多数研究工作中，深度学习架构的表现优于CMLA，这证明深度学习架构比CMLA更高效和稳健。

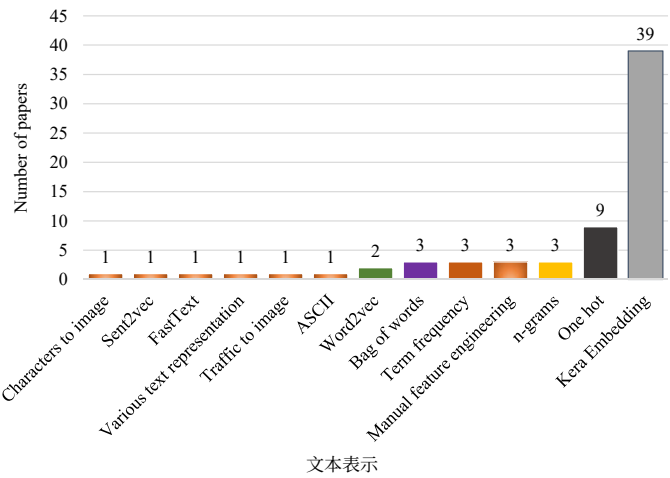


图9. 基于与自然语言处理文本表示的比较，网络安全应用深度学习方法统计（针对DGA、电子邮件、URL及安全日志数据分析）。

自然语言处理是一个重要的领域，在网络安全中有许多重要的应用。它处理将文本转换为数值表示。存在许多文本表示方法，它们的性能隐含地取决于文本表示。在过去的几年里，各种基于深度学习的已发表论文使用了各种文本表示方法。我们展示了针对DGA、电子邮件、URL及安全日志数据分析的已发表工作的各种文本表示方法的统计数据，如图9所示。大多数研究工作利用了Keras嵌入，因为它有助于保留文本中单词或字符的序列信息。然而，基于词嵌入模型的工作不多，因为大多数网络安全文本数据不涉及语义和上下文表示。我们考虑了以下文本表示方法：1) 词袋模型，2) n-gram，3) 独热编码，4) ASCII表示，5) Keras嵌入，6) Word2vec（Sent2vec），7) FastText，8) 将字符转换为图像，以及9) 手动特征工程。图9表明，Keras嵌入文本表示在基于深度学习的已发表研究研究中大量使用。这可能是因为网络安全文本数据中序列特征更重要，而Keras嵌入能够捕获序列信息，而其他文本表示方法（如词袋模型、词频和独热编码）无法捕获序列特征。词嵌入也可以学习文本中的序列特征，但与Keras嵌入相比，它在计算上更昂贵。

数据集非常重要，在基于深度学习的网络安全应用开发中发挥着重要作用。我们在上一节详细讨论了各种可用数据集中存在的问题。现有基于深度学习的网络安全应用中使用的数据集通常根据以下分类：1) 基准数据集可用于评估现有和新引入算法的性能，2) 从公开来源收集的数据集，3) 从实时环境中收集的数据被视为实时数据集，以及4) 私有数据集是那些不属于这三种类型的数据集。

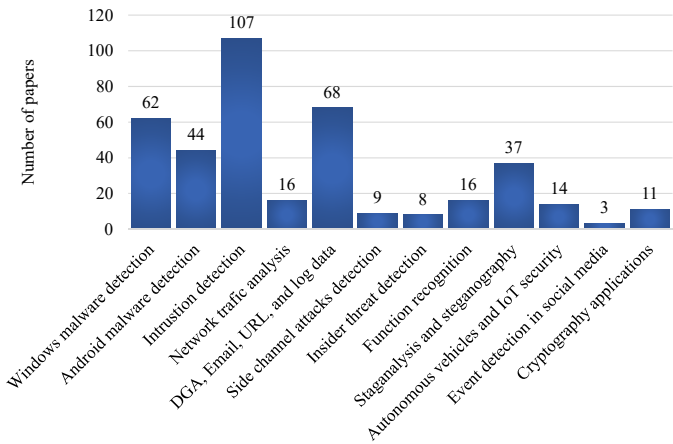


Fig. 7. Statistics of the DL based Cyber Security Applications.

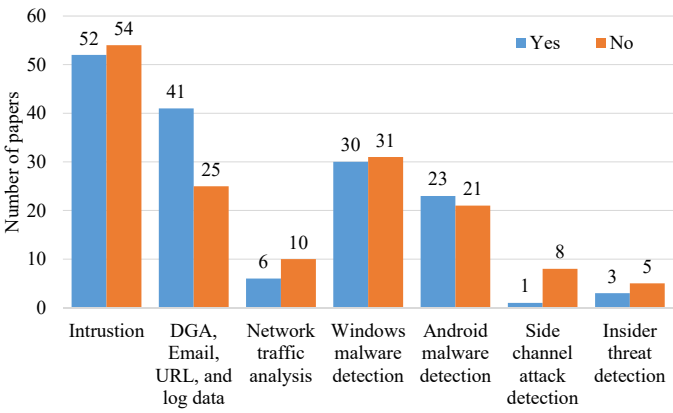


Fig. 8. Statistics of DL approach for Cyber Security Applications based on comparison with classical ML algorithms.

insider threat detection, 11) attacks detection in autonomous vehicles, 12) event detection in social media, 13) cryptography applications. Among these, we find that DL architectures are largely used in intrusion detection with highest number of studies reported in literature. Next comes Windows malware detection, Android malware detection and DGA, Email, URL and Security log data analysis.

Among the research works on DL based cyber security applications, the main significant published studies have not compared the performance of DL architecture with the existing CMLAs. This is very much required because for certain cyber security applications the CMLAs are more sufficient than the DL architectures. To identify this, we have shown the statistics of the various DL based cyber security applications in Fig. 8. We have considered the following cyber security applications: 1) intrusion detection, 2) DGA, Email, URL and Security log Data analysis, 3) network traffic analysis, 4) Windows malware detection, 5) Android malware detection, 6) side channel attacks detection, and 7) insider threat detection. The figure indicates that fewer published research works based on DL architectures have compared the results with CMLAs. Most importantly, the DL architectures outperformed the CMLAs in most of the research works justifying that DL architectures are more efficient and robust than the CMLAs.

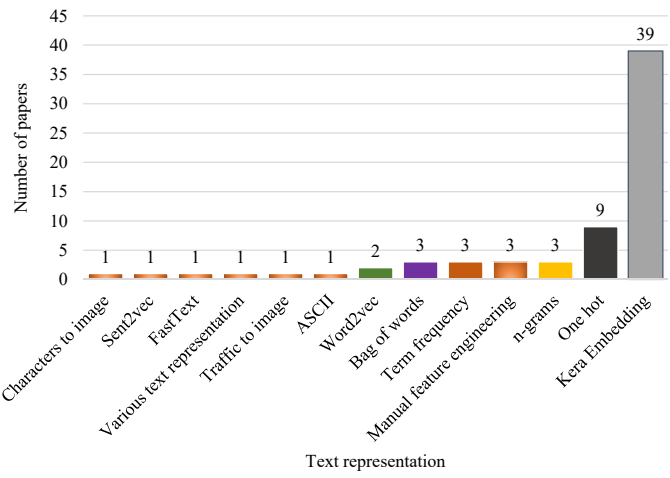


Fig. 9. Statistics of DL approach for DGA, Email, URL and Security log Data analysis based on comparison with NLP text representation.

NLP is an important domain which has many important applications in cyber security. It deals with conversion of text to numerical representation. There exist many text representations and their performance implicitly depend on the text representation. In the last years, various DL based published papers have used various text representations. We have shown the statistics of various text representation methods of published works specific to DGA, Email, URL, and Security log Data analysis as shown in Fig. 9. Most of the research works have utilized Keras embedding as it helps to preserve the sequence information of words or characters in the texts. However, not much work exists based on the word embedding models because most of the cyber security text data does not involve semantic and contextual representation. We have considered the following text representations: 1) Bag of words, 2) n-grams, 3) One hot, 4) ASCII representations, 5) Keras embedding, 6) Word2vec (Sent2vec), 7) FastText, 8) Characters converted into image, and 9) Manual feature engineering. Fig. 9 shows that the Keras embedding text representation is largely used in published research studies on DL. This may be due to the reason that sequential features are more important in cyber security text data and Keras embedding has the capability to capture sequential information while the other text representations such as Bag of Words, term frequency, and one hot encoding are not capable of capturing sequential features. Word embedding can also learn sequential features in the text but it is computationally expensive compared to Keras embedding.

The datasets are important and play an important role in the development of DL based cyber security applications. We have discussed that major issues exists in the various available datasets in detail in the above section. The datasets used in the existing DL based cyber security applications typically are categorized based on the following: 1) benchmark datasets can be used for performance evaluation of existing and as well as newly introduced algorithms, 2) datasets collected from publicly available sources, 3) real-time datasets are collected from real-time environment are considered as real-time datasets, and 4) private datasets are ones that do not

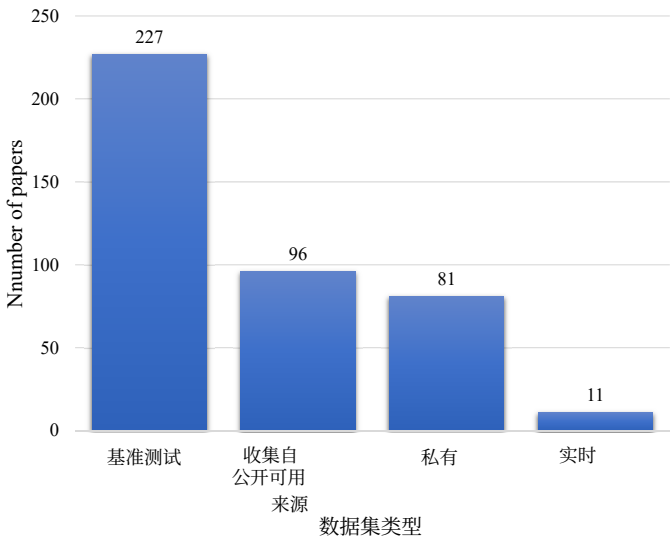


图10. 基于数据集分类的基于深度学习的网络安全应用统计。

这些数据集不属于这三种类型。基于数据集类型的深度学习网络安全应用详细统计数据如图10所示。虽然大多数研究工作使用了基准数据集，但有些研究工作使用属于其余类别的数据集。最重要的是，基于基准数据集的研究工作获得的结果可以在未来进行复现、比较和改进。因此，基准数据集对于未来研究中推进深度学习在网络安全中的应用至关重要。

X. 建议的组织网络安全混合系统

网络犯罪分子在不断改进他们的攻击技能，因此从互联网不同来源收集的大量用户行为数据使用深度学习方法进行实时监控和追踪。为了对海量数据进行预处理和应用深度学习方法，需要新的工具和技术。为此，我们提出一个高度可扩展的分布式计算平台，并采用基于深度学习的解决方案，以分布式方式收集数据，并使用分布式算法分析数据，旨在准确检测和分类安全事件。这有助于用户了解管理员是否需要采取行动。在该解决方案中，深度学习架构不仅用于处理和发现模式，还用于解释数据，旨在找出每个威胁的风险程度。所提出的解决方案监控网络和连接的设备，以识别与恶意攻击相关的异常事件。所提出的模型旨在为组织提供所需的态势感知能力，以应对其最紧迫的信任和安全问题。

基于从调研论文中获得的认知，我们提出的通用深度学习框架用于网络安全应用，包含多层安全防护，旨在更准确地检测恶意活动。该框架被设计得尽可能通用，是多种深度学习模型的混合体，旨在应对当今的网络安全挑战。

主要而言，该深度学习框架包含数据收集、数据预处理和基于深度学习的分类模块。在数据收集阶段，数据样本被动地从各种传感器中收集，并存储在NoSQL数据库中。这些原始数据样本随后被传递到预处理模块，该模块使用分布式日志解析器提取重要信息。最后，信息将被传递到基于深度学习的分类模块，用于检测和分类恶意活动。该框架更通用，能够应对现代社会中的各种网络安全挑战。它包含以下子模块：

- 基于DNS数据分析的深度学习网络威胁态势感知子模块。
- 用于深度学习的网络安全威胁检测的全球BGP更新分析子模块。
- 基于深度学习的垃圾邮件和网络钓鱼检测子模块，使用URL、电子邮件和社交媒体数据分析。
- 基于深度学习的混合入侵检测子模块，可在网络和主机层面检测攻击。
- 用于网络流量分析的深度学习方法子模块。
- 用于深度学习的恶意软件二进制分析子模块，用于识别恶意软件结构和行为的详细信息。

- 基于深度学习的勒索软件识别子模块。
- 基于深度学习和可视化的物联网智慧城市僵尸网络检测子模块。
- 基于深度学习的Android恶意软件分析子模块。
- 基于深度学习的云应用操作日志异常检测子模块。

- 基于深度学习和科学计算模型的恶意软件传播建模子模块。

- 基于深度学习的网络异常检测因果方法子模块。

- 使用图像处理和深度学习的恶意软件可视化子模块。

- 使用区块链技术的隐私保护深度学习子模块。

- 适用于云环境的基于深度学习的网络安全子模块。

- 适用于雾计算的基于深度学习的网络安全子模块。
- 适用于密码学的基于深度学习的网络安全子模块。
- 基于生成对抗网络的对抗环境下增强深度学习模型鲁棒性的子模块。

- 基于强化学习的子模块，通过预训练来提升系统性能。

由上述提到的子模块共同组成的组织网络安全系统，能够更准确地检测攻击，并快速阻断恶意活动与目标主机之间的通信点。

XI. 结论与未来研究

随着大数据和物联网的技术进步，未来将迎来工业4.0时代，本文讨论了需要全面回顾研究以应对CPS中恶意攻击的需求，从而引出本研究的核心重点。

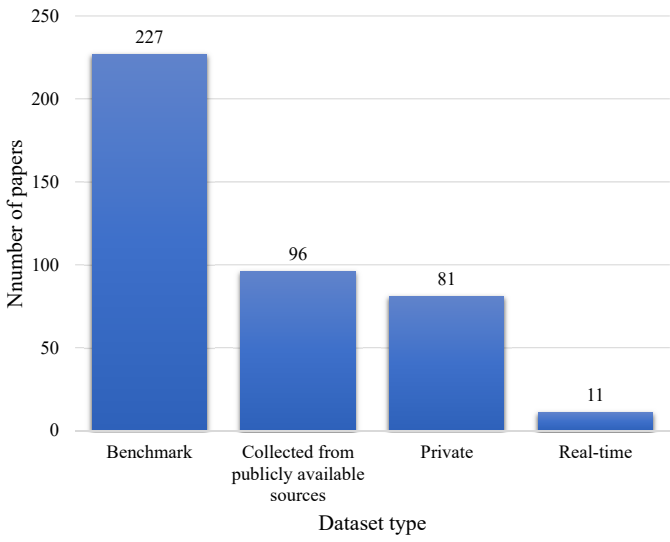


Fig. 10. Statistics of DL based Cyber Security Applications based on the dataset classification.

belong to these three types. The detailed statistics of the DL application in cyber security based on dataset type is shown in Fig. 10. While most of the research works have used the benchmark datasets, some research works use datasets which belong to the remaining categories. Most importantly, the results obtained in research works based on the benchmark datasets can be reproduced, compared, and enhanced in future. Thus, the need for benchmark datasets is critical for advancing DL applications in cyber security in future research studies.

X. SUGGESTED HYBRID SYSTEMS FOR ORGANIZATIONAL CYBER SECURITY

Cybercriminals are in constant improvement of their attack skills and hence huge amount of data about user behaviors from different sources over the Internet is processed using DL methods to monitor and trace them in real-time. In order to preprocess and apply DL methods on the massive amount of data, new tools and technologies are warranted. To achieve this, we propose a highly scalable distributed computing platform with a DL based solution to collect data in a distributed manner and use distributed algorithms to analyze data with the aim to accurately detect and classify security events. This facilitates the user to know whether the administrator has to take an action. In this solution, DL architectures are used to not only process and find patterns but also to interpret data with the aim to find the degree of risk in each threat. The proposed solution monitors the network and the connected devices to identify variations from normal events that are associated with malicious attacks. The proposed model is designed to provide organizations with the situational awareness needed to deal with their most pressing issues of trust and security.

Based on the knowledge obtained from the surveyed papers, our proposed general DL framework for cyber security applications consists of multiple layer of security with the aim to detect malicious activities more accurately. The proposed framework is considered to be as generic as possible and it is a hybrid of many DL models with the aim to meet

today’s cyber security challenges. Primarily the proposed DL framework contains data collection, data preprocessing, and DL based classification modules. In the Data collection phase, the data samples are collected passively from various sensors and stored in NoSQL database. These raw data samples are further passed into preprocessing module which extracts important information using distributed log parser. Finally, the information will be passed into DL based classification module to detect as well as classify the malicious activities. The proposed framework is more general to handle various cyber security challenges in the modern society. It contains the following sub modules;

- Cyber threat situational awareness sub module based on DNS data analysis using DL.
- A sub module to analyze the global BGP updates for cyber security threat detection using DL.
- DL based sub module for Spam and Phishing detection using URL, Email, and social media data analysis.
- DL based hybrid intrusion detection sub module which can detect attacks at network and host level.
- DL approach sub module for network traffic analysis.
- A sub module for identification of detailed information on the structure and behavior of the malware using malware binary analysis using DL.
- DL based sub module for ransomware identification.
- DL and Visualization sub module for Botnet Detection in the Internet of Things of Smart Cities.
- DL based sub module for Android malware analysis.
- DL based anomaly detection sub module using operational logs in cloud applications.
- Malware spread modeling sub module using DL and scientific computing models.
- A Casual approach with DL sub module for the network anomaly detection.
- Malware visualization sub module using image processing and DL.
- Privacy preserved DL sub module using block chain technologies.
- DL based cyber security sub module for Cloud environment.
- DL based cyber security sub module for fog computing.
- DL based cyber security sub module for Cryptography.
- GAN based sub module for enhancement of robustness of DL models in an adversarial environment.
- Reinforcement learning based sub module for enhancing the system performance through pretraining.

An organisation’s cyber security system composed of the above mentioned sub modules collectively is capable of detecting attacks more accurately and results in blocking the communication point between the malicious activities and the target host very quickly.

XI. CONCLUSION AND FUTURE RESEARCH

With technological advancements of BD and IoT resulting in Industry 4.0 of the future, the need for a well-rounded review of studies to overcome malicious attacks in CPS was discussed in this paper leading to the main focus of the study.

深度学习技术能够从大量数据中捕捉模式以区分合法和恶意活动，这构成了本研究进行广泛回顾适合网络安全的深度学习架构的关键动机。因此，本综述论文采用教程风格呈现，首先介绍了现有的经典机器学习算法和深度学习架构，以应对零日攻击。我们总结了文献中报道的关于网络安全深度学习应用的主要目标、成就和局限性。此外，本文独特地根据深度学习架构类型、使用的数据集以及网络安全应用类型等标准，回顾了基于深度学习的网络安全应用。接下来，讨论了自然语言处理、信号与图像处理以及大数据分析在网络安全的应用中的重要性。然后，我们对应用于网络安全的各种深度学习架构进行了全面的文献综述，包括采用可解释人工智能、迁移学习、强化学习和对抗深度学习的最新研究。最近，由于攻击者遵循对抗性机器学习来绕过基于深度学习的模型，并且由于文献中研究的匮乏，本文回顾了深度学习方法在对抗环境中的鲁棒性，为未来的研究工作开辟了重要途径。

我们考虑了网络安全在智慧城市、物联网、云计算和边缘计算、生物识别、普适计算、区块链以及因果理论等新兴领域中的重要性，并讨论了深度学习在这些下一代技术中的应用。我们提出了一种基于深度学习的混合框架，具有不同层次的安全机制，以更准确地学习恶意软件和合法活动的特征。该框架将实时进化，以检测和预防高级攻击。

总之，由于大数据和物联网在CPS中的兴起，深度学习是所有网络安全应用的重要方法。大数据中的数据分布高度非线性、嘈杂且不完整。经典机器学习算法不足以处理大数据，其性能是网络安全应用中需要解决的问题。最后，通过我们的综述我们发现，深度学习在网络安全问题中的应用是合适的，因为它们非常适合学习具有大量特征和高阶多项式项的复杂非线性假设，这是由于大数据的维度所致。总之，这项关于深度学习在网络安全中应用的详细综述为广泛的研究领域提供了深刻的见解，将激励研究人员推进深度学习在网络安全中的应用。

当多种学习方法涌现时，未来需要开展良好的研究来识别最优的深度学习架构。共享任务是推动网络安全研究中深度学习应用的前沿方式之一，预计未来将会有更多共享任务的举办。另一个新兴的方面是，由于基准数据集的获取有限，许多已发表的研究未能使用私有数据集和基准数据集的多种修改版本。由于私有基准数据集的修改版本无法公开用于进一步研究，它们的大多数解决方案

无法直接进行比较。这可能会削弱未来针对实验评估以确定任何实际环境最佳深度学习架构的研究。在选择有效且实用的深度学习架构时，必须考虑多个检查清单，例如准确率、空间和时间复杂度、检测新恶意软件的能力、在实时系统中易于集成和部署，以及在对抗环境中的鲁棒性。另一个需要考虑的重要因素是，目前大多数用于网络安全应用的深度学习模型都是使用监督学习方法训练的，这需要标记数据集，而收集它们成本高昂。因此，当代深度学习网络安全应用的学习方法至少需要半监督，最多需要无监督。为了能够自适应地检测新攻击，深度学习架构需要在网络安全领域进行重新训练，这意味着它依赖于增量学习和终身学习。探索深度学习在网络安全中应用的增量学习和终身学习类型，是未来研究在为工业4.0加强安全方面的重要方向。

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The ability of DL techniques to capture patterns from large volume of data to distinguish the legitimate and malicious activities formed the key motivation of the study to conduct an extensive review of DL architectures suitable for cyber security. Hence, this survey paper was presented in a tutorial-style, initially with a description of existing classical ML algorithms and DL architectures to address zero-day attacks. We summarized the major objectives, achievements and limitations of studies on DL applications in cyber security that were reported in literature. In addition, the paper uniquely reviewed the DL based cyber security applications according to various criteria such as the type of DL architecture, datasets used and the type of cyber security applications. Next, the importance of NLP, signal and image processing, and big data analytics in cyber security applications were discussed. Then, we carried out a comprehensive literature review of various DL architectures applied in cyber security, including state-of-the-art studies conducted with explainable AI, transfer learning, reinforcement learning, and adversarial deep learning. Recently, since attackers follow adversarial ML to bypass the DL based models and due to the paucity of studies in literature, in this paper we reviewed the robustness of DL methods in adversarial environments towards significant avenues for future research work.

We considered the importance of cyber security in emerging areas such as smart cities, IoT, cloud and edge computing, biometrics, pervasive computing, blockchain, and causal theory and discussed DL applications for such next generation technologies. We proposed a DL based hybrid framework with different layers of security to learn the characteristics of malware and legitimate activities more accurately. The proposed framework would evolve in real-time to detect and prevent from advanced attacks.

To conclude, DL is an important method for all the cyber security applications due to the rise of big data and IoT in CPS. The data distribution in big data is highly non-linear, noisy and dirty. Classical ML algorithms are not sufficient to deal with big data and the performance is of concern to be addressed in cyber security applications. Finally, through our review we find that DL applications for cyber security problems are suitable as they are well-suited to learn complex non-linear hypotheses with large number of features and high-order polynomial terms due to the dimensions of big data. In summary, this detailed survey on DL applications in cyber security provides a deep insight into a wide spectrum of research studies that would motivate researchers to advance the state of DL applications in cyber security.

When several types of learning approaches emerge, good research studies in future are required to identify an optimal DL architecture. Shared tasks are one of the prominent ways to push DL applications in cyber security research forward and organisation of shared tasks is anticipated to be more in the near future. Another emerging aspect is that due to limited availability of benchmark datasets, few of the published research studies have utilized the private datasets and various modified versions of benchmark datasets. Since private and modified version of benchmark datasets are not publicly available for further research, most of their solutions

are not directly comparable. This can undermine possible future research towards experimental evaluation in arriving at the best-fit DL architecture for any real-life environment. When choosing an effective and practical DL architecture, several checklists have to be considered such as accuracy, space and time complexity, ability to detect new malwares, easy integration and deployment in real-time system, and robustness in an adversarial environment. Another important factor to be considered is that most of the existing DL based models for cyber security applications are trained using supervised learning approaches, which require the labeled datasets and it is expensive to collect them. Therefore, there is a dire need for the learning methodology of contemporary DL based cyber security applications to be at least semi-supervised and at most unsupervised. in order to self-adapt to detect new attacks, DL architectures require retraining in cyber security domain, which implies that it relies on incremental learning and continual lifelong learning. Exploring the types of incremental learning and continual lifelong learning with DL applications in cyber security is another significant direction of future research towards enforcing security for Industry 4.0.

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Vinayakumar Ravi 于印度阿默拉巴德阿默拉瓦蒂希瓦维迪佩瑟姆大学阿默拉巴德工程学院计算机科学专业获得博士学位。他目前是辛辛那提儿童医院医学中心（辛辛那提，俄亥俄州，美国）的一名博士后研究员，从事开发和实现用于大数据集成和数据分析的新型计算和机器学习算法与应用。他于2014年在阿默拉瓦蒂希瓦维迪佩瑟姆大学迈索尔分校获得MCA学位，并于2011年在奥蒂斯JSS学院艺术、商业与科学学院获得BCA学位。他在机器学习应用于网络安全方面发表了多篇论文。他的博士研究工作集中于机器学习（有时是深度学习）在网络安全中的应用，并讨论了自然语言处理、图像处理和网络安全大数据分析的重要性。他参与了多项国际共享任务，并在SSCC’ 18和ICACCI’ 18框架下组织了检测恶意域名（DMD 2018）的共享任务。他的研究兴趣包括用于网络安全的机器学习和深度学习应用，以及自然语言处理和图像处理。更多详情请访问 <https://vinayakumarr.github.io/>。

他在机器学习应用于网络安全方面发表了多篇论文。他的博士研究工作集中于机器学习（有时是深度学习）在网络安全中的应用，并讨论了自然语言处理、图像处理和网络安全大数据分析的重要性。他参与了多项国际共享任务，并在SSCC’ 18和ICACCI’ 18框架下组织了检测恶意域名（DMD 2018）的共享任务。他的研究兴趣包括用于网络安全的机器学习和深度学习应用，以及自然语言处理和图像处理。更多详情请访问 <https://vinayakumarr.github.io/>。



Mamoun Alazab 获得澳大利亚联邦大学科学学院、信息技术与工程学院的计算机科学博士学位。他是澳大利亚查尔斯·达尔文大学工程、IT与环境学院的副教授。Alazab是一位拥有行业 and 学术经验的网络安全研究人员和实践者。他的研究是多学科的，专注于计算机系统的网络安全和数字取证，包括网络环境中的当前和新兴问题，如

物联网和 cyber-physical systems, 重点关注 cyber crime 检测和预防。他发表了 100 多篇研究论文。他发表了多次受邀和主题演讲，仅在 2018 年就举办了 22 场活动。他主持和召集了 50 多次会议和研讨会。他与政府和行业在许多项目上密切合作。他是多个编辑委员会的编委，包括 IEEE Access 的副主编、Security and Communication Networks Journal 的主编，以及 Journal of Digital Forensics, Security and Law (JDFSL) 的书评部分主编。他是 IEEE 的高级会员。

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Vinayakumar Ravi received the Ph.D. degree in computer science from Computational Engineering & Networking, Amrita School of Engineering, Coimbatore, Amrita Vishwa Vidyapeetham, India. He is currently a Postdoctoral research fellow in developing and implementing novel computational and machine learning algorithms and applications for big data integration and data mining with Cincinnati Children’s Hospital Medical Center, Cincinnati, OH, USA. He has received MCA from Amrita Vishwa Vidyapeetham, Mysore in 2014 and BCA from JSS College of Arts, Commerce and Sciences, Ooty road, Mysore in 2011. He has several papers in Machine Learning applied to Cyber Security. His Ph.D. work centers on Application of Machine learning (some times Deep learning) for Cyber Security and discusses the importance of Natural language processing, Image processing and Big data analytics for Cyber Security. He has participated in several international shared tasks and organized a shared task on detecting malicious domain names (DMD 2018) as part of SSCC’18 and ICACCI’18. His research interests include machine learning and deep learning applications with natural language processing and image processing for Cyber Security. More details available at <https://vinayakumarr.github.io/>.



Mamoun Alazab received his PhD degree in Computer Science from the Federation University of Australia, School of Science, Information Technology and Engineering. He is an Associate Professor in the College of Engineering, IT and Environment at Charles Darwin University, Australia. He is a Cyber Security researcher and practitioner with industry and academic experience. Alazab’s research is multi-disciplinary that focuses on Cyber Security and digital forensics of computer systems including current and emerging issues in the cyber environment like cyber-physical systems and internet of things with a focus on cyber crime detection and prevention. He has more than 100 research papers. He delivered many invited and keynote speeches, 22 events in 2018 alone. He convened and chaired more than 50 conferences and workshops. He works closely with government and industry on many projects. He is an editor on multiple editorial boards including Associate Editor of IEEE Access, Editor of the Security and Communication Networks Journal, and Book Review Section Editor: Journal of Digital Forensics, Security and Law (JDFSL). He is a Senior Member of the IEEE.



K. P. Soman 在阿鲁米塔工程学院科钦拥有25年的研究和教学经验。他在国内外期刊和会议论文集中发表了约150篇论文。他组织了一系列关于高级信号处理（使用小波）、模式分类的核方法、深度学习和大数据分析的研讨会和暑期学校，面向工业界和学术界。他著有《小波洞察》、《数据挖掘洞察》和《支持向量机和其他核》等书籍。

“深度学习方法”和“信号与图像处理——稀疏方法”，由普赖斯霍尔出版社、新德里和 Elsevier 出版。更多详情请访问 <https://nlp.amrita.edu/somankp/>。



Simran K 是印度阿默里塔工程学院计算工程与网络专业的M.Tech学生。她于2018年在印度海得拉巴的MVSR工程学院获得了信息技术学士。她的兴趣领域包括机器学习、深度学习、自然语言处理、网络安全、图像处理和物联网（IoT）。更多详情请访问<https://simranketha.github.io/>。



K. P. Soman has 25 years of research and teaching experience at Amrita School of Engineering, Coimbatore. He has around 150 publications in national and international journals and conference proceedings. He has organized a series of workshops and summer schools in Advanced signal processing using wavelets, Kernel Methods for pattern classification, Deep Learning, and Big-data Analytics for industry and academia. He authored books on "Insight into Wavelets", "Insight into Data mining", "Support Vector Machines and Other Kernel Methods" and "Signal and Image processing-the sparse way", published by Prentice Hall, New Delhi, and Elsevier. More details available at <https://nlp.amrita.edu/somankp/>.



Simran K is a M.Tech student in the Computational Engineering & Networking, Amrita School of Engineering, Coimbatore, Amrita Vishwa Vidyapeetham, India. She has received B.E in Information Technology from MVSR engineering college, Hyderabad, India in 2018. Her areas of interests are Machine Learning, Deep Learning, Natural language processing, Cyber Security, Image processing and Internet of Things (IoT). More details available at <https://simranketha.github.io/>.



Sriram Srinivasan 是印度阿默拉巴特工程学院计算工程与网络专业的M.Tech学生。他在2016年从印度库姆贝科南萨斯特拉大学斯里那瓦萨·拉马努金中心获得了计算机科学与工程学士学位。他的兴趣领域包括机器学习、深度学习、网络安全和物联网（IoT）。更多详情请访问<https://thecodinglama.github.io/>。



Sriram Srinivasan is a M.Tech student in the Computational Engineering & Networking, Amrita School of Engineering, Coimbatore, Amrita Vishwa Vidyapeetham, India. He has received B.Tech in Computer Science and Engineering from Sastra University, Srinivasa Ramanujan Center, Kumbakonam, India in 2016. His areas of interests are Machine Learning, Deep Learning, Cyber Security and Internet of Things (IoT). More details available at <https://thecodinglama.github.io/>.



Sitalakshmi Venkatraman 博士在印度、新加坡、新西兰和自2007年起在澳大利亚的工业和学术界拥有超过30年的工作经验。她专门研究将高效的计算模型和数据挖掘技术应用于各种行业问题，最近通过与澳大利亚的工业和大学合作，在电子健康、电子安全和电子商业领域。目前，她领导着墨尔本理工学院的商业分析团队进行教学和研究。她已出版九本图书章节

以及在国际知名期刊和会议上发表的130余篇研究论文。她是专业协会和国际期刊编辑委员会的资深成员，并担任多个国际会议的程序委员会成员。



Dr. Sitalakshmi Venkatraman has more than 30 years of work experience both in industry and academics within India, Singapore, New Zealand, and in Australia since 2007. She specialises in applying efficient computing models and data mining techniques for various industry problems and recently in the e-health, e-security and e-business domains through collaborations with industry and universities in Australia. Currently she leads the Business Analytics team for teaching and research in Melbourne Polytechnic. She has published nine book chapters and more than 130 research papers in internationally well-known refereed journals and conferences. She is a Senior Member of professional societies and editorial boards of international journals and serves as Program Committee Member of several international conferences.



Quoc-Viet Pham (M’ 18)于2013年在越南河内科技大学获得电子与通信工程学士学位，2015年和2017年分别在日本仁济大学获得通信工程硕士和博士学位。他目前是韩国釜山国立大学计算机、信息和通信研究所的研究教授。2017年9月至2019年12月期间，他曾在庆熙大学和昌原国立大学工作，

金海大学担任多个学术职位。他于2017年获得金海大学工程领域最佳博士论文奖。他的研究兴趣包括凸优化、博弈论、机器学习以及移动边缘/云计算，以及5G无线网络和未来的资源分配。他是IEEE的成员。



Quoc-Viet Pham (M’18) received the B.S. degree in electronics and telecommunications engineering from Hanoi University of Science and Technology, Vietnam, in 2013, and the M.S. and Ph.D. degrees, both in telecommunications engineering, from Inje University, South Korea, in 2015 and 2017 respectively. He is currently a research professor at Research Institute of Computer, Information and Communication, Pusan National University, South Korea. From Sept. 2017 to Dec. 2019, he was with Kyung Hee University, Changwon National University, and Inje University on various academic positions. He received the best PhD thesis award in Engineering from Inje University in 2017. His research interests include convex optimization, game theory, and machine learning to mobile edge/cloud computing, and resource allocation for 5G wireless networks and beyond. He is a member of the IEEE.